MIL-M-28787/213A(EC)
7 August 1979
SUPERSEDING
MIL-M-28787/213(EC)
25 October 1977

MILITARY SPECIFICATION

* MODULES, STANDARD ELECTRONIC

CONVERTER, PARALLEL/SERIAL, KEY CODE EQG

This specification is approved for use by Naval Electronic Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document establishes the requirements and test procedures for the procurement of an electronic parallel/serial converter module.

1.2 Module description. The following electrical parameters reflect the 0° to 60°C end-of-life limits from table II.

	Name	Converter, parallel/serial
	Family	Digital
	Key code	EQG
	Size	1A
	Weight	35 grams maximum
	Failure rate	0.513 failures/10^6 hours maximum
*	Environment	Class I as specified in MIL-STD-1389
	Supply voltage	5.0 Vdc ±10 percent
	Maximum power dissipation	1,848 mW (outputs not loaded)
	Input voltage	
	Low level	-0.5 V minimum, 0.7 V maximum
	High level	2.0 V minimum, V_{CC} maximum
	Input current	
	D0-D31	
	Low level (V_{IL} = 0.0 V)	-0.40 mA maximum
	(V_{IL} = 0.4 V)	-0.36 mA maximum
	High level (V_{IH} = 2.5 V)	20 µA maximum
	(V_{IH} = 5.5 V)	100 µA maximum

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Input current -Continued		
SD		
Low level	(V_{IL} = 0.0 V)	-1.60 mA maximum
	(V_{IL} = 0.4 V)	-1.44 mA maximum
High level	(V_{IH} = 2.5 V)	80 µA maximum
	(V_{IH} = 5.5 V)	400 µA maximum
URC and LRC		
Low level	(V_{IL} = 0.0 V)	-3.91 mA maximum
	(V_{IL} = 0.4 V)	-3.52 mA maximum
High level	(V_{IH} = 2.5 V)	160 µA maximum
	(V_{IH} = 5.5 V)	800 µA maximum
HRM and SRM		
Low level	(V_{IL} = 0.0 V)	-3.20 mA maximum
	(V_{IL} = 0.4 V)	-2.88 mA maximum
High level	(V_{IH} = 2.5 V)	160 µA maximum
	(V_{IH} = 5.5 V)	800 µA maximum
Output voltage, SDO		
Low level	(I_{OL} = 4.0 mA)	0.4 V maximum
High level	(I_{OH} = -400 µA)	2.5 V minimum
Output drive current maximum capability, SDO		
Low level	(V_{OL} = 0.4 V)	4.0 mA
High level	(V_{OH} = 2.5 V)	-400 µA
Propagation delay time, t_{PLH}		
LRC to SDO	(C_L = 50 pF)	10 ns minimum, 45 ns maximum
	(C_L = 150 pF)	10 ns minimum, 60 ns maximum
Propagation delay time, t_{PHL}		
LRC to SDO	(C_L = 50 pF)	10 ns minimum, 50 ns maximum
	(C_L = 150 pF)	10 ns minimum, 60 ns maximum
Setup time, high level and low level		
D0-D31 and SD		20 ns minimum
HRM and SRM		40 ns minimum
Hold time		
D0-D31 and SDO		5 ns minimum
HRM and SRM		5 ns minimum
Pulse width		
URC and LRC		20 ns minimum
Maximum clock frequency		
URC and LRC	(C_L = 50 pF)	15 MHz
Clock delay		
URC to LRC		65 ns minimum

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* 1.3 Module function. This module contains two 32-bit ranks to perform parallel to serial conversion. Parallel loading of the upper rank is accomplished by taking the holding register mode (HRM) low. The parallel data (D0-D31) is loaded into the upper rank and appears at the input of the lower rank after the positive transition of the upper rank clock (URC). During loading, serial data flow is inhibited in the upper rank. Synchronous shifting of serial data in the upper rank occurs after the positive transition of the upper rank clock (URC) when the holding register mode (HRM) is held high. The lower rank clock (LRC) loads the contents of the upper holding resister when the shift register mode (SRM) is low or shifts data on SD into the lower register when SRM is high. Serial output data appears at the output SDO.

1.3.1 Input to output relationship. Each 32-bit parallel/serial converter has the input to output relationship of table I.

1.3.2 Parallel outputs. Paralleling of outputs is not permissible.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATION

MILITARY

* MIL-M-28787 - Modules, Standard Electronic, General Specification for.

STANDARDS

MILITARY

* MIL-STD-1389 - Design Requirements for Standard Electronic Modules.

* MIL-STD-1665 - Test Equipment for the Standard Electronic Modules Program.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 General requirements. The module shall comply with the requirements of MIL-M-28787 except as modified herein.

3.2 Correlation requirements. Before any production modules are tested, all testers and test equipment shall be correlated in accordance with MIL-M-28787.

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TABLE I. Input to output relationship.

UPPER RANK

Inputs				Upper rank contents at t_n	Upper rank contents at t_n +1
HRM	URC	SD	Parallel D0 ------ D31	Q0 Q1 Q2 ------ Q31	Q0 Q1 Q2 ------ Q31
L	↑	X	d0 ------ d31	X X X ------ X	d0 ---------- d31
H	↑	L	X ------ X	b0 b1 b2 ------ b31	L b0 b1 ------ b30
H	↑	H	X ------ X	b0 b1 b2 ------ b31	H b0 b1 ------ b30

LOWER RANK

Inputs			Upper rank contents	Output - SDO	
SRM	LRC	SD	Q0 ---------- Q31	t_n +1	t_n +32
L	↑	X	q0 ---------- q31	q31	---
H	↑	L	X ---------- X	---	L
H	↑	H	X ---------- X	---	H

NOTES:
1. H = High level (steady state).
2. L = Low level (steady state).
3. X = Irrelevant.
4. ↑ = Transition from low to high level.
5. d0-d31 = The level of steady state input at D0-D31 respectively.
6. b0-b31 = The level of steady state upper rank contents.
7. q0-q31 = The level of steady state lower rank input at Q0-Q31 respectively.
8. t_n = Bit time before transition of clock.
9. t_n +1 = Bit time after 1 low to high-clock transition.
10. t_n +32 = Bit time after 32 low to high-clock transitions.

* 3.3 Failure rate requirements. All modules procured to this specification shall have a predicted failure rate no greater than 0.513 failures per million operating hours when predicted in accordance with MIL-STD-1389.

* 3.4 Performance requirements. The module shall conform to the electrical requirements specified in table II and shall operate within the environment of design class I as specified in MIL-STD-1389.

3.5 Module configuration requirements. The module configuration requirements shall be in accordance with MIL-STD-1389 and as specified herein.

3.5.1 Key code. The module key code shall be EQG.

3.5.2 Size. The module size shall be 1A.

3.5.3 Weight. The module shall have a maximum weight of 35 grams.

3.5.4 Contact pins. The module shall have 40 contact pins positioned A1 through A40.

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3.5.5 Fin position. The module shall have an insertion and extraction and heat dissipation fin on increment A. The cross sectional area of the entire fin shall not exceed 57 percent of the area specified in the MIL-STD-1389 fin configuration requirements.

3.5.6 Depth. The module shall have no additional depth requirements.

3.5.7 Holddown holes. The module shall have no additional requirements for location or quantity of holddown holes.

3.5.8 Marking. Marking shall be as specified in MIL-STD-1389, and the module shall be marked with the military part number M28787/213-1, except no marking shall interfere with heat dissipation from the fin (for example, affixing a label which may be a heat barrier).

* 3.6 Circuit pin configuration. The function diagram and pin assignments shall conform to figure 1.

3.7 Acceptance test. All production modules shall be tested in accordance with notes 1 and 2 of table II.

3.8 End-of-life limits. Modules which have been plugged into an operating system, or exceed 2 years from date of manufacture and are to be tested, shall be capable of conforming to the 25°C end-of-life limits of table II.

3.9 Power supply voltage. The module shall be capable of conforming to the requirements of table II for a power supply voltage (V_{CC}) of 5.0 $\pm$0.5 Vdc, except that the limits for output short circuit current apply only for a power supply voltage of 5.5 Vdc $\pm$1 percent and the limits for V_{out} (low-loaded) (a maximum stress test) apply only for a power supply voltage of 7.0 Vdc $\pm$0, -1 percent.

3.10 Functional tests. All modules shall be subjected to functional tests which conform to the requirements specified in 3.10.1 through 3.10.3.

3.10.1 Test patterns. This specification includes a set of unique input patterns (table VIII) capable of detecting all nonredundant failures in the module logic networks. These test patterns are generated by assuming "stuck at H state" and "stuck at L state" in each of the basic logic functions and applying the necessary input patterns to detect these failures. A "stuck at H state" indicates a failure resulting in a steady high-level output. A "stuck at L state" indicates a failure resulting in a steady low-level output. Unique test patterns necessary to satisfy these requirements shall be submitted with the qualification samples. If the test patterns submitted do not meet these criteria, the additional tests required shall be added to table VIII. Prior to any change in module design which would necessitate a different set of test patterns to detect a "stuck at L state" or "stuck at H state", the vendor shall be required to submit to the procuring activity and Standard Electronic Module Program Quality Assurance Activity (SEMP-QAA) the proposed new set of test patterns for approval.

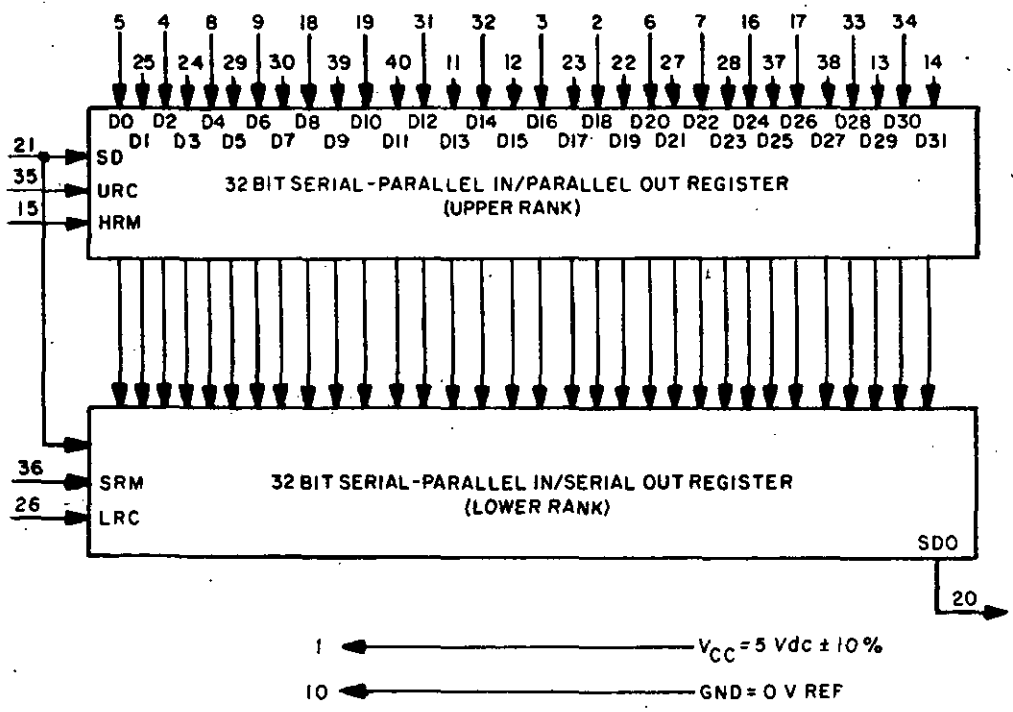
3.10.2 Input requirements. The following input voltage levels shall be used for test patterns:

Low level - Input voltage shall be 0.7 V $\pm$1%.
High level - Input voltage shall be 2.0 V $\pm$1%.

3.10.3 Output requirements. The output requirements for the test patterns shall be as follows (per figure 15 output loads):

Low level - Output voltage shall be 0 V minimum, 0.4 V maximum.
High level - Output voltage shall be 2.5 V minimum, V_{CC} maximum.

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NOTES:
1. D0-D31 = Parallel inputs.
2. SD = Serial data input.
3. URC = Upper rank clock input.
4. LRC = Lower rank clock input.
5. HRM = Holding register mode input.
6. SRM = Shift register mode input.
7. SDO = Serial data output.

* FIGURE 1. DQG module function diagram, 32-bit parallel/serial converter.

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TABLE II. Module electrical requirements.

Test condition	Parameter	Ref test table	Test type (see notes)	Acceptance requirements 25° ±5°C ambient temperature				Operating temperature 0° to 60°C				Unit
				Initial		End-of-life		Initial		End-of-life		
				Min	Max	Min	Max	Min	Max	Min	Max	
A	Isolation	III	S	10	∞	1.0	∞	10	∞	1.0	∞	MΩ
B	V_{out} (low-loaded)	IV	S	0.01	0.80	0.01	0.80	0.01	0.80	0.01	0.80	V
C1-C32	Input breakdown current (DO-D31)	V	S	0	100	0	100	0	100	0	100	µA
C33	Input breakdown current (SD)	V	S	0	400	0	400	0	400	0	400	µA
C34-C37	Input breakdown current (URC, LRC, HRM, and SRM)	V	S	0	800	0	800	0	800	0	800	µA
D	Output leakage current (SDO)	VI	X	0	1	0	1	0	1	0	1	mA
E	Output short circuit current	VII	X	-6	-130	-6	-130	-6	-130	-6	-130	mA
F1-F109	Functional	VIII	X	Test vectors must produce output vectors as specified in table VIII.								
G1-G37	Input clamp diode voltage	IX	X	-	-1.5	-	-1.5	-	-1.5	-	-1.5	V
H	Static output voltage, low level	X	X	0.0	0.4	0.0	0.4	0.0	0.4	0.0	0.4	V
J	Static output voltage, high level	XI	X	2.5	4.5	2.5	4.5	2.5	4.5	2.5	4.5	V
K1-K32	Input current, high level (DO-D31)	XII	X	0	20	0	20	0	20	0	20	µA
K33	Input current, high level (SD)	XII	X	0	80	0	80	0	80	0	80	µA
K34-K37	Input current, high level (URC, LRC, HRM, and SRM)	XII	X	0	160	0	160	0	160	0	160	µA
L1-L32	Input current, low level (DO-D31)	XIII	X	-0.1	-0.4	-0.1	-0.4	-0.1	-0.4	-0.1	-0.4	mA
L33	Input current, low level (SD)	XIII	X	-0.1	-1.6	-0.1	-1.6	-0.1	-1.6	-0.1	-1.6	mA

See notes at end of table.

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TABLE II. Module electrical requirements - Continued.

Test condition	Parameter	Ref test table	Test type (see notes)	Acceptance requirements 25° ±5°C ambient temperature				Operating temperature 0° to 60°C				Unit
				Initial		End-of-life		Initial		End-of-life		
				Min	Max	Min	Max	Min	Max	Min	Max	
L34,L35	Input current, low level (URC and LRC)	XIII	X	-0.10	-3.91	-0.10	-3.91	-0.10	-3.91	-0.10	-3.91	mA
L36,L37	Input current, low level (HRM and SRM)	XIII	X	-0.1	-3.2	-0.1	-3.2	-0.1	-3.2	-0.1	-3.2	mA
M	Power supply current	XIV	X	50	340	50	340	50	340	50	340	mA
N1	Transition time (t_{TLH}), C_L = 50 pF	XV	S	5	25	5	25	5	28	5	28	ns
N2	Transition time (t_{TLH}), C_L = 150 pF	XV	Q	6	35	6	35	6	40	6	40	ns
* P1	Transition time (t_{THL}), C_L = 50 pF	XVI	S	1	10	1	10	1	12	1	12	ns
P2	Transition time (t_{THL}), C_L = 150 pF	XVI	Q	2	15	2	15	2	18	2	18	ns
Q1	Propagation delay time (t_{PLH}) and SD input setup and hold time check, C_L = 50 pF	XVII	S	10	40	10	40	10	45	10	45	ns
Q2	Propagation delay time (t_{PLH}) and SD input setup and hold time check, C_L = 150 pF	XVII	Q	10	55	10	55	10	60	10	60	ns
R1	Propagation delay time (t_{PHL}) and SD input setup and hold time check, C_L = 50 pF	XVIII	S	10	45	10	45	10	50	10	50	ns

See notes at end of table.

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TABLE II. Module electrical requirements - Continued.

Test condition	Parameter	Ref test table	Test type (see notes)	Acceptance requirements 25° ±5°C ambient temperature				Operating temperature 0° to 60°C				Unit
				Initial		End-of-life		Initial		End-of-life		
				Min	Max	Min	Max	Min	Max	Min	Max	
R2	Propagation delay time (t_{PHL}) and SD input setup and hold time check, C_L = 150 pF	XVIII	Q	10	55	10	55	10	60	10	60	ns
S	Maximum operating frequency check and minimum LRC clock pulse width check (measuring output pulse period, t_{PD})	XIX	S	130	150	130	150	130	150	130	150	ns
T1	HRM setup and hold time check for parallel load	XX	S	0.0	0.4	0.0	0.4	0.0	0.4	0.0	0.4	V
T2	HRM setup and hold time check for serial shift	XX	S	2.5	4.5	2.5	4.5	2.5	4.5	2.5	4.5	V
T3	SRM setup and hold time check for parallel load	XX	S	0.0	0.4	0.0	0.4	0.0	0.4	0.0	0.4	V
T4	SRM setup and hold time check for serial shift	XX	S	2.5	4.5	2.5	4.5	2.5	4.5	2.5	4.5	V
U1	Serial data, high level, setup and hold time check (measuring output pulse period, t_{PD})	XXI	S	130	150	130	150	130	150	130	150	ns
U2	Serial data, high level, setup and hold time check (measuring output pulse width, t_{PW})	XXI	S	60	80	60	80	60	80	60	80	ns
U3	Serial data, low level, setup and hold time check (measuring output pulse period, t_{PD})	XXI	S	130	150	130	150	130	150	130	150	ns
U4	Serial data, low level, setup and hold time check (measuring output pulse width, t_{PW})	XXI	S	60	80	60	80	60	80	60	80	ns

See notes at end of table.

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TABLE II. Module electrical requirements - Continued.

Test condition	Parameter	Ref test table	Test type (see notes)	Acceptance requirements 25° ±5°C ambient temperature				Operating temperature 0° to 60°C				Unit
				Initial		End-of-life		Initial		End-of-life		
				Min	Max	Min	Max	Min	Max	Min	Max	
V1	Upper rank clock (URC) and lower rank clock (LRC) minimum pulse width check (measuring output pulse period, t_{PD})	XXII	S	130	150	130	150	130	150	130	150	ns
V2	Upper rank clock (URC) and lower rank clock (LRC) minimum pulse width check (measuring output pulse width, t_{PW})	XXII	S	60	80	60	80	60	80	60	80	ns
W1	D31 input setup and hold time check and URC to LRC clock delay check (measuring output pulse period, t_{PD})	XXIII	S	180	220	180	220	180	220	180	220	ns
W2	D31 input setup and hold time check and URC to LRC clock delay check (measuring output pulse width, t_{PW})	XXIII	S	85	115	85	115	85	115	85	115	ns
X	Output crosstalk, low level	XXIV	Q	0	400	0	400	0	400	0	400	mV
Y	Output crosstalk, high level	XXV	Q	0	400	0	400	0	400	0	400	mV
Z	Power supply transient amplitude	XXVI	Q	0	600	0	600	0	600	0	600	mV
AA	Power supply filter capacitance	XXVII	S	0.14	0.26	0.14	0.26	0.14	0.26	0.14	0.26	µF
* BB1-BB32	Input capacitance (D0-D31)	XXVIII	Q	2	20	2	20	2	20	2	20	pF
BB33	Input capacitance (SD)	XXVIII	Q	2	50	2	50	2	50	2	50	pF
BB34-BB37	Input capacitance (URC, LRC, HRM and SRM)	XXVIII	Q	2	100	2	100	2	100	2	100	pF

NOTES:
1. (X) parameters shall be 100 percent tested during module acceptance tests.
2. (S) parameters shall be sample acceptance tested (1 percent or two modules from each inspection lot, whichever is greater) at 25°C.
3. (Q) parameters shall be tested during qualification testing only.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Quality assurance. The quality assurance provisions of MIL-M-28787 shall apply.

4.2 Acceptance tests. Acceptance tests of all modules shall be in accordance with procedures and equipment specified in MIL-M-28787.

* 4.3 Test equipment. The following test equipment is necessary to conduct acceptance tests for this module. Test equipment shall be as specified in MIL-STD-1665 and that listed below:

Equipment	MIL-STD-1665 designator	No. required (maximum per test figure)
DC power supply	Item P01	4
Pulse generator	Item J01	3
Oscilloscope	Item D05	1
Oscilloscope probe	Item Y01	2
Megohm bridge	Item R06	1
DC voltmeter	Item V03	1
DC ammeter	Item X01	1
Capacitance bridge	Item C01	1
Capacitance meter	Item C05	1
Resistors		
1 kΩ ±1%, 1/4 W		1
51.1Ω ±1%, 1/4 W		3
1.3 kΩ ±1%, 1/8 W		1
51Ω ±5%, 1/2 W		2
750Ω ±1%, 1/4 W		1
Value as required for figure 15, note 2, 1/8 W		1
6.19 kΩ ±1%, 1/8 W		1
Capacitors		
Value as required for figure 15, note 4		1
Value as required for figure 15, note 5		1
Diode		
1N3600, 1N4150, or equivalent		1
Integrated circuit		
5476, or equivalent		1
Switch		
SPDT		1

All resistors shall be metal film and all capacitors shall be ceramic unless otherwise specified.

4.4 Test procedures. The module shall be tested as specified in table II.

* 4.4.1 Alternate test methods. Alternate test methods may be employed if approved by the Standard Electronic Modules Program Quality Assurance Activity (SEMP-QAA) and they conform to the requirements of MIL-M-28787.

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* 4.4.2 Order of testing. Test conditions A through BB shall be performed in any order with the following exceptions. If test conditions A, B, C, D, or E are part of a sequence of tests, the order of testing shall be as follows: A, B, C, D, E where applicable, followed by the remaining tests in the sequence.

4.5 Qualification testing. Initial qualification tests and periodic-check tests shall be performed in accordance with the test plan specified in MIL-M-28787 with the following modifications or additions.

4.5.1 Operating temperature tests. The operating temperature tests shall be conducted in accordance with MIL-M-28787. Before starting these tests, test conditions A through BB in table II shall be performed in the order specified in 4.4.2 using the initial limits at 25° ±5°C ambient. When electrical testing is called for at the end of each temperature stabilization period, test conditions D through AA shall be performed in the order specified in 4.4.2 using the initial limits for applicable operating temperature. At all other times during the operating temperature tests, the module shall be operated in the test circuit shown on figure 2.

4.5.1.1 Component temperature test. The component temperature test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 4.

4.5.2 Vibration tests. The vibration tests shall be conducted in accordance with MIL-M-28787. During the vibration tests, the module shall be operated in the test circuit shown on figure 5. Following the vibration tests, the electrical tests of table II which are marked with an (X) in the test type column and test condition AA shall be performed using the initial limits at 25° ±5°C ambient.

4.5.3 Life test. The life test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 2. Following the life test, electrical tests D through AA shall be performed in the order specified in 4.4.2 using the 0° to 60°C operating temperature end-of-life requirements of table II. After the module has been returned to 25°C, electrical tests D through BB shall be performed in the order specified in 4.4.2 using the 25° ±5°C ambient end-of-life requirements of table II.

5. PACKAGING

* 5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-28787.

6. NOTES

6.1 Definitions. Definitions are as specified in MIL-STD-1331.

6.2 End-of-life. When any individual module circuit parameter fails to conform to the end-of-life requirements of table II, the module is considered to have completed its useful life.

6.3 Cross reference of classification. Part number M28787/213-1 supersedes part number M28787/213. The two parts are interchangeable.

6.4 Changes from previous issue. The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Preparing activity:
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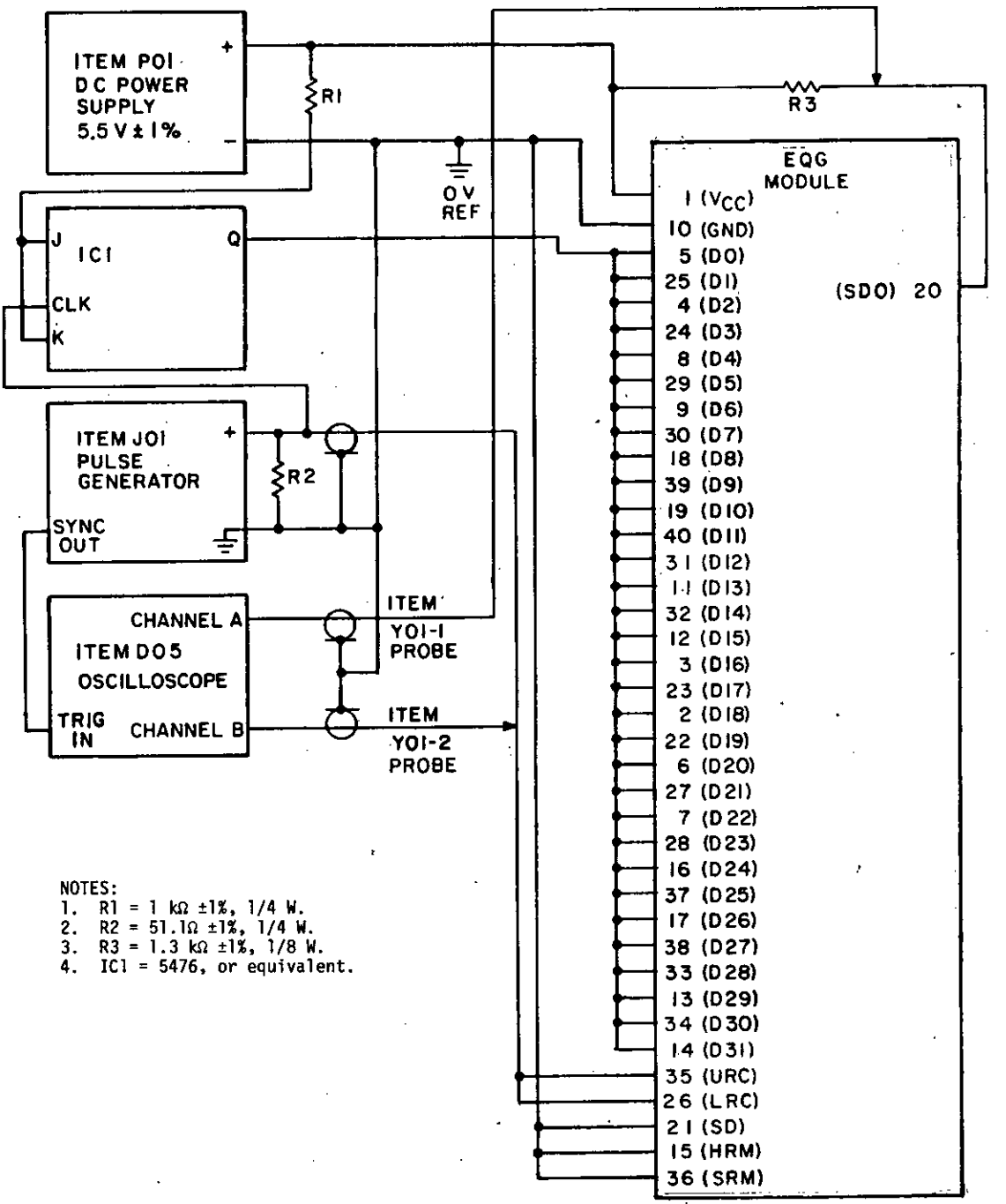
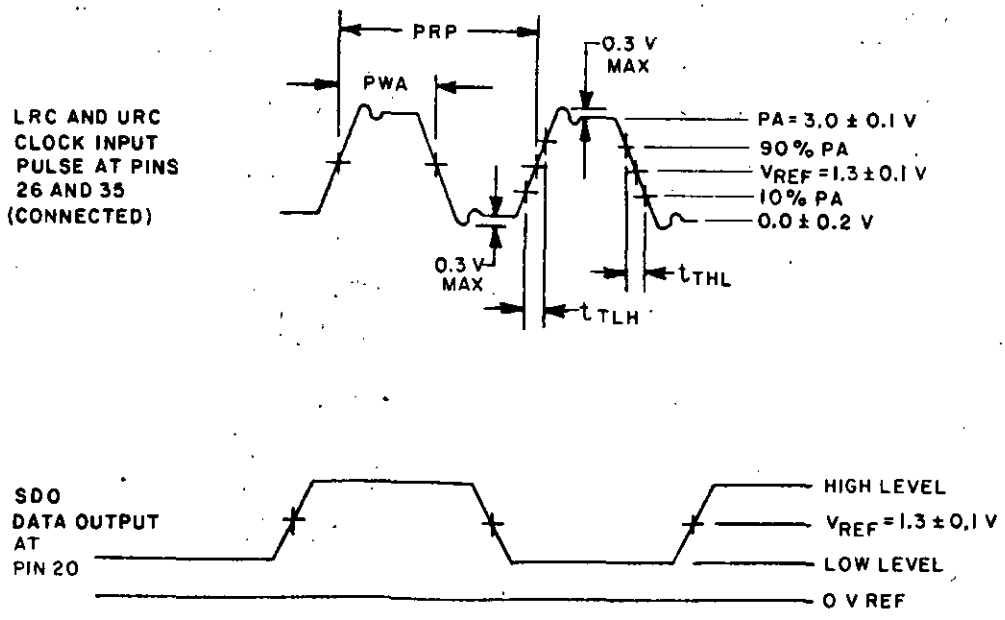


FIGURE 2. Operating temperature and life test circuit.

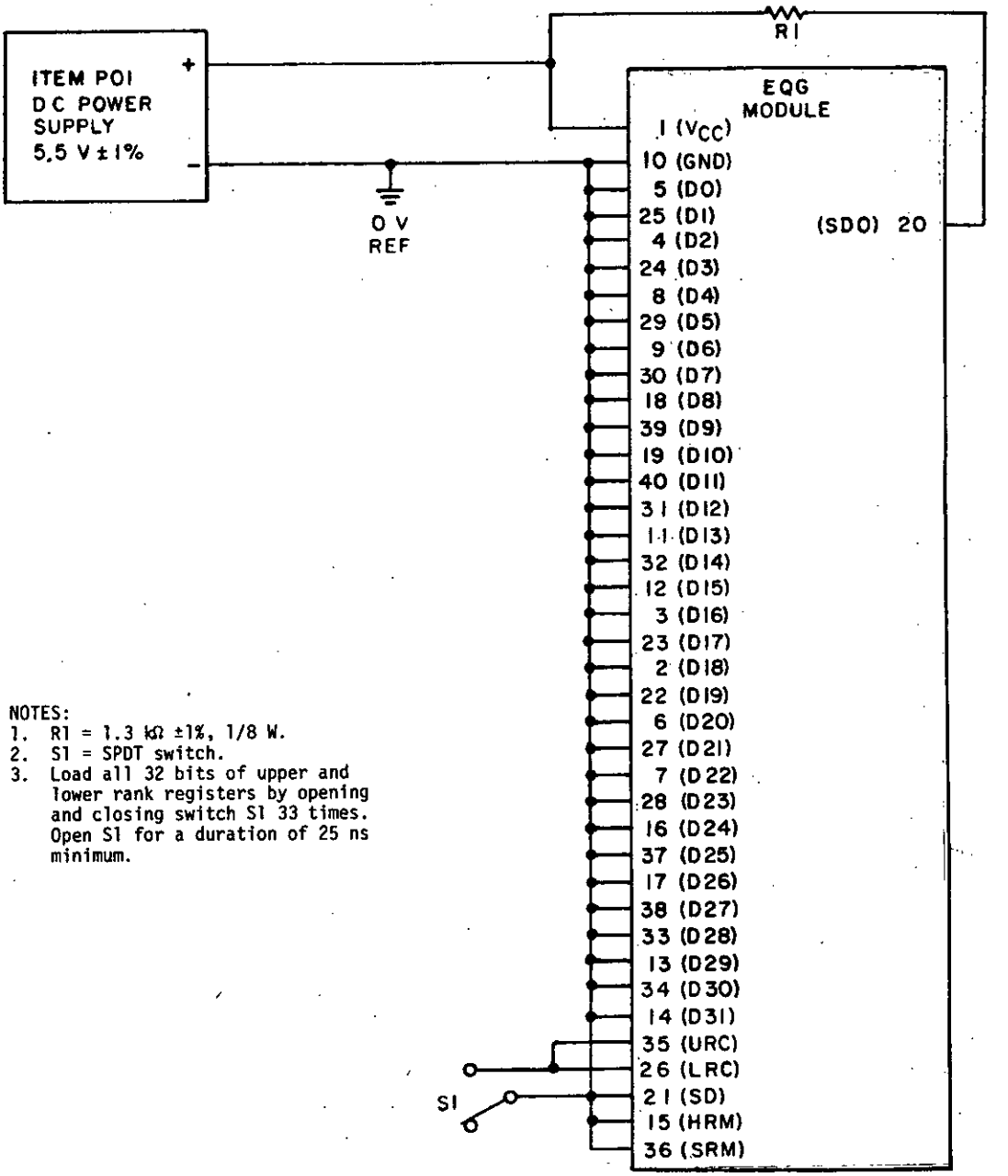
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NOTES:
1. Clock input pulse parameters:
   PRP = 500 ±50 ns.
   PWA = 200 ±20 ns.
   $t_{TLH} = t_{THL}$ = 15 ±10 ns.
2. Verify that the output has a high level ≥ 2.5 V and a low level ≤ 0.4 V.
3. All voltages are referenced to 0 V unless otherwise specified.

FIGURE 3. Operating temperature and life test circuit waveforms.

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NOTES:
1. R1 = 1.3 kΩ ±1%, 1/8 W.
2. S1 = SPDT switch.
3. Load all 32 bits of upper and lower rank registers by opening and closing switch S1 33 times. Open S1 for a duration of 25 ns minimum.

FIGURE 4. Component temperature test circuit.

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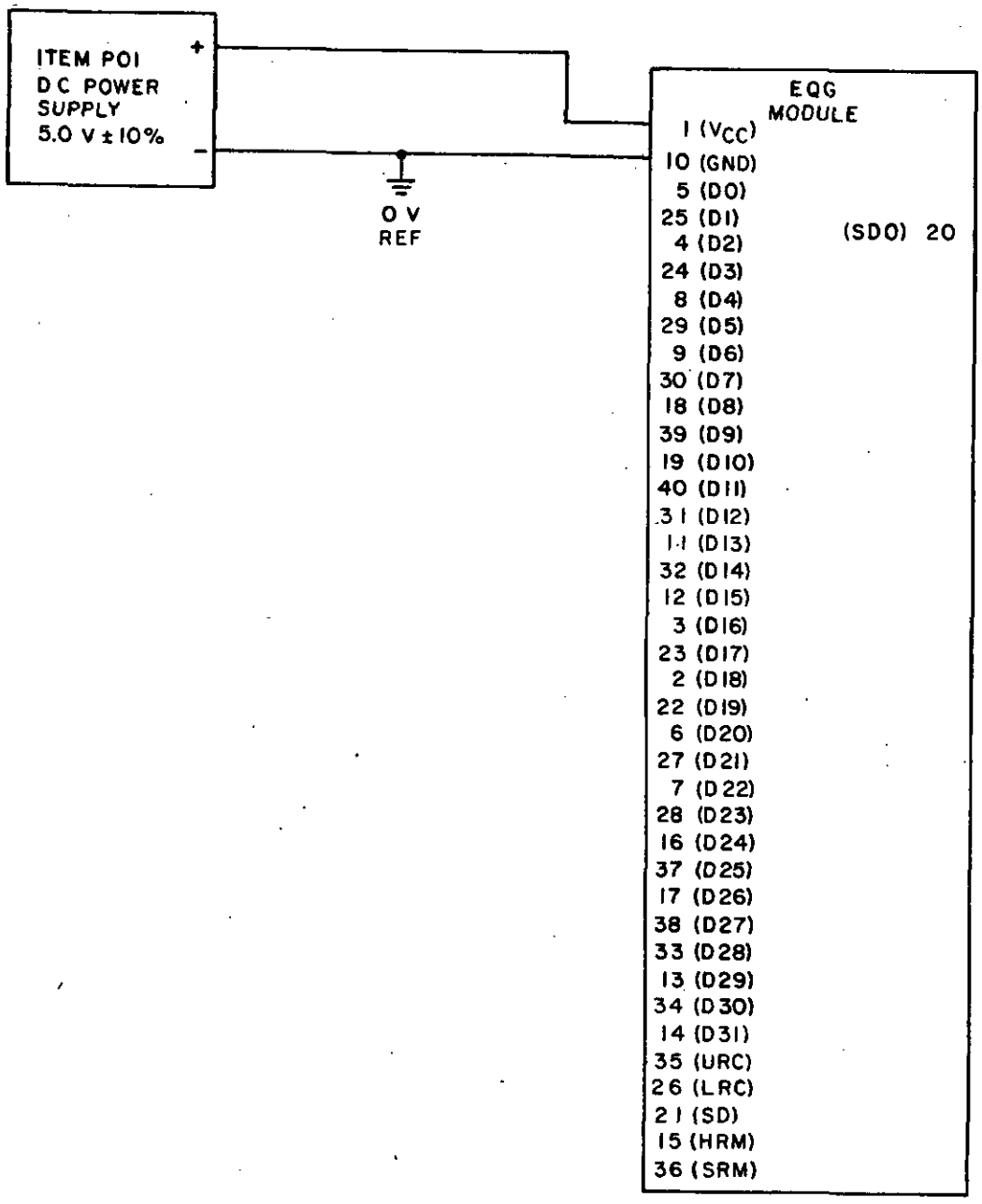


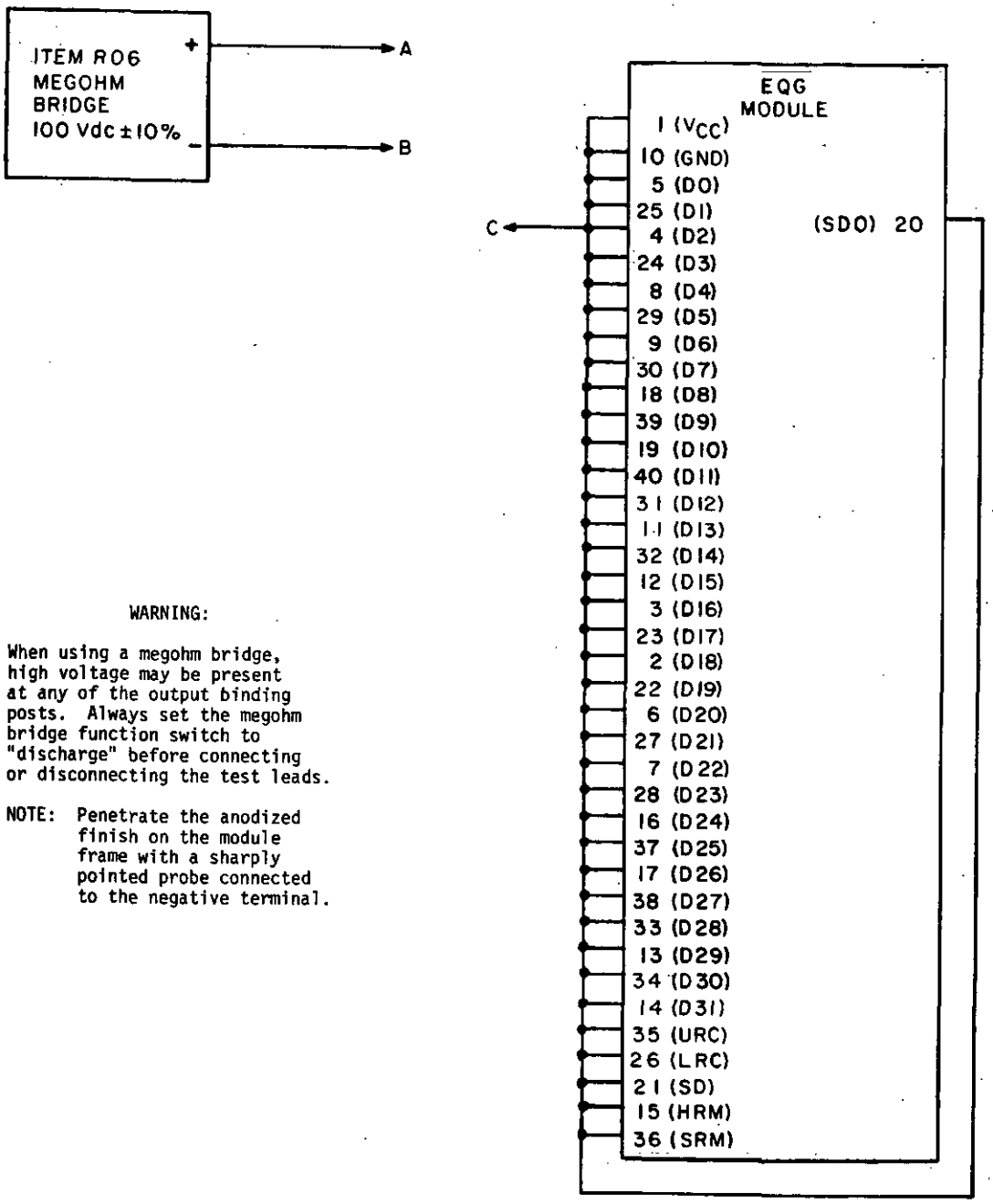
FIGURE 5. Vibration test circuit.

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TABLE III. Isolation test.

Test condition	Parameter	Connect		Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
A	Isolation	C	Frame	10	∞	MΩ	6

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WARNING:

When using a megohm bridge, high voltage may be present at any of the output binding posts. Always set the megohm bridge function switch to "discharge" before connecting or disconnecting the test leads.

NOTE: Penetrate the anodized finish on the module frame with a sharply pointed probe connected to the negative terminal.

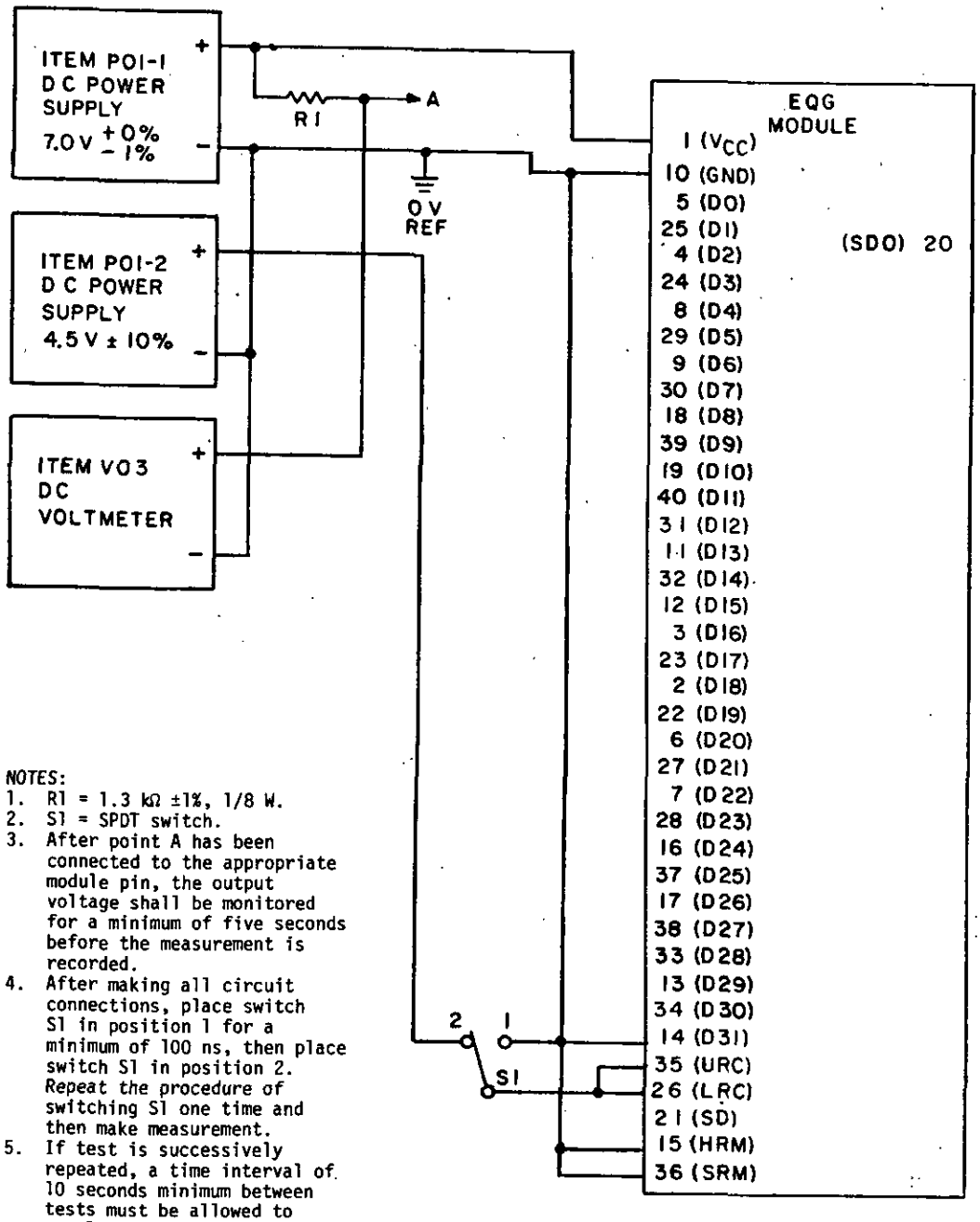
FIGURE 6. Isolation test circuit for table III.

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TABLE IV. V_{out} (low-loaded) test.

Test condition	Parameter	Connect	Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	Min	Max		
B	SDO . V_{out} (low-loaded)	20	0.01	0.80	V	7

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NOTES:
1. R1 = 1.3 kΩ ±1%, 1/8 W.
2. S1 = SPDT switch.
3. After point A has been connected to the appropriate module pin, the output voltage shall be monitored for a minimum of five seconds before the measurement is recorded.
4. After making all circuit connections, place switch S1 in position 1 for a minimum of 100 ns, then place switch S1 in position 2. Repeat the procedure of switching S1 one time and then make measurement.
5. If test is successively repeated, a time interval of 10 seconds minimum between tests must be allowed to preclude detrimental effects due to module heating.

FIGURE 7. V_{out} (low-loaded) test circuit for table IV.

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TABLE V. Input breakdown current test.

Test condition	Parameter		Connect		Initial limits 25° ±5°C Ambient		Unit	Test figure
			A to pin	B to pin	Min	Max		
C1	D0	Input breakdown current	25,4,24,8,29,9,30,18,39,19,40,31,11,32,12,3,23,2,22,6,27,7,28,16,37,17,38,33,13,34,14,35,26,21,15,36	5	0	100	μA	8
C2	D1		5,4,24,8,29,9,30,18,39,19,40,31,11,32,12,3,23,2,22,6,27,7,28,16,37,17,38,33,13,34,14,35,26,21,15,36	25				
C3	D2		5,25,24,8,29,9,30,18,39,19,40,31,11,32,12,3,23,2,22,6,27,7,28,16,37,17,38,33,13,34,14,35,26,21,15,36	4				
C4	D3		5,25,4,8,29,9,30,18,39,19,40,31,11,32,12,3,23,2,22,6,27,7,28,16,37,17,38,33,13,34,14,35,26,21,15,36	24				
C5	D4		5,25,4,24,29,9,30,18,39,19,40,31,11,32,12,3,23,2,22,6,27,7,28,16,37,17,38,33,13,34,14,35,26,21,15,36	8				
C6	D5		5,25,4,24,8,9,30,18,39,19,40,31,11,32,12,3,23,2,22,6,27,7,28,16,37,17,38,33,13,34,14,35,26,21,15,36	29				
C7	D6		5,25,4,24,8,29,30,18,39,19,40,31,11,32,12,3,23,2,22,6,27,7,28,16,37,17,38,33,13,34,14,35,26,21,15,36	9				
C8	D7	Input breakdown current	5,25,4,24,8,29,9,18,39,19,40,31,11,32,12,3,23,2,22,6,27,7,28,16,37,17,38,33,13,34,14,35,26,21,15,36	30	0	100	μA	8

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TABLE V. Input breakdown current test - Continued.

Test condition	Parameter		Connect		Initial limits 25° ±5°C Ambient		Unit	Test figure
			A to pin	B to pin	Min	Max		
C9	D8	Input breakdown current	5,25,4,24,8,29,9,30,39,19,40, 31,11,32,12,3,23,2,22,6,27,7, 28,16,37,17,38,33,13,34,14, 35,26,21,15,36	18	0	100	µA	8
C10	D9		5,25,4,24,8,29,9,30,18,19,40, 31,11,32,12,3,23,2,22,6,27,7, 28,16,37,17,38,33,13,34,14, 35,26,21,15,36	39				
C11	D10		5,25,4,24,8,29,9,30,18,39,40, 31,11,32,12,3,23,2,22,6,27,7, 28,16,37,17,38,33,13,34,14, 35,26,21,15,36	19				
C12	D11		5,25,4,24,8,29,9,30,18,39,19, 31,11,32,12,3,23,2,22,6,27,7, 28,16,37,17,38,33,13,34,14, 35,26,21,15,36	40				
C13	D12		5,25,4,24,8,29,9,30,18,39,19, 40,11,32,12,3,23,2,22,6,27,7, 28,16,37,17,38,33,13,34,14, 35,26,21,15,36	31				
C14	D13		5,25,4,24,8,29,9,30,18,39,19, 40,31,32,12,3,23,2,22,6,27,7, 28,16,37,17,38,33,13,34,14, 35,26,21,15,36	11				
C15	D14		5,25,4,24,8,29,9,30,18,39,19, 40,31,11,12,3,23,2,22,6,27,7, 28,16,37,17,38,33,13,34,14, 35,26,21,15,36	32				
C16	D15	Input breakdown current	5,25,4,24,8,29,9,30,18,39,19, 40,31,11,32,3,23,2,22,6,27,7, 28,16,37,17,38,33,13,34,14, 35,26,21,15,36	12	0	100	µA	8

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TABLE V. Input breakdown current test - Continued.

Test condition	Parameter		Connect		Initial limits 25° ±5°C Ambient		Unit	Test figure
			A to pin	B to pin	Min	Max		
C17	D16	Input breakdown current	5,25,4,24,8,29,9,30,18,39,19,40,31,11,32,12,23,2,22,6,27,7,28,16,37,17,38,33,13,34,14,35,26,21,15,36	3	0	100	µA	8
C18	D17		5,25,4,24,8,29,9,30,18,39,19,40,31,11,32,12,3,2,22,6,27,7,28,16,37,17,38,33,13,34,14,35,26,21,15,36	23				
C19	D18		5,25,4,24,8,29,9,30,18,39,19,40,31,11,32,12,3,23,22,6,27,7,28,16,37,17,38,33,13,34,14,35,26,21,15,36	2				
C20	D19		5,25,4,24,8,29,9,30,18,39,19,40,31,11,32,12,3,23,2,6,27,7,28,16,37,17,38,33,13,34,14,35,26,21,15,36	22				
C21	D20		5,25,4,24,8,29,9,30,18,39,19,40,31,11,32,12,3,23,2,22,27,7,28,16,37,17,38,33,13,34,14,35,26,21,15,36	6				
C22	D21		5,25,4,24,8,29,9,30,18,39,19,40,31,11,32,12,3,23,2,22,6,7,28,16,37,17,38,33,13,34,14,35,26,21,15,36	27				
C23	D22		5,25,4,24,8,29,9,30,18,39,19,40,31,11,32,12,3,23,2,22,6,27,28,16,37,17,38,33,13,34,14,35,26,21,15,36	7				
C24	D23	Input breakdown current	5,25,4,24,8,29,9,30,18,39,19,40,31,11,32,12,3,23,2,22,6,27,7,16,37,17,38,33,13,34,14,35,26,21,15,36	28	0	100	µA	8

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TABLE Input breakdown current test - Continued.

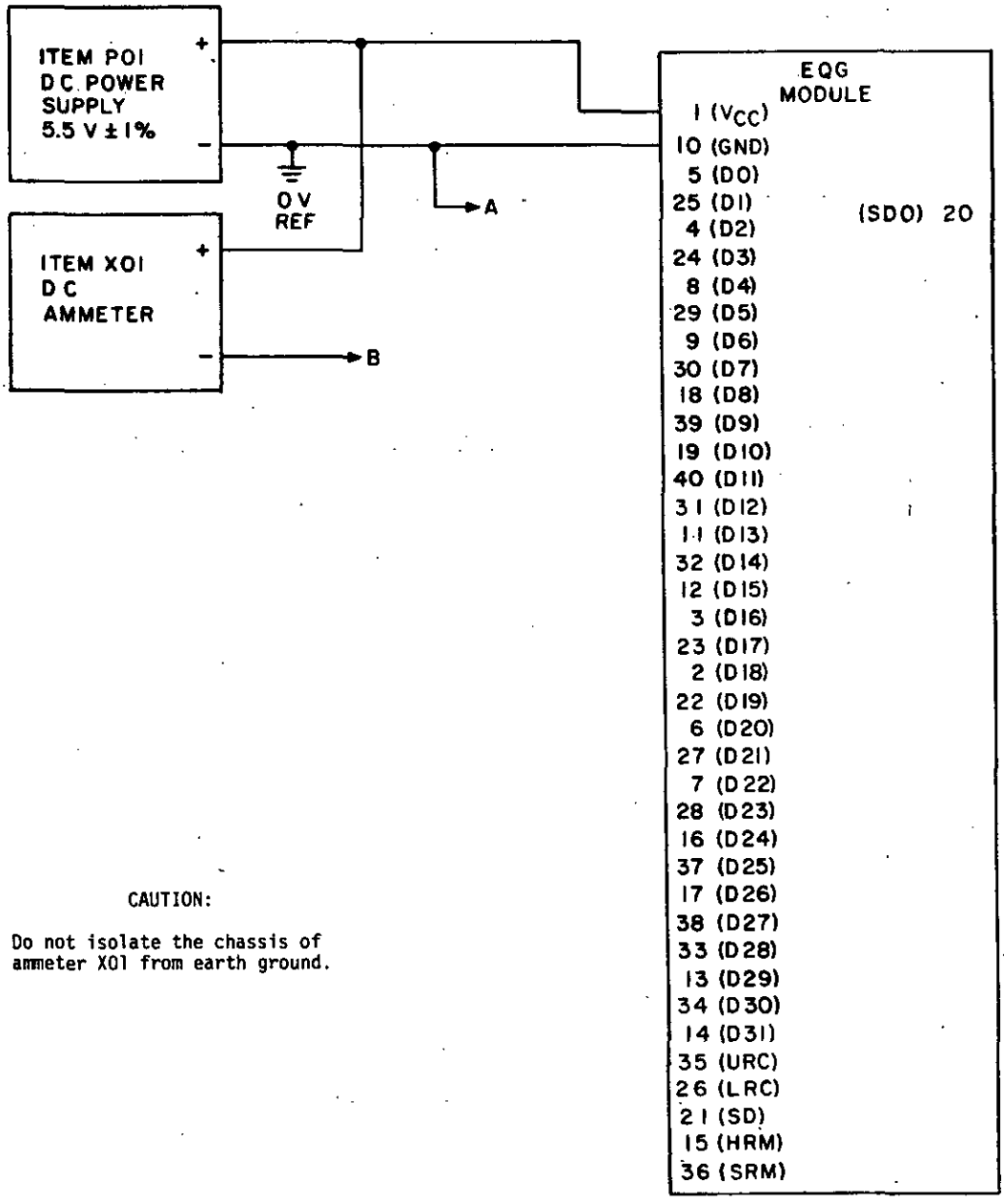
Test condition	Parameter		Connect		Initial limits 25° ±5°C Ambient		Unit	Test figure
			A to pin	B to pin	Min	Max		
C25	D24	Input breakdown current	5,25,4,24,8,29,9,30,18,39,19, 40,31,11,32,12,3,23,2,22,6, 27,7,28,37,17,38,33,13,34,14, 35,26,21,15,36	16	0	100	µA	8
C26	D25		5,25,4,24,8,29,9,30,18,39,19, 40,31,11,32,12,3,23,2,22,6, 27,7,28,16,17,38,33,13,34,14, 35,26,21,15,36	37				
C27	D26		5,25,4,24,8,29,9,30,18,39,19, 40,31,11,32,12,3,23,2,22,6, 27,7,28,16,37,38,33,13,34,14, 35,26,21,15,36	17				
C28	D27		5,25,4,24,8,29,9,30,18,39,19, 40,31,11,32,12,3,23,2,22,6, 27,7,28,16,37,17,33,13,34,14, 35,26,21,15,36	38				
C29	D28		5,25,4,24,8,29,9,30,18,39,19, 40,31,11,32,12,3,23,2,22,6, 27,7,28,16,37,17,38,13,34,14, 35,26,21,15,36	33				
C30	D29		5,25,4,24,8,29,9,30,18,39,19, 40,31,11,32,12,3,23,2,22,6, 27,7,28,16,37,17,38,33,34,14, 35,26,21,15,36	13				
C31	D30		5,25,4,24,8,29,9,30,18,39,19, 40,31,11,32,12,3,23,2,22,6, 27,7,28,16,37,17,38,33,13,14, 35,26,21,15,36	34				
C32	D31	Input breakdown current	5,25,4,24,8,29,9,30,18,39,19, 40,31,11,32,12,3,23,2,22,6, 27,7,28,16,37,17,38,33,13,34, 35,26,21,15,36	14	0	100	µA	8

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TABLE V. Input breakdown current test - Continued.

Test condition	Parameter		Connect		Initial limits 25° ±5°C Ambient		Unit	Test figure
			A to pin	B to pin	Min	Max		
C33	SD	Input breakdown current	5,25,4,24,8,29,9,30,18,39,19, 40,31,11,32,12,3,23,2,22,6, 27,7,28,16,37,17,38,33,13,34, 14,35,26,15,36	21	0	400	µA	8
C34	URC		5,25,4,24,8,29,9,30,18,39,19, 40,31,11,32,12,3,23,2,22,6, 27,7,28,16,37,17,38,33,13,34, 14,26,21,15,36	35		800		
C35	LRC		5,25,4,24,8,29,9,30,18,39,19, 40,31,11,32,12,3,23,2,22,6, 27,7,28,16,37,17,38,33,13,34, 14,35,21,15,36	26				
C36	HRM		5,25,4,24,8,29,9,30,18,39,19, 40,31,11,32,12,3,23,2,22,6, 27,7,28,16,37,17,38,33,13,34, 14,35,26,21,36	15				
C37	SRM	Input breakdown current	5,25,4,24,8,29,9,30,18,39,19, 40,31,11,32,12,3,23,2,22,6, 27,7,28,16,37,17,38,33,13,34, 14,35,26,21,15	36	0	800	µA	8

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CAUTION:

Do not isolate the chassis of ammeter X01 from earth ground.

FIGURE 8. Input breakdown current test circuit for table V.

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TABLE VI. Output leakage current test.

Test condition	Parameter	Initial limits 25° ±5°C Ambient		Unit	Test figure
		Min	Max		
D	SDO Output leakage current	0	1	mA	9

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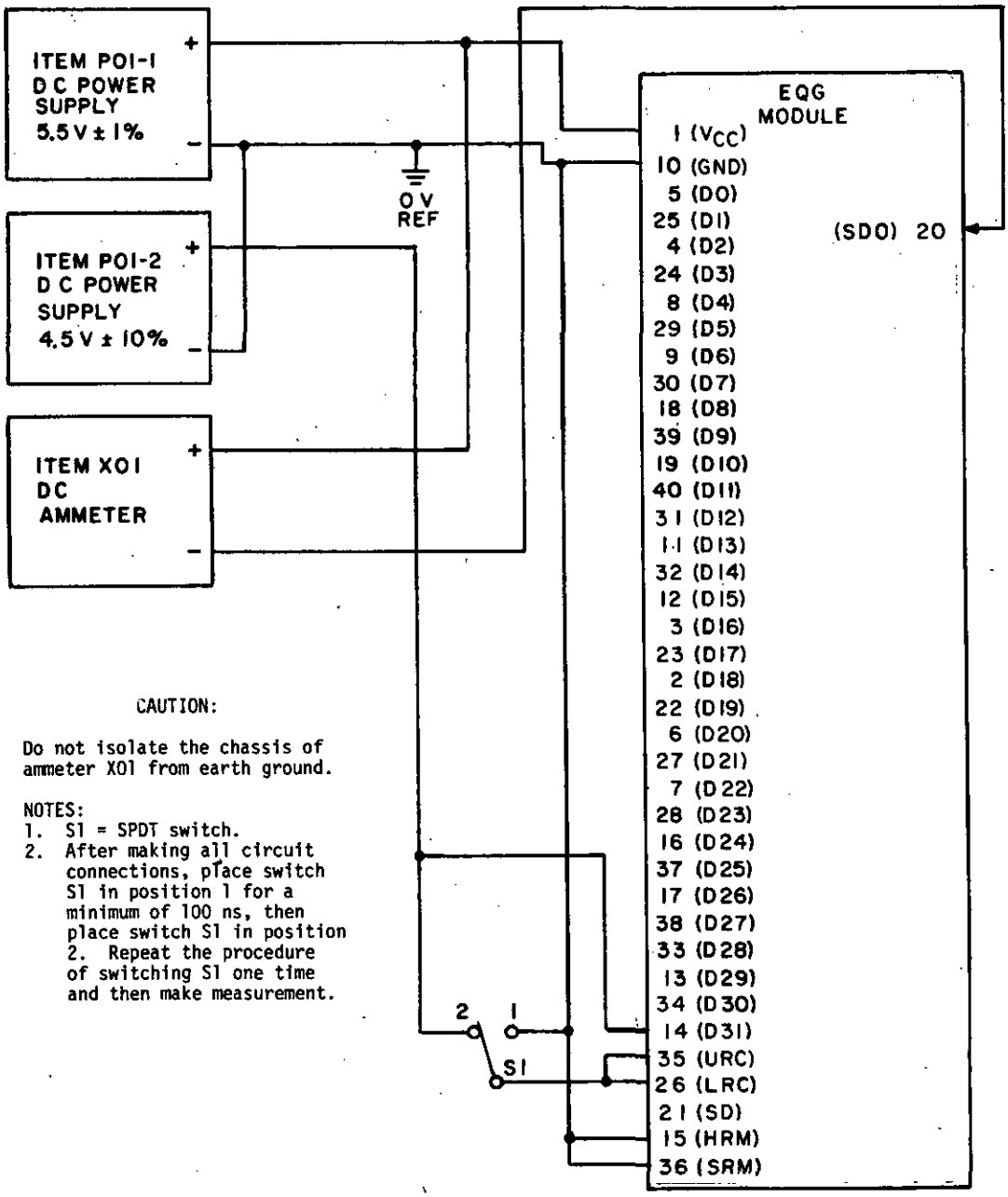


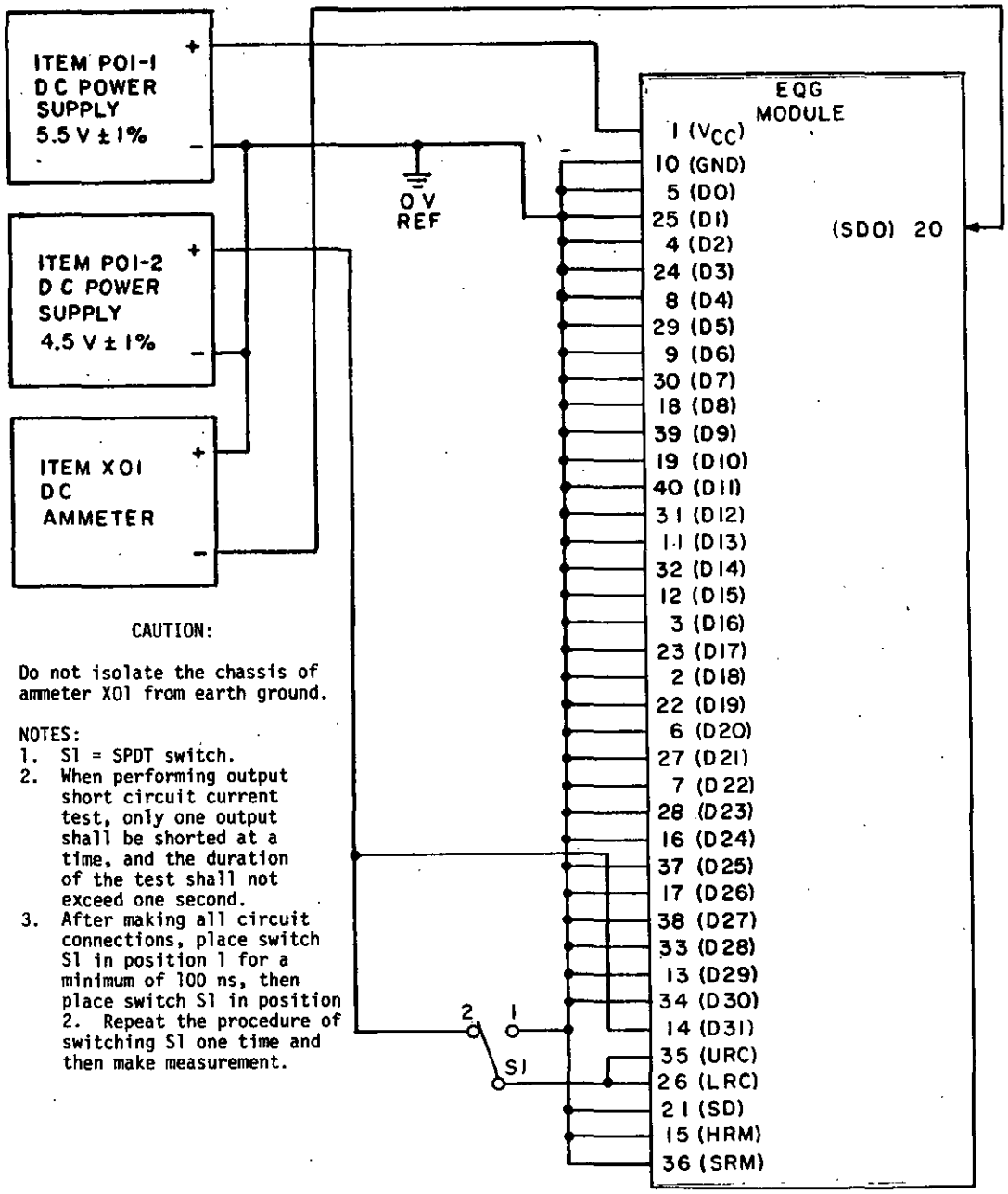
FIGURE 9. Output leakage current test circuit for table VI.

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TABLE VII. Output short circuit current test.

Test condition	Parameter	Initial limits 25° ±5°C Ambient		Unit	Test figure
		Min	Max		
E	SDO Output short circuit current	-6	-130	mA	10

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CAUTION:

Do not isolate the chassis of ammeter XOI from earth ground.

NOTES:
1. SI = SPDT switch.
2. When performing output short circuit current test, only one output shall be shorted at a time, and the duration of the test shall not exceed one second.
3. After making all circuit connections, place switch SI in position 1 for a minimum of 100 ns, then place switch SI in position 2. Repeat the procedure of switching SI one time and then make measurement.

FIGURE 10. Output short circuit current test circuit for table VII.

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TABLE VIII. Functional test.

Test condition	Parameter	Connect																																				Output logic state (see notes)	Test figure
		Pin 21 to	Pin 15 to	Pin 36 to	Pin 35 to	Pin 26 to	Pin 5 to	Pin 25 to	Pin 4 to	Pin 24 to	Pin 8 to	Pin 29 to	Pin 9 to	Pin 30 to	Pin 18 to	Pin 39 to	Pin 19 to	Pin 40 to	Pin 31 to	Pin 11 to	Pin 32 to	Pin 12 to	Pin 3 to	Pin 23 to	Pin 2 to	Pin 22 to	Pin 6 to	Pin 27 to	Pin 7 to	Pin 28 to	Pin 16 to	Pin 37 to	Pin 17 to	Pin 38 to	Pin 33 to	Pin 13 to	Pin 34 to	Pin 14 to	Pin 20	
	Functional	B	B	A	A	A	A	B	A	B	A	B	A	B	A	B	A	B	A	B	A	B	A	B	A	B	A	B	A	B	A	B	A	B	A	B	X	11, 12
		B	B	A	B	A	A	B	A	B	A	B	A	B	A	B	A	B	A	B	A	B	A	B	A	B	A	B	A	B	A	B	A	B	A	B	X	
		B	B	A	A	A	A	B	A	B	A	B	A	B	A	B	A	B	A	B	A	B	A	B	A	B	A	B	A	B	A	B	A	B	A	B	X	
		B	A	A	A	A	A	B	A	B	A	B	A	B	A	B	A	B	A	B	A	B	A	B	A	B	A	B	A	B	A	B	A	B	A	B	X	
		B	A	A	B	A	A	B	A	B	A	B	A	B	A	B	A	B	A	B	A	B	A	B	A	B	A	B	A	B	A	B	A	B	A	B	X	
		B	A	A	A	A	A	B	A	B	A	B	A	B	A	B	A	B	A	B	A	B	A	B	A	B	A	B	A	B	A	B	A	B	A	B	X	
		A	B	A	A	A	B	A	B	A	B	A	B	A	B	A	B	A	B	A	B	A	B	A	B	A	B	A	B	A	B	A	B	A	B	A	X	
		A	B	A	B	A	B	A	B	A	B	A	B	A	B	A	B	A	B	A	B	A	B	A	B	A	B	A	B	A	B	A	B	A	B	A	X	
		A	B	A	A	A	B	A	B	A	B	A	B	A	B	A	B	A	B	A	B	A	B	A	B	A	B	A	B	A	B	A	B	A	B	A	X	
		A	A	B	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	X	
		A	A	B	B	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	X	
F1		A	A	B	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	H	
		B	A	B	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	X	
		B	A	B	A	B	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	X	
		B	A	B	B	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	X	
F2		B	A	B	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	L	
		B	A	B	A	B	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	X	
F3		B	A	B	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	H	
		A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	X	
		A	A	A	A	B	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	X	
F4		A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	L	
		B	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	X	
		B	A	A	A	B	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	X	
F5	Functional	B	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	H	11, 12
		A	B	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	X	
		A	B	B	A	B	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	X	
		A	B	B	B	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	X	

See notes at end of table.

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TABLE VIII. Functional test - Continued.

Test condition	Parameter	Connect																																		Output logic state (see notes)	Test figure
		Pin 21 to	Pin 15 to	Pin 36 to	Pin 35 to	Pin 26 to	Pin 5 to	Pin 25 to	Pin 4 to	Pin 24 to	Pin 8 to	Pin 29 to	Pin 9 to	Pin 30 to	Pin 18 to	Pin 39 to	Pin 19 to	Pin 40 to	Pin 31 to	Pin 11 to	Pin 32 to	Pin 12 to	Pin 3 to	Pin 23 to	Pin 2 to	Pin 22 to	Pin 6 to	Pin 27 to	Pin 7 to	Pin 28 to	Pin 16 to	Pin 37 to	Pin 17 to	Pin 38 to	Pin 33 to	Pin 13 to	Pin 34 to	Pin 14 to	Pin 20	
F6	Functional	A	B	B	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	H	11, 12
		A	A	B	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	X	
		A	A	B	B	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	X	
		A	A	B	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	X	
F7		A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	H	
		A	A	A	A	B	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	X	
		A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	X	
F8		B	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	L	
		B	A	A	A	B	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	X	
		B	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	X	
F9		A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	H	
		A	A	A	A	B	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	X	
		A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	X	
F10		B	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	L	
		B	A	A	A	B	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	X	
		B	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	X	
F11		A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	H	
		A	A	A	A	B	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	X	
		A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	X	
F12		B	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	L	
		B	A	A	A	B	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	X	
		B	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	X	
F13	Functional	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	H	11, 12
		A	A	A	A	B	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	X	
		A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	X	

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TABLE VIII. Functional test - Continued.

Test condition	Parameter	Connect																																						Output logic state (see notes)	Test figure
		Pin 21 to	Pin 15 to	Pin 36 to	Pin 35 to	Pin 26 to	Pin 5 to	Pin 25 to	Pin 4 to	Pin 24 to	Pin 8 to	Pin 29 to	Pin 9 to	Pin 30 to	Pin 18 to	Pin 39 to	Pin 19 to	Pin 40 to	Pin 31 to	Pin 11 to	Pin 32 to	Pin 12 to	Pin 3 to	Pin 23 to	Pin 2 to	Pin 22 to	Pin 6 to	Pin 27 to	Pin 7 to	Pin 28 to	Pin 16 to	Pin 37 to	Pin 17 to	Pin 38 to	Pin 33 to	Pin 13 to	Pin 34 to	Pin 14 to	Pin 20	
F14	Functional	B	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	L	11, 12
		B	A	A	A	B	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	X	
		B	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	X	
F15		A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	H	
		A	A	A	A	B	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	X	
		A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	X	
F16		B	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	L	
		B	A	A	A	B	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	X	
		B	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	X	
F17		A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	H	
		A	A	A	A	B	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	X	
		A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	X	
F18		B	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	L	
		B	A	A	A	B	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	X	
		B	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	X	
F19		A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	H	
		A	A	A	A	B	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	X	
		A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	X	
F20		B	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	L	
		B	A	A	A	B	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	X	
		B	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	X	
F21	Functional	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	H	11, 12
		A	A	A	A	B	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	X	
		A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	X	

See notes at end of table.

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TABLE VIII. Functional test - Continued.

Test condition	Parameter	Connect																																				Output logic state (see notes)	Test figure
		Pin 21 to	Pin 15 to	Pin 36 to	Pin 35 to	Pin 26 to	Pin 5 to	Pin 25 to	Pin 4 to	Pin 24 to	Pin 8 to	Pin 29 to	Pin 9 to	Pin 30 to	Pin 18 to	Pin 39 to	Pin 19 to	Pin 40 to	Pin 31 to	Pin 11 to	Pin 32 to	Pin 12 to	Pin 3 to	Pin 23 to	Pin 2 to	Pin 22 to	Pin 6 to	Pin 27 to	Pin 7 to	Pin 28 to	Pin 16 to	Pin 37 to	Pin 17 to	Pin 38 to	Pin 33 to	Pin 13 to	Pin 34 to	Pin 14 to	Pin 20	
F22	Functional	B	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	L	11, 12
		B	A	A	A	B	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	X	
		B	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	X	
F23		A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	H	
		A	A	A	A	B	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	X	
		A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	X	
F24		B	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	L	
		B	A	A	A	B	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	X	
		B	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	X	
F25		A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	H	
		A	A	A	A	B	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	X	
		A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	X	
F26		B	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	L	
		B	A	A	A	B	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	X	
		B	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	X	
F27		A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	H	
		A	A	A	A	B	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	X	
		A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	X	
F28		B	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	L	
		B	A	A	A	B	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	X	
		B	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	X	
F29	Functional	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	H	11, 12
		A	A	A	A	B	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	X	
		A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	X	

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Test condition	Parameter	Connect																																				Output logic state (see notes)	Test figure
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F30	Functional	B	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	L	11, 12
		B	A	A	A	B	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	X	
		B	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	X	
F31		A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	H	
		A	A	A	A	B	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	X	
		A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	X	
F32		B	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	L	
		B	A	A	A	B	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	X	
		B	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	X	
F33		A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	H	
		A	A	A	A	B	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	X	
		A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	X	
F34		B	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	L	
		B	A	A	A	B	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	X	
		B	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	X	
F35		A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	H	
		A	A	A	A	B	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	X	
		A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	X	
F36		B	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	L	
		B	A	A	A	B	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	X	
		B	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	X	
F37	Functional	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	H	11, 12
		A	A	A	A	B	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	X	
		A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	X	

See notes at end of table.

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TABLE VIII. Functional test - Continued.

Test condition	Parameter	Connect																																					Output logic state (see notes)	Test figure
		Pin 21 to	Pin 15 to	Pin 36 to	Pin 35 to	Pin 26 to	Pin 5 to	Pin 25 to	Pin 4 to	Pin 24 to	Pin 8 to	Pin 29 to	Pin 9 to	Pin 30 to	Pin 18 to	Pin 39 to	Pin 19 to	Pin 40 to	Pin 31 to	Pin 11 to	Pin 32 to	Pin 12 to	Pin 3 to	Pin 23 to	Pin 2 to	Pin 22 to	Pin 6 to	Pin 27 to	Pin 7 to	Pin 28 to	Pin 16 to	Pin 37 to	Pin 17 to	Pin 38 to	Pin 33 to	Pin 13 to	Pin 34 to	Pin 14 to	Pin 20	
F38	Functional	B	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	L	11, 12
		B	A	A	A	B	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	X	
		B	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	X	
F39		A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	H	
		A	A	A	A	B	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	X	
		A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	X	
F40		B	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	L	
		B	A	A	A	B	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	X	
		B	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	X	
F41		A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	H	
		A	A	A	A	B	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	X	
F42		A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	L	
		A	B	B	A	A	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	X	
		A	B	B	A	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	X	
		A	B	B	B	A	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	X	
		A	B	B	A	A	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	X	
F43		A	B	A	A	A	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	H	
		A	B	A	A	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	X	
F44		A	B	A	A	A	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	H	
		A	B	A	A	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	X	
F45		A	B	A	A	A	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	H	
		A	B	A	A	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	X	
F46	Functional	A	B	A	A	A	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	H	11, 12
		A	B	A	A	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	X	

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TABLE VIII. Functional test - Continued.

Test condition	Parameter	Connect																																					Output logic state (see notes)	Test figure
		Pin 21 to	Pin 15 to	Pin 36 to	Pin 35 to	Pin 26 to	Pin 5 to	Pin 25 to	Pin 4 to	Pin 24 to	Pin 8 to	Pin 29 to	Pin 9 to	Pin 30 to	Pin 18 to	Pin 39 to	Pin 19 to	Pin 40 to	Pin 31 to	Pin 11 to	Pin 32 to	Pin 12 to	Pin 3 to	Pin 23 to	Pin 2 to	Pin 22 to	Pin 6 to	Pin 27 to	Pin 7 to	Pin 28 to	Pin 16 to	Pin 37 to	Pin 17 to	Pin 38 to	Pin 33 to	Pin 13 to	Pin 34 to	Pin 14 to	Pin 20	
F47	Functional	A	B	A	A	A	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	H	11, 12
		A	B	A	A	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	X	
F48		A	B	A	A	A	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	H	
		A	B	A	A	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	X	
F49		A	B	A	A	A	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	H	
		A	B	A	A	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	X	
F50		A	B	A	A	A	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	H	
		A	B	A	A	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	X	
F51		A	B	A	A	A	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	H	
		A	B	A	A	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	X	
F52		A	B	A	A	A	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	H	
		A	B	A	A	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	X	
F53		A	B	A	A	A	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	H	
		A	B	A	A	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	X	
F54		A	B	A	A	A	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	H	
		A	B	A	A	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	X	
F55		A	B	A	A	A	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	H	
		A	B	A	A	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	X	
F56		A	B	A	A	A	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	H	
		A	B	A	A	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	X	
F57	Functional	A	B	A	A	A	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	H	11, 12
		A	B	A	A	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	X	

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TABLE VIII. Functional test - Continued.

Test condition	Parameter	Connect																																					Output logic state (see notes)	Test figure
		Pin 21 to	Pin 15 to	Pin 36 to	Pin 35 to	Pin 26 to	Pin 5 to	Pin 25 to	Pin 4 to	Pin 24 to	Pin 8 to	Pin 29 to	Pin 9 to	Pin 30 to	Pin 18 to	Pin 39 to	Pin 19 to	Pin 40 to	Pin 31 to	Pin 11 to	Pin 32 to	Pin 12 to	Pin 3 to	Pin 23 to	Pin 2 to	Pin 22 to	Pin 6 to	Pin 27 to	Pin 7 to	Pin 28 to	Pin 16 to	Pin 37 to	Pin 17 to	Pin 38 to	Pin 33 to	Pin 13 to	Pin 34 to	Pin 14 to	Pin 20	
F58	Functional	A	B	A	A	A	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	H	11, 12
		A	B	A	A	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	X	
F59		A	B	A	A	A	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	H	
		A	B	A	A	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	X	
F60		A	B	A	A	A	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	H	
		A	B	A	A	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	X	
F61		A	B	A	A	A	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	H	
		A	B	A	A	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	X	
F62		A	B	A	A	A	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	H	
		A	B	A	A	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	X	
F63		A	B	A	A	A	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	H	
		A	B	A	A	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	X	
F64		A	B	A	A	A	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	H	
		A	B	A	A	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	X	
F65		A	B	A	A	A	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	H	
		A	B	A	A	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	X	
F66		A	B	A	A	A	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	H	
		A	B	A	A	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	X	
F67		A	B	A	A	A	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	H	
		A	B	A	A	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	X	
F68	Functional	A	B	A	A	A	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	H	11, 12
		A	B	A	A	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	X	

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TABLE VIII. Functional test - Continued.

Test condition	Parameter	Connect																																						Output logic state (see notes)	Test figure
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F69	Functional	A	B	A	A	A	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	H	11, 12
		A	B	A	A	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	X	
F70		A	B	A	A	A	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	H	
		A	B	A	A	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	X	
F71		A	B	A	A	A	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	H	
		A	B	A	A	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	X	
F72		A	B	A	A	A	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	H	
		A	B	A	A	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	X	
F73		A	B	A	A	A	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	H	
		A	B	A	A	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	X	
F74		A	B	A	A	A	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	H	
		A	B	A	A	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	X	
F75		A	B	A	A	A	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	B	H	
		B	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	X	
		B	A	A	B	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	X	
		B	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	X	
		A	B	B	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	X	
		A	B	B	A	B	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	X	
		A	B	B	B	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	X	
F76		A	B	B	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	L	
		B	B	B	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	B	
		B	B	B	B	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	X	
		B	B	B	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	B	
		B	B	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	X	
		B	B	A	A	B	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	B	
F77	Functional	B	B	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	L	11, 12
		B	B	A	A	B	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	X	

See notes at end of table.

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TABLE VIII. Functional test - Continued.

Test condition	Parameter	Connect																																					Output logic state (see notes)	Test figure
		Pin 21 to	Pin 15 to	Pin 36 to	Pin 35 to	Pin 26 to	Pin 5 to	Pin 25 to	Pin 4 to	Pin 24 to	Pin 8 to	Pin 29 to	Pin 9 to	Pin 30 to	Pin 18 to	Pin 39 to	Pin 19 to	Pin 40 to	Pin 31 to	Pin 11 to	Pin 32 to	Pin 12 to	Pin 3 to	Pin 23 to	Pin 2 to	Pin 22 to	Pin 6 to	Pin 27 to	Pin 7 to	Pin 28 to	Pin 16 to	Pin 37 to	Pin 17 to	Pin 38 to	Pin 33 to	Pin 13 to	Pin 34 to	Pin 14 to	Pin 20	
F78	Functional	B	B	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	B	L	11, 12
		B	B	A	A	B	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	B	X	
F79		B	B	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	B	L	
		B	B	A	A	B	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	B	X	
F80		B	B	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	B	L	
		B	B	A	A	B	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	B	X	
F81		B	B	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	B	L	
		B	B	A	A	B	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	B	X	
F82		B	B	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	B	L	
		B	B	A	A	B	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	B	X	
F83		B	B	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	B	L	
		B	B	A	A	B	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	B	X	
F84		B	B	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	B	L	
		B	B	A	A	B	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	B	X	
F85		B	B	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	B	L	
		B	B	A	A	B	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	B	X	
F86		B	B	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	B	L	
		B	B	A	A	B	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	B	X	
F87		B	B	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	B	L	
		B	B	A	A	B	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	B	X	
F88	Functional	B	B	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	B	L	11, 12
		B	B	A	A	B	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	B	X	

See notes at end of table.

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TABLE VIII. Functional test - Continued.

Test condition	Parameter	Connect																																				Output logic state (see notes)	Test figure
		Pin 21 to	Pin 15 to	Pin 36 to	Pin 35 to	Pin 26 to	Pin 5 to	Pin 25 to	Pin 4 to	Pin 24 to	Pin 8 to	Pin 29 to	Pin 9 to	Pin 30 to	Pin 18 to	Pin 39 to	Pin 19 to	Pin 40 to	Pin 31 to	Pin 11 to	Pin 32 to	Pin 12 to	Pin 3 to	Pin 23 to	Pin 2 to	Pin 22 to	Pin 6 to	Pin 27 to	Pin 7 to	Pin 28 to	Pin 16 to	Pin 37 to	Pin 17 to	Pin 38 to	Pin 33 to	Pin 13 to	Pin 34 to	Pin 14 to	Pin 20	
F89	Functional	B	B	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	B	L	11, 12
		B	B	A	A	B	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	B	X	
F90		B	B	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	B	L	
		B	B	A	A	B	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	B	X	
F91		B	B	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	B	L	
		B	B	A	A	B	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	B	X	
F92		B	B	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	B	L	
		B	B	A	A	B	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	B	X	
F93		B	B	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	B	L	
		B	B	A	A	B	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	B	X	
F94		B	B	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	B	L	
		B	B	A	A	B	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	B	X	
F95		B	B	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	B	L	
		B	B	A	A	B	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	B	X	
F96		B	B	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	B	L	
		B	B	A	A	B	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	B	X	
F97		B	B	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	B	L	
		B	B	A	A	B	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	B	X	
F98		B	B	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	B	L	
		B	B	A	A	B	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	B	X	
F99	Functional	B	B	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	B	L	11, 12
		B	B	A	A	B	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	B	X	

See notes at end of table.

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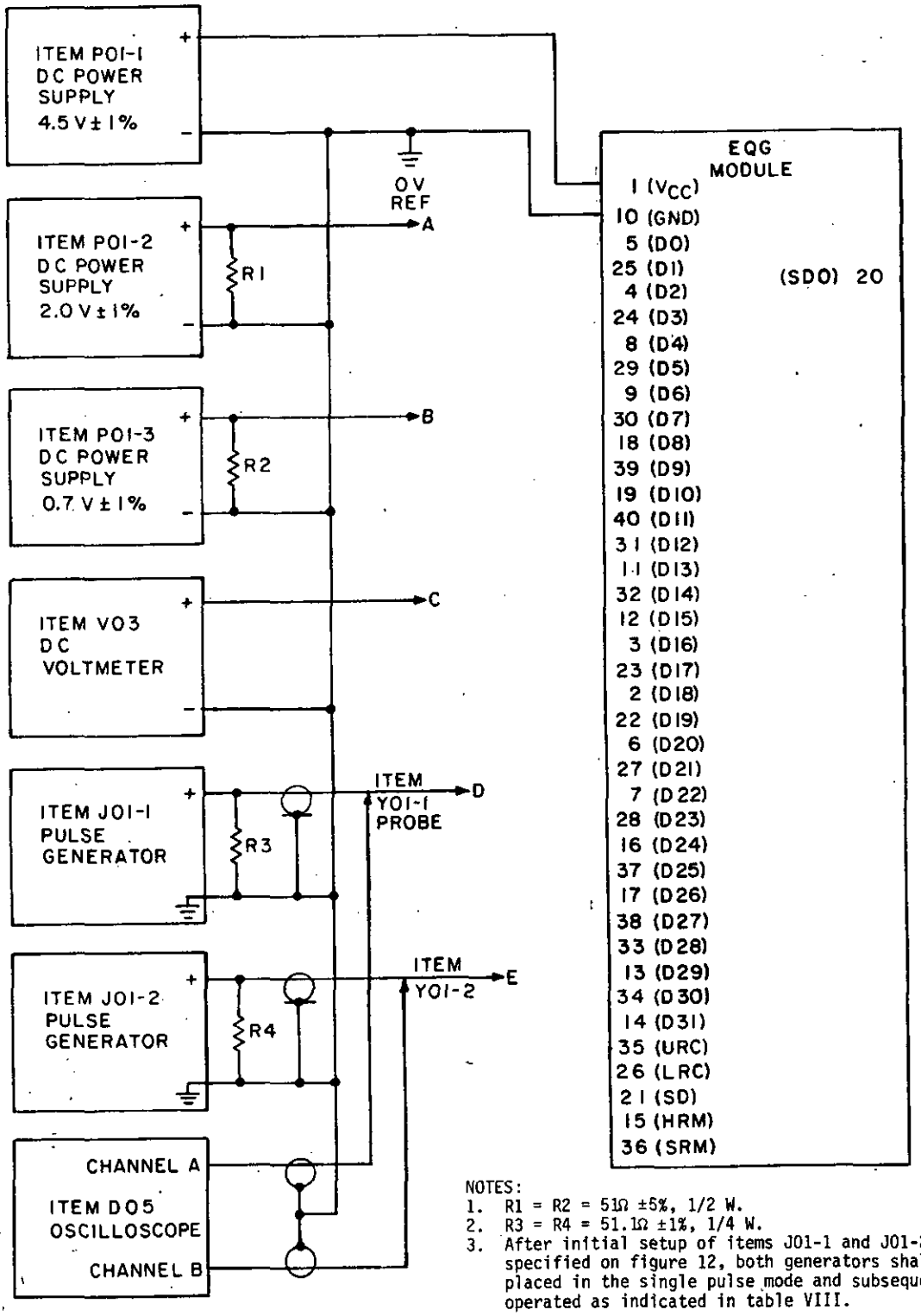
TABLE VIII. Functional test - Continued.

Test condition	Parameter	Connect																																			Output logic state (see notes)	Test figure
		Pin 21 to	Pin 15 to	Pin 36 to	Pin 35 to	Pin 26 to	Pin 5 to	Pin 25 to	Pin 4 to	Pin 24 to	Pin 8 to	Pin 29 to	Pin 9 to	Pin 30 to	Pin 18 to	Pin 39 to	Pin 19 to	Pin 40 to	Pin 31 to	Pin 11 to	Pin 32 to	Pin 12 to	Pin 3 to	Pin 23 to	Pin 2 to	Pin 22 to	Pin 6 to	Pin 27 to	Pin 7 to	Pin 28 to	Pin 16 to	Pin 37 to	Pin 17 to	Pin 38 to	Pin 33 to	Pin 13 to	Pin 34 to	Pin 14 to	Pin 20	
F100	Functional	B	B	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	B	L	11, 12
		B	B	A	A	B	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	B	X	
F101		B	B	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	B	L	
		B	B	A	A	B	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	B	X	
F102		B	B	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	B	L	
		B	B	A	A	B	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	B	X	
F103		B	B	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	B	L	
		B	B	A	A	B	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	B	X	
F104		B	B	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	B	L	
		B	B	A	A	B	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	B	X	
F105		B	B	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	B	L	
		B	B	A	A	B	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	B	X	
F106		B	B	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	B	L	
		B	B	A	A	B	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	B	X	
F107		B	B	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	B	L	
		B	B	A	A	B	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	B	X	
F108		B	B	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	B	L	
		B	B	B	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	B	X	
		B	B	B	A	B	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	B	X	
F109	Functional	B	B	B	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	B	L	11, 12

NOTES:

1. Initial and end-of-life limits at 25° ±5°C ambient and 0° to 60°C operating temperature:
   L = 0.0 V minimum, 0.4 V maximum.
   H = 2.5 V minimum, 4.5 V maximum.
   X = Don't care.
2. Verification of output logic states may be performed using a logic state indicator in place of item V03 shown on figure 11.

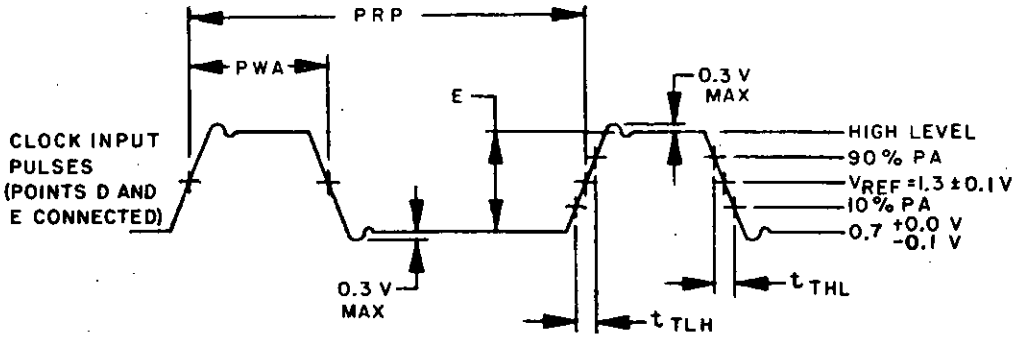
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NOTES:
1. R1 = R2 = 51Ω ±5%, 1/2 W.
2. R3 = R4 = 51.1Ω ±1%, 1/4 W.
3. After initial setup of items J01-1 and J01-2 as specified on figure 12, both generators shall be placed in the single pulse mode and subsequently operated as indicated in table VIII.

FIGURE 11. Functional test circuit for table VIII.

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NOTES:
1. Clock input pulse parameters:
   PWA = 500 ±100 ns.
   PRP = 1,000 ±200 ns.
   $t_{TLH} = t_{THL}$ = 25 ±20 ns.
2. Clock threshold level setup:
   Make circuit connections per table VIII. Adjust items J01-1 and J01-2 to obtain the required PWA and PRP. Use item V03 to adjust the dc offset of items J01-1 and J01-2 for a low level of 0.7 +0.0 V, -0.1 V (clock low level threshold). Using item D05 adjust the pulse generator's amplitude for a high level of 2.0 +0.1 V, -0.0 V (clock high level threshold). The *high level amplitude is adjusted to the required voltage by combining* the dc offset as measured by item V03 with pulse amplitude as measured with item D05. After the pulse generator adjustments have been completed, the pulse generators shall be placed in the single pulse mode.
3. All voltages are referenced to 0 V unless otherwise specified.

FIGURE 12. Functional test circuit waveforms for table VIII.

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TABLE IX. Input clamp diode voltage test.

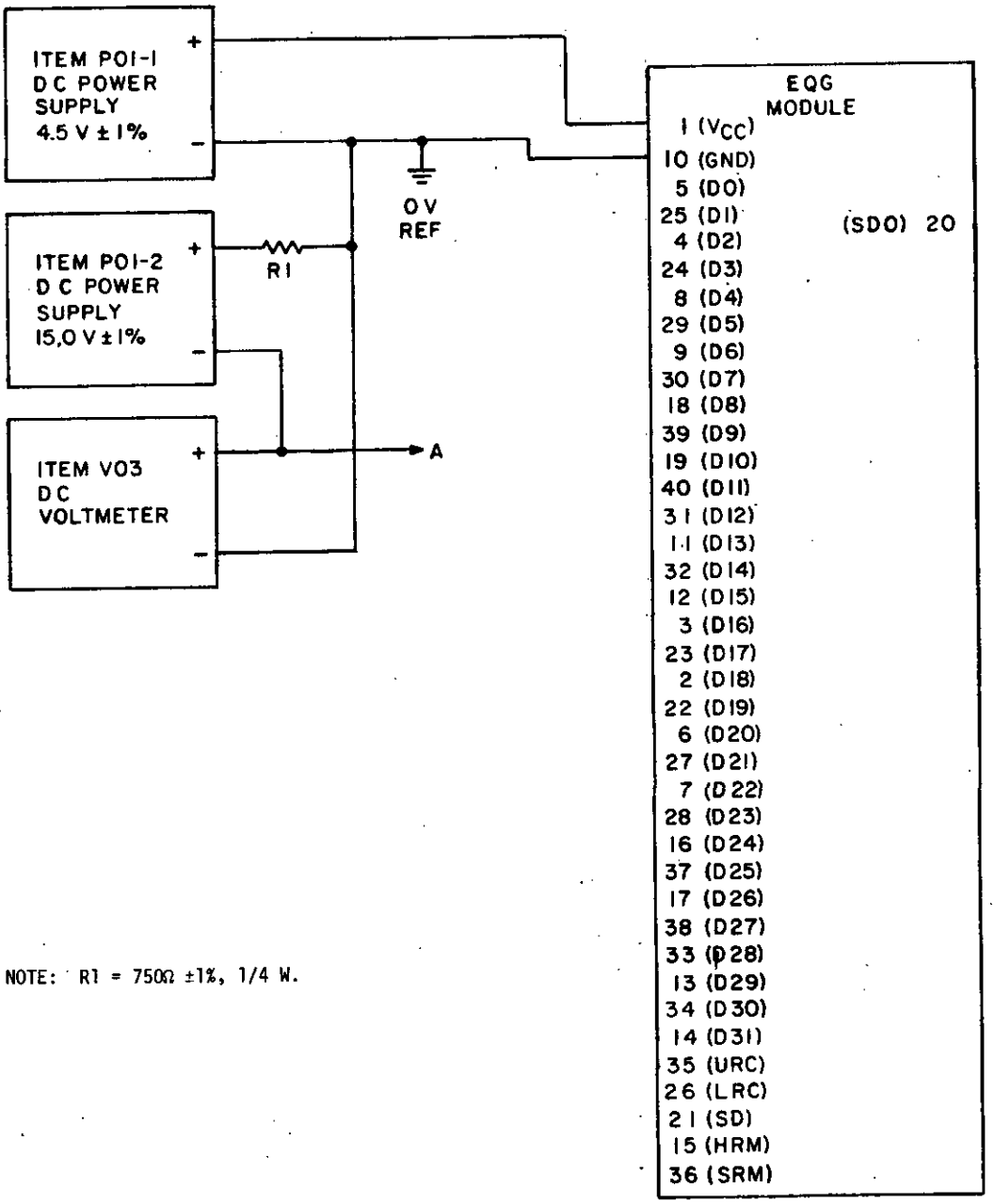
Test condition	Parameter		Connect	Initial limits 25° ±5°C Ambient		Unit	Test figure
			A to pin	Min	Max		
G1	D0	Input clamp diode voltage	5	-	-1.5	V	13
G2	D1		25				
G3	D2		4				
G4	D3		24				
G5	D4		8				
G6	D5		29				
G7	D6		9				
G8	D7		30				
G9	D8		18				
G10	D9		39				
G11	D10		19				
G12	D11		40				
G13	D12		31				
G14	D13		11				
G15	D14		32				
G16	D15		12				
G17	D16		3				
G18	D17		23				
G19	D18	Input clamp diode voltage	2	-	-1.5	V	13

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TABLE IX. Input clamp diode voltage test - Continued.

Test condition	Parameter		Connect	Initial limits 25° ±5°C Ambient		Unit	Test figure
			A to pin	Min	Max		
G20	D19	Input clamp diode voltage	22	-	-1.5	V	13
G21	D20		6				
G22	D21		27				
G23	D22		7				
G24	D23		28				
G25	D24		16				
G26	D25		37				
G27	D26		17				
G28	D27		38				
G29	D28		33				
G30	D29		13				
G31	D30		34				
G32	D31		14				
G33	URC		35				
G34	LRC		26				
G35	SD		21				
G36	HRM		15				
G37	SRM	Input clamp diode voltage	36	-	-1.5	V	13

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T16:42Z
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NOTE: R1 = 750Ω ±1%, 1/4 W.

FIGURE 13. Input clamp diode voltage test circuit for table IX.

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TABLE X. Static output voltage (low-level) test.

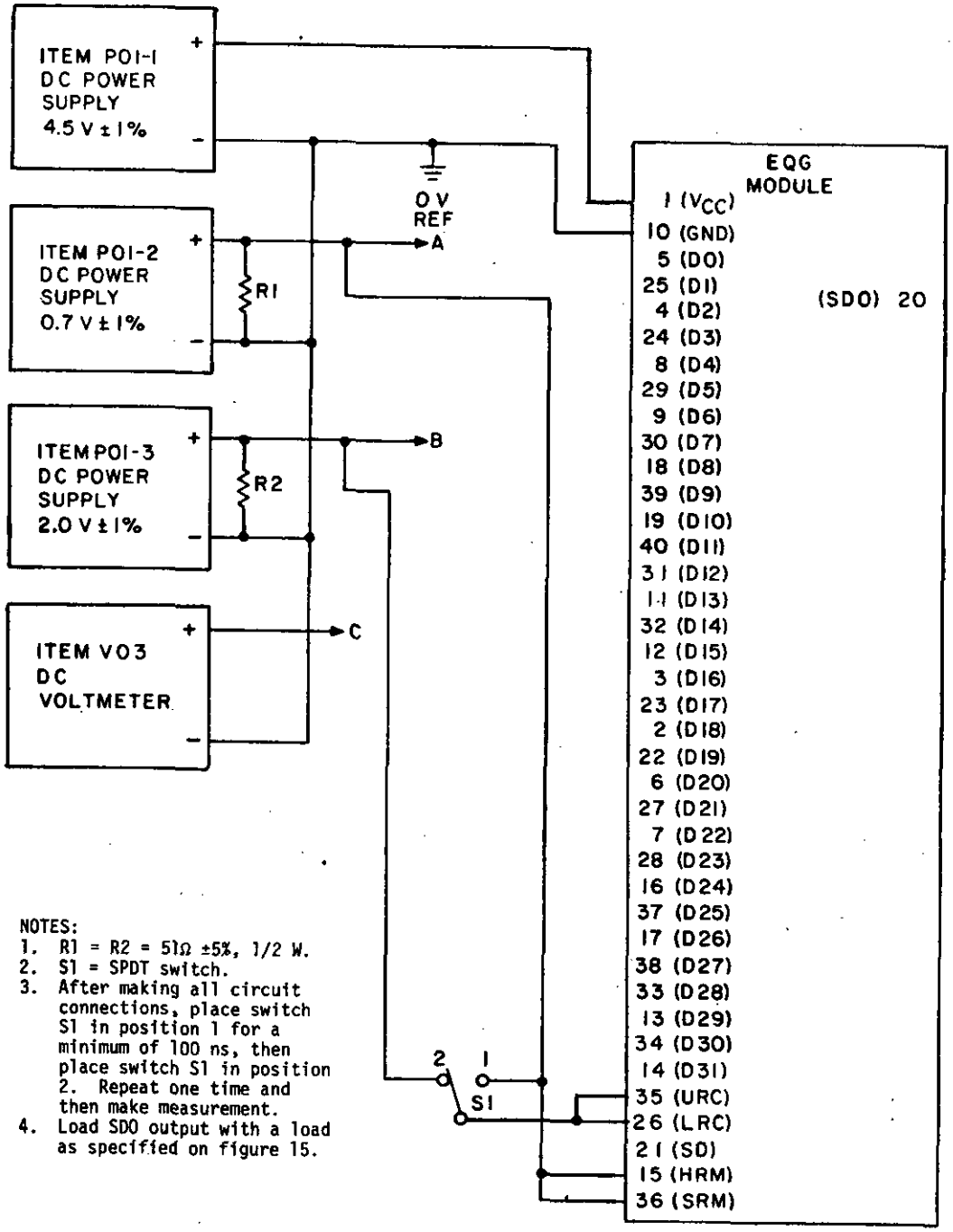
Test condition	Parameter	Connect			Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
H	SDO Static output voltage, low level	14	Open	20	0.0	0.4	V	14

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TABLE XI. Static output voltage (high-level) test.

Test condition	Parameter	Connect			Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
J	SDO Static output voltage, high level	Open	14	20	2.5	4.5	V	14

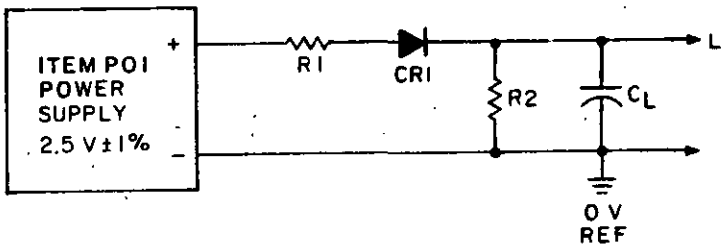
Source: https://assist.dla.mil -- Downloaded: 202610-20T16:42Z
Check the source to verify that this is the current version before use.



NOTES:
1. R1 = R2 = 51Ω ±5%, 1/2 W.
2. S1 = SPDT switch.
3. After making all circuit connections, place switch S1 in position 1 for a minimum of 100 ns, then place switch S1 in position 2. Repeat one time and then make measurement.
4. Load SDO output with a load as specified on figure 15.

FIGURE 14. Static output voltage (low-level and high-level) test circuit for tables X and XI.

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NOTES:
1. CR1 = 1N3600, 1N4150, or equivalent.
2. R1 = Select value for current flow of 4.0 mA ±5% out of point L with point L held at 0.4 V.
3. R2 = 6.19 kΩ ±1%, 1/8 W.
4. For test conditions N1, P1, Q1, R1, S, T1-T4, U1-U4, V1-V2, W1, W2, X, and Y: C_L = 50 ±5 pF (including probe and parasitic capacitance).
5. For test conditions N2, P2, Q2, and R2: C_L = 150 ±5 pF (including probe and parasitic capacitance).
6. Capacitor C_L is not required for dc electrical test conditions H and J.
7. Lead length from load circuit to module output pin under test shall be less than 1 inch (25.4 mm).

FIGURE 15. EQG logic load circuit.

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TABLE XII. Input current (high-level) test.

Test condition	Parameter		Connect		Initial limits 25° ±5°C Ambient		Unit	Test figure
			A to pin	B to pin	Min	Max		
K1	D0	Input current, high level	5	25,4,24,8,29,9,30,18,39,19,40, 31,11,32,12,3,23,2,22,6,27,7, 28,16,37,17,38,33,13,34,14,35, 26,21,15,36	0	20	µA	16
K2	D1		25	5,4,24,8,29,9,30,18,39,19,40, 31,11,32,12,3,23,2,22,6,27,7, 28,16,37,17,38,33,13,34,14, 35,26,21,15,36				
K3	D2		4	5,25,24,8,29,9,30,18,39,19,40, 31,11,32,12,3,23,2,22,6,27,7, 28,16,37,17,38,33,13,34,14,35, 26,21,15,36				
K4	D3		24	5,25,4,8,29,9,30,18,39,19,40, 31,11,32,12,3,23,2,22,6,27,7, 28,16,37,17,38,33,13,34,14,35, 26,21,15,36				
K5	D4		8	5,25,4,24,29,9,30,18,39,19,40, 31,11,32,12,3,23,2,22,6,27,7, 28,16,37,17,38,33,13,34,14,35, 26,21,15,36				
K6	D5		29	5,25,4,24,8,9,30,18,39,19,40, 31,11,32,12,3,23,2,22,6,27,7, 28,16,37,17,38,33,13,34,14,35, 26,21,15,36				
K7	D6		9	5,25,4,24,8,29,30,18,39,19,40, 31,11,32,12,3,23,2,22,6,27,7, 28,16,37,17,38,33,13,34,14,35, 26,21,15,36				
K8	D7	Input current, high level	30	5,25,4,24,8,29,9,18,39,19,40, 31,11,32,12,3,23,2,22,6,27,7, 28,16,37,17,38,33,13,34,14,35, 26,21,15,36	0	20	µA	16

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T16:42Z
Check the source to verify that this is the current version before use.

TABLE XII. Input current (high-level) test - Continued.

Test condition	Parameter		Connect		Initial limits 25° ±5°C Ambient		Unit	Test figure
			A to pin	B to pin	Min	Max		
K9	D8	Input current, high level	18	5,25,4,24,8,29,9,30,39,19,40, 31,11,32,12,3,23,2,22,6,27,7, 28,16,37,17,38,33,13,34,14,35, 26,21,15,36	0	20	µA	16
K10	D9		39	5,25,4,24,8,29,9,30,18,19,40, 31,11,32,12,3,23,2,22,6,27,7, 28,16,37,17,38,33,13,34,14,35, 26,21,15,36				
K11	D10		19	5,25,4,24,8,29,9,30,18,39,40, 31,11,32,12,3,23,2,22,6,27,7, 28,16,37,17,38,33,13,34,14,35, 26,21,15,36				
K12	D11		40	5,25,4,24,8,29,9,30,18,39,19, 31,11,32,12,3,23,2,22,6,27,7, 28,16,37,17,38,33,13,34,14,35, 26,21,15,36				
K13	D12		31	5,25,4,24,8,29,9,30,18,39,19, 40,11,32,12,3,23,2,22,6,27,7, 28,16,37,17,38,33,13,34,14,35, 26,21,15,36				
K14	D13		11	5,25,4,24,8,29,9,30,18,39,19, 40,31,32,12,3,23,2,22,6,27,7, 28,16,37,17,38,33,13,34,14,35, 26,21,15,36				
K15	D14		32	5,25,4,24,8,29,9,30,18,39,19, 40,31,11,12,3,23,2,22,6,27,7, 28,16,37,17,38,33,13,34,14,35, 26,21,15,36				
K16	D15	Input current, high level	12	5,25,4,24,8,29,9,30,18,39,19, 40,31,11,32,3,23,2,22,6,27,7, 28,16,37,17,38,33,13,34,14,35, 26,21,15,36	0	20	µA	16

Source: https://assist.dla.mil -- Downloaded: 202610-20T16:42Z
Check the source to verify that this is the current version before use.

TABLE XII. Input current (high-level) test - Continued.

Test condition	Parameter	Connect		Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
K17	D16 Input current, high level	3	5,25,4,24,8,29,9,30,18,39,19,40,31,11,32,12,23,2,22,6,27,7,28,16,37,17,38,33,13,34,14,35,26,21,15,36	0	20	µA	16
K18	D17	23	5,25,4,24,8,29,9,30,18,39,19,40,31,11,32,12,3,2,22,6,27,7,28,16,37,17,38,33,13,34,14,35,26,21,15,36				
K19	D18	2	5,25,4,24,8,29,9,30,18,39,19,40,31,11,32,12,3,23,22,6,27,7,28,16,37,17,38,33,13,34,14,35,26,21,15,36				
K20	D19	22	5,25,4,24,8,29,9,30,18,39,19,40,31,11,32,12,3,23,2,6,27,7,28,16,37,17,38,33,13,34,14,35,26,21,15,36				
K21	D20	6	5,25,4,24,8,29,9,30,18,39,19,40,31,11,32,12,3,23,2,22,27,7,28,16,37,17,38,33,13,34,14,35,26,21,15,36				
K22	D21	27	5,25,4,24,8,29,9,30,18,39,19,40,31,11,32,12,3,23,2,22,6,7,28,16,37,17,38,33,13,34,14,35,26,21,15,36				
K23	D22	7	5,25,4,24,8,29,9,30,18,39,19,40,31,11,32,12,3,23,2,22,6,27,28,16,37,17,38,33,13,34,14,35,26,21,15,36				
K24	D23 Input current, high level	28	5,25,4,24,8,29,9,30,18,39,19,40,31,11,32,12,3,23,2,22,6,27,7,16,37,17,38,33,13,34,14,35,26,21,15,36	0	20	µA	16

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T16:42Z
Check the source to verify that this is the current version before use.

TABLE XII. Input current (high-level) test - Continued.

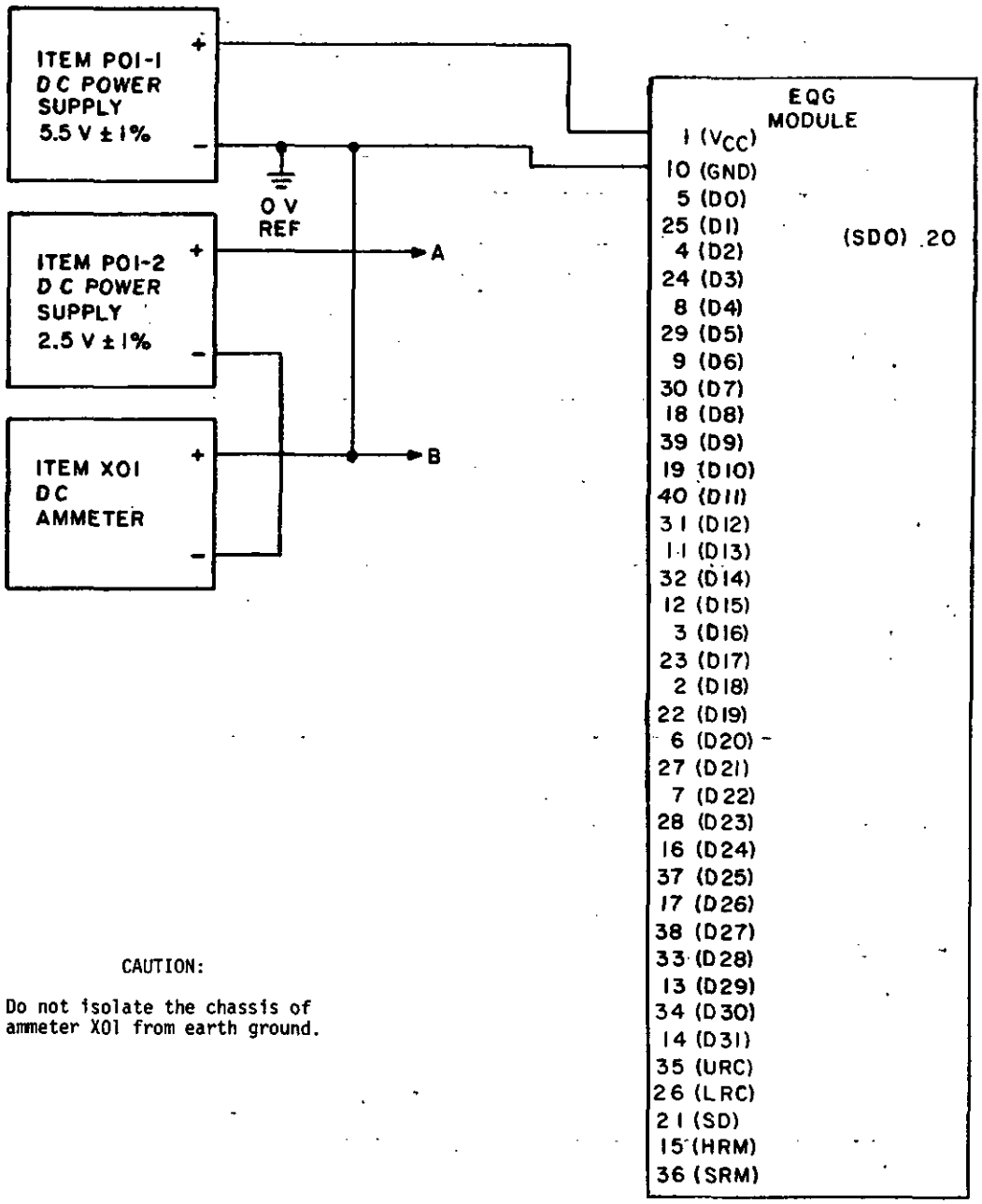
Test condition	Parameter		Connect		Initial limits 25° +5°C Ambient		Unit	Test figure
			A to pin	B to pin	Min	Max		
K25	D24	Input current, high level	16	5,25,4,24,8,29,9,30,18,39,19, 40,31,11,32,12,3,23,2,22,6,27, 7,28,37,17,38,33,13,34,14,35, 26,21,15,36	0	20	µA	16
K26	D25		37	5,25,4,24,8,29,9,30,18,39,19, 40,31,11,32,12,3,23,2,22,6,27, 7,28,16,17,38,33,13,34,14,35, 26,21,15,36				
K27	D26		17	5,25,4,24,8,29,9,30,18,39,19, 40,31,11,32,13,3,23,2,22,6,27, 7,28,16,37,38,33,13,34,14,35, 26,21,15,36				
K28	D27		38	5,25,4,24,8,29,9,30,18,39,19, 40,31,11,32,12,3,23,2,22,6,27, 7,28,16,37,17,33,13,34,14,35, 26,21,15,36				
K29	D28		33	5,25,4,24,8,29,9,30,18,39,19, 40,31,11,32,12,3,23,2,22,6,27, 7,28,16,37,17,38,13,34,14,35, 26,21,15,36				
K30	D29		13	5,25,4,24,8,29,9,30,18,39,19, 40,31,11,32,12,3,23,2,22,6,27, 7,28,16,37,17,38,33,34,14,35, 26,21,15,36				
K31	D30		34	5,25,4,24,8,29,9,30,18,39,19, 40,31,11,32,12,3,23,2,22,6,27, 7,28,16,37,17,38,33,13,14,35, 26,21,15,36				
K32	D31	Input current, high level	14	5,25,4,24,8,29,9,30,18,39,19, 40,31,11,32,12,3,23,2,22,6,27, 7,28,16,37,17,38,33,13,34,35, 26,21,15,36	0	20	µA	16

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TABLE XII. Input current (high-level) test - Continued.

Test condition	Parameter	Connect		Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
K33	SD Input current, high level	21	5,25,4,24,8,29,9,30,18,39,19, 40,31,11,32,12,3,23,2,22,6,27, 7,28,16,37,17,38,33,13,34,14, 35,26,15,36	0	80	µA	16
K34	URC	35	5,25,4,24,8,29,9,30,18,39,19, 40,31,11,32,12,3,23,2,22,6,27, 7,28,16,37,17,38,33,13,34,14, 26,21,15,36		160		
K35	LRC	26	5,25,4,24,8,29,9,30,18,39,19, 40,31,11,32,12,3,23,2,22,6,27, 7,28,16,37,17,38,33,13,34,14, 35,21,15,36				
K36	HRM	15	5,25,4,24,8,29,9,30,18,39,19, 40,31,11,32,12,3,23,2,22,6,27, 7,28,16,37,17,38,33,13,34,14, 35,26,21,36				
K37	SRM Input current, high level	36	5,25,4,24,8,29,9,30,18,39,19, 40,31,11,32,12,3,23,2,22,6,27, 7,28,16,37,17,38,33,13,34,14, 35,26,21,15	0	160	µA	16

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CAUTION:

Do not isolate the chassis of ammeter XO1 from earth ground.

FIGURE 16. Input current (high-level) test circuit for table XII.

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TABLE XIII. Input current (low-level) test.

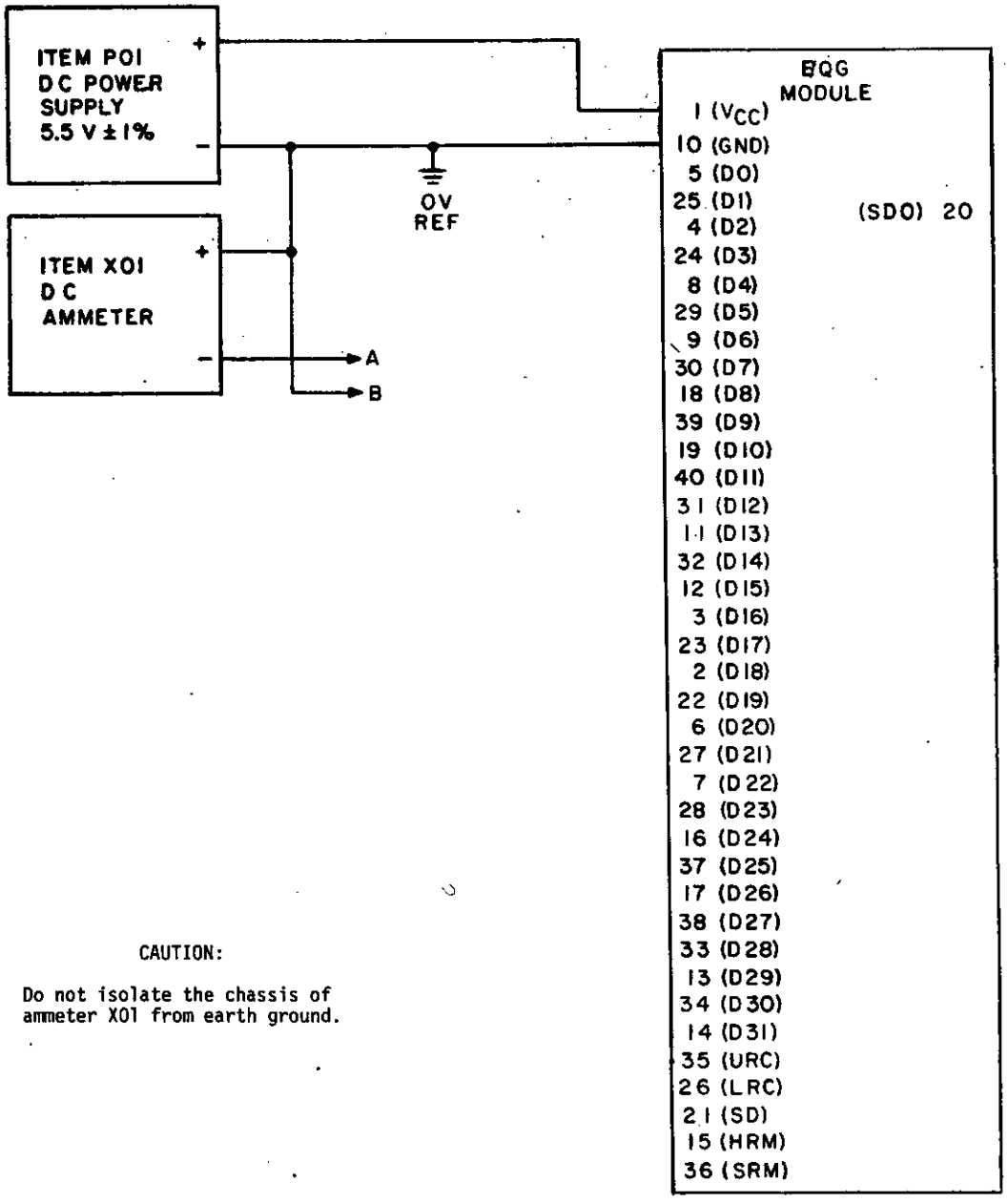
Test condition	Parameter		Connect		Initial limits 25° ±5°C Ambient		Unit	Test figure
			A to pin	B to pin	Min	Max		
L1	D0	Input current, low level	5	15	-0.1	-0.4	mA	17
L2	D1		25					
L3	D2		4					
L4	D3		24					
L5	D4		8					
L6	D5		29					
L7	D6		9					
L8	D7		30					
L9	D8		18					
L10	D9		39					
L11	D10		19					
L12	D11		40					
L13	D12		31					
L14	D13		11					
L15	D14		32					
L16	D15		12					
L17	D16		3					
L18	D17		23					
L19	D18	Input current, low level	2	15	-0.1	-0.4	mA	17

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TABLE XIII. Input current (low-level) test - Continued.

Test condition	Parameter		Connect		Initial limits 25° ±5°C Ambient		Unit	Test figure
			A to pin	B to pin	Min	Max		
L20	D19	Input current, low level	22	15	-0.1	-0.4	mA	17
L21	D20		6					
L22	D21		27					
L23	D22		7					
L24	D23		28					
L25	D24		16					
L26	D25		37					
L27	D26		17					
L28	D27		38					
L29	D28		33					
L30	D29		13					
L31	D30		34					
L32	D31		14			-0.4		
L33	SD		21		-0.1	-1.6		
L34	URC		35		-0.10	-3.91		
L35	LRC		26	15	-0.10	-3.91		
L36	HRM		15	Open	-0.1	-3.2		
L37	SRM	Input current, low level	36	15	-0.1	-3.2	mA	17

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CAUTION:

Do not isolate the chassis of ammeter X01 from earth ground.

FIGURE 17. Input current (low-level) test circuit for table XIII.

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TABLE XIV. Power supply current test.

Test condition	Parameter	Initial limits 25° ±5°C Ambient		Unit	Test figure
		Min	Max		
M	Power supply current	50	340	mA	18

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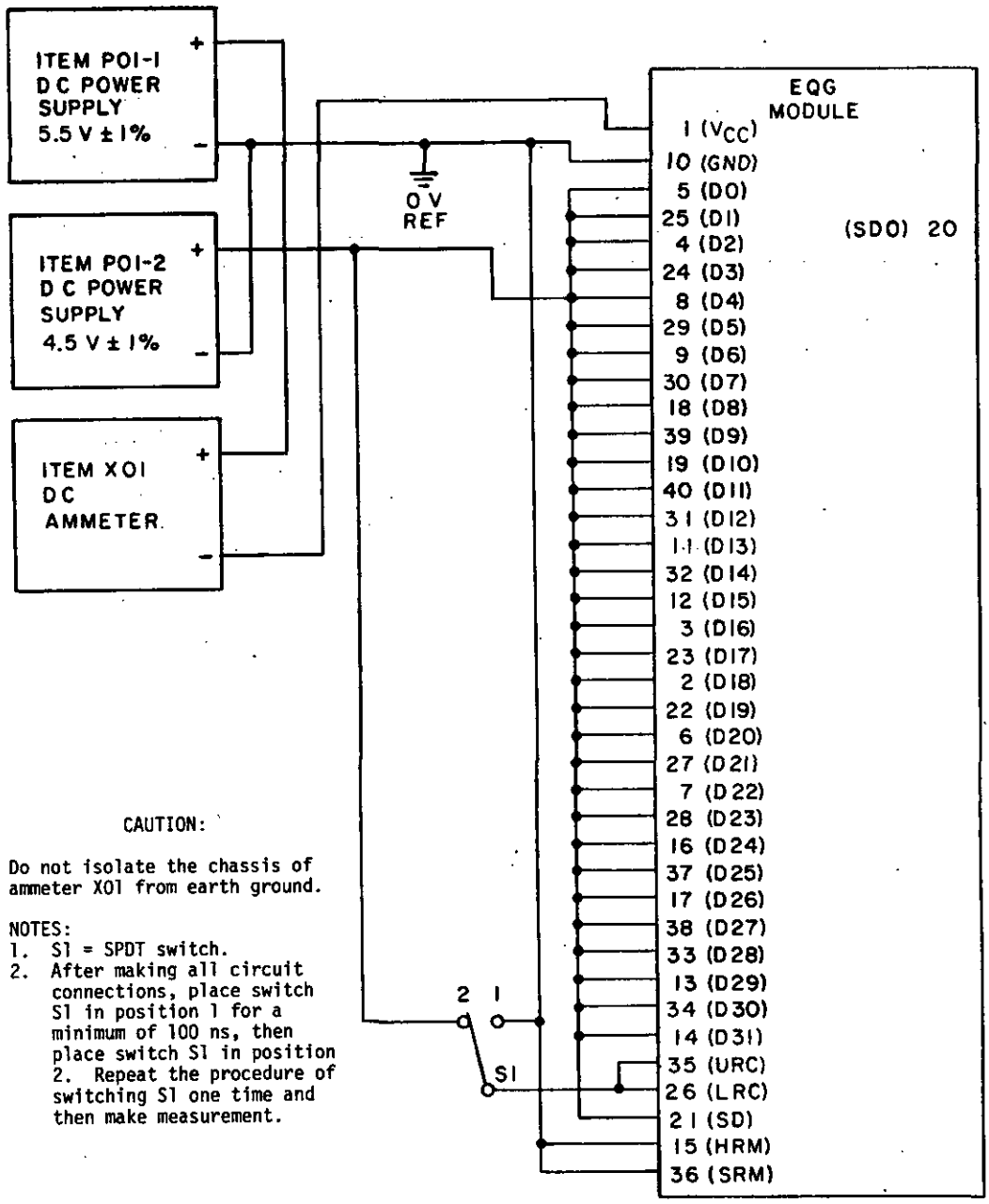


FIGURE 18. Power supply current test circuit for table XIV.

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Check the source to verify that this is the current version before use.

TABLE XV. Transition time (t_{TLH}) test.

Test condition	Parameter	Initial limits 25° ±5°C Ambient		Unit	Test figure
		Min	Max		
N1	SDO Transition time (t_{TLH}), C_L = 50 pF	5	25	ns	19,20
N2	SDO Transition time (t_{TLH}), C_L = 150 pF	6	35	ns	19,20

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TABLE XVI. Transition time (t_{THL}) test.

Test condition	Parameter	Initial limits 25° ±5°C Ambient		Unit	Test figure
		Min	Max		
P1	SDO Transition time (t_{THL}), C_L = 50 pF	1	10	ns	19,20
P2	SDO Transition time (t_{THL}), C_L = 150 pF	2	15	ns	19,20

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T16:42Z
Check the source to verify that this is the current version before use.

TABLE XVII. Propagation delay time (t_{PLH}) test and SD input setup and hold time check.

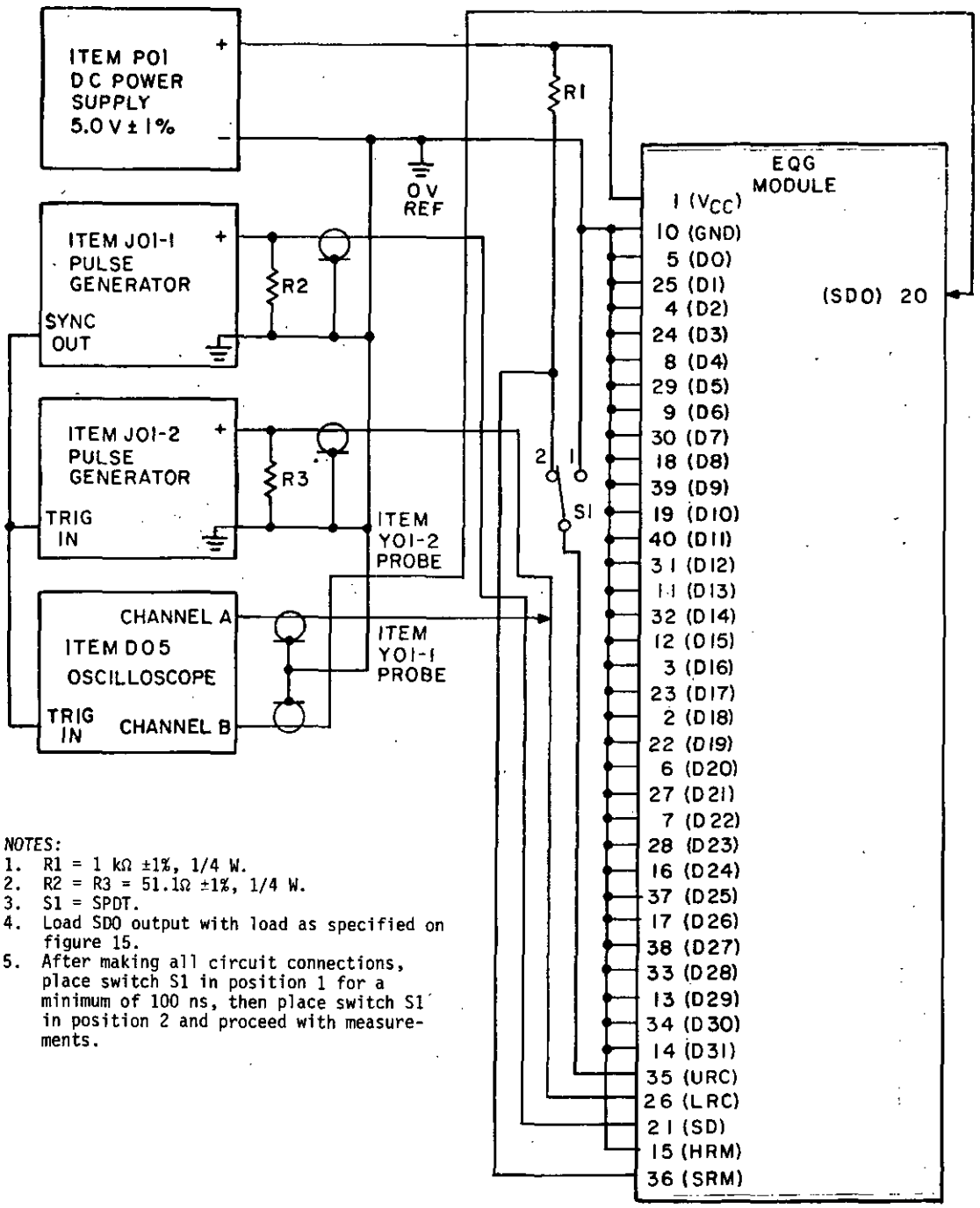
Test condition	Parameter		Initial limits 25° ±5°C Ambient		Unit	Test figure
			Min	Max		
Q1	LRC to SDO	Propagation delay time (t_{PLH}) and SD input setup and hold time check, C_L = 50 pF	10	40	ns	19,20
Q2	LRC to SDO	Propagation delay time (t_{PLH}) and SD input setup and hold time check, C_L = 150 pF	10	55	ns	19,20

Source: https://assist.dla.mil -- Downloaded: 202610-20T16:42Z
Check the source to verify that this is the current version before use.

TABLE XVIII. Propagation delay time (t_{PHL}) test and SD input setup and hold time check.

Test condition	Parameter		Initial limits 25° ±5°C Ambient		Unit	Test figure
			Min	Max		
R1	LRC to SDO	Propagation delay time (t_{PHL}) and SD input setup and hold time check, C_L = 50 pF	10	45	ns	19,20
R2	LRC to SDO	Propagation delay time (t_{PHL}) and SD input setup and hold time check, C_L = 150 pF	10	55	ns	19,20

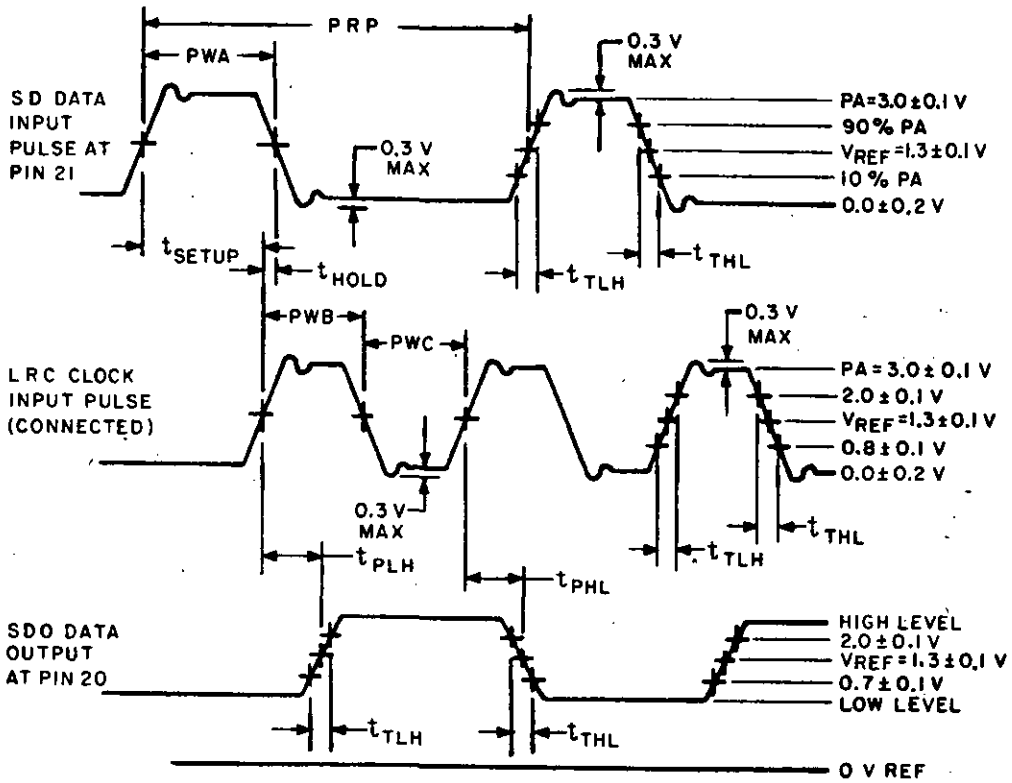
Source: https://assist.dla.mil -- Downloaded: 2026-10-20T16:42Z
Check the source to verify that this is the current version before use.



NOTES:
1. R1 = 1 kΩ ±1%, 1/4 W.
2. R2 = R3 = 51.1Ω ±1%, 1/4 W.
3. S1 = SPDT.
4. Load SDO output with load as specified on figure 15.
5. After making all circuit connections, place switch S1 in position 1 for a minimum of 100 ns, then place switch S1 in position 2 and proceed with measurements.

FIGURE 19. Transition time (t_{TLH} and t_{THL}), propagation delay time (t_{PLH} and t_{PHL}), and SD input setup and hold time check test circuit for tables XV, XVI, XVII, and XVIII.

Source: https://assist.dla.mil -- Downloaded: 202610-20T16:42Z
Check the source to verify that this is the current version before use.



NOTES:

1. Data input pulse parameters:
   PRP = 500 ±50 ns.
   PWA = 25 ±1 ns.
   $t_{TLH} = t_{THL} = 4 \pm 2$ ns.
2. Clock input pulse parameters:
   PWB = 50 ±3 ns.
   PWC = 200 ±5 ns.
   $t_{TLH} = t_{THL} = 4 \pm 1$ ns.
3. Item J01-2 is operated in the double pulse mode and uses the positive output. Using item D05 adjust the amplitude and dc offset controls to obtain the required input logic levels of the data and clock input pulses. Using the delay and pulse width controls adjust the pulse relationship of the data input and clock input pulses so that:

   $t_{SETUP} = 20 \begin{smallmatrix} +1.0 \text{ ns} \\ -0.0 \text{ ns} \end{smallmatrix}$

   $t_{HOLD} = 5 \begin{smallmatrix} +1.0 \text{ ns} \\ -0.0 \text{ ns} \end{smallmatrix}$

   $PWA = t_{SETUP} + t_{HOLD}$.
4. Verify that the SDO output has a high level $\geq$ 2.5 V and a low level $\leq$ 0.4 V before making measurements.
5. The physical length of electrical conductors between resistors R2 and R3 and input pins; between items Y01-1 and Y01-2 and monitored pins; and between the load circuit and the output pin, shall be less than 1 inch (25.4 mm).
6. All voltages are referenced to 0 V unless otherwise specified.

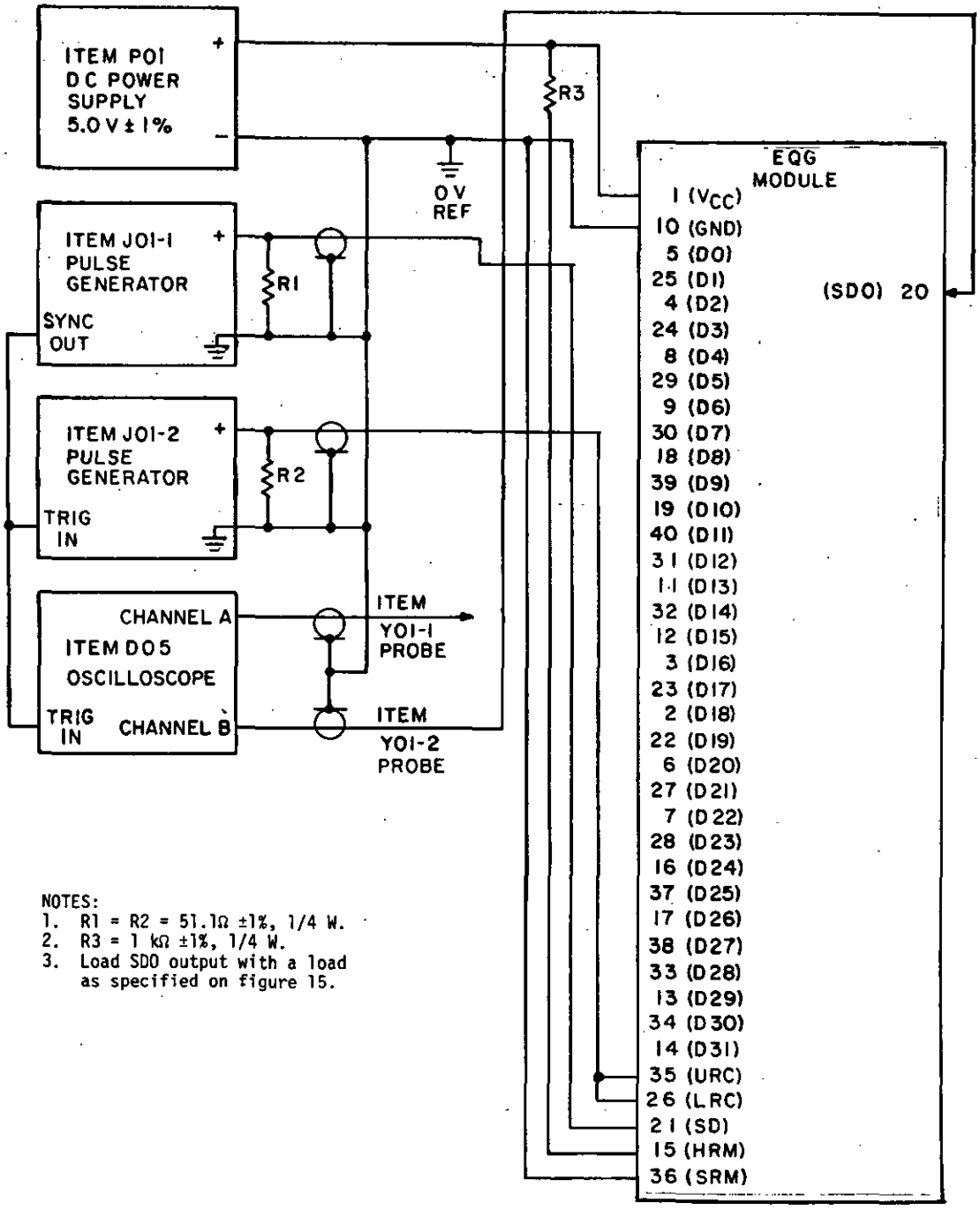
FIGURE 20. Transition time (t_{TLH} and t_{THL}), propagation delay time (t_{PLH} and t_{PHL}), and SD input setup and hold time check test circuit waveforms for tables XV, XVI, XVII, and XVIII.

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TABLE XIX. Maximum operating frequency check and minimum LRC clock pulse width check (measuring output pulse period t_{PD}).

Test condition	Parameter	Initial limits 25° ±5°C Ambient		Unit	Test figure
		Min	Max		
S	Maximum operating frequency check and minimum LRC clock pulse width check (measuring output pulse period t_{PD}), C_L = 50 pF	130	150	ns	21,22

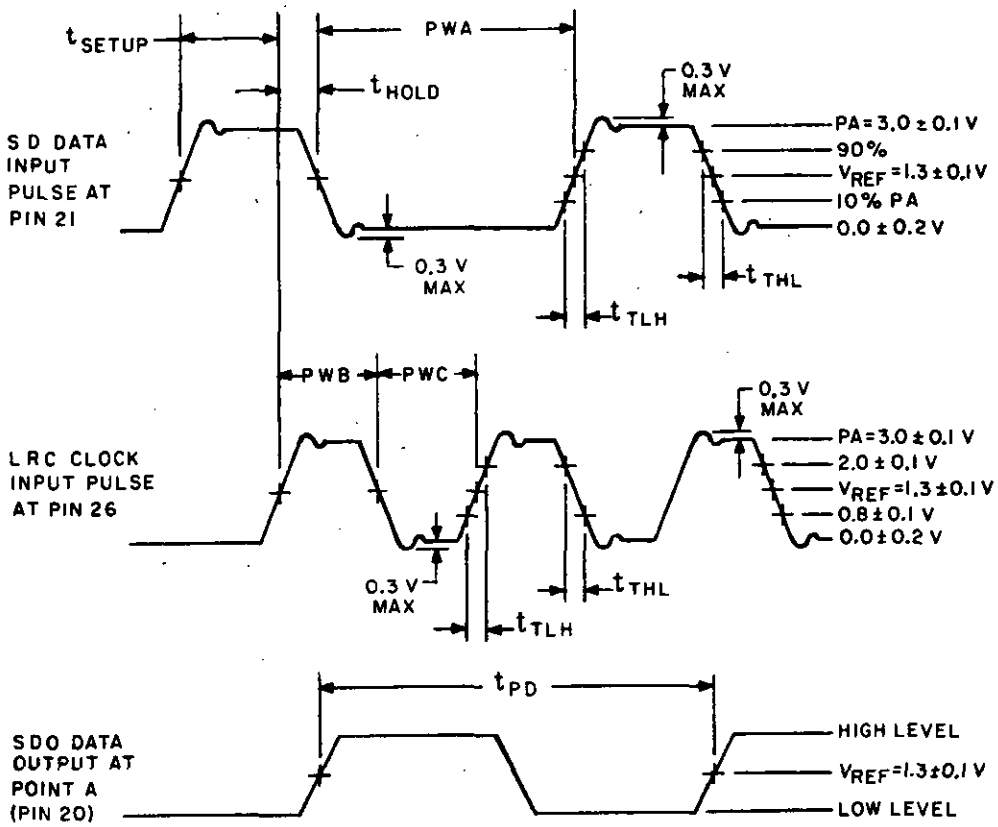
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NOTES:
1. R1 = R2 = 51.1Ω ±1%, 1/4 W.
2. R3 = 1 kΩ ±1%, 1/4 W.
3. Load SDO output with a load as specified on figure 15.

FIGURE 21. Maximum operating frequency check and minimum LRC clock pulse width check (measuring output pulse period t_{PD}) test circuit for table XIX.

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NOTES:

1. Data input pulse parameters:
   PWA = 115 ±3 ns.
   $t_{TLH} = t_{THL} = 4 \pm 2$ ns.
   $t_{SETUP} = 20 \pm 3$ ns.
   $t_{HOLD} = 5 \pm 2$ ns.
2. Clock input pulse parameters:
   PWB = $20 \begin{smallmatrix} +1 \text{ ns} \\ -0 \text{ ns} \end{smallmatrix}$ (minimum clock pulse width).
   PWC = 50 ±3 ns.
   $t_{TLH} = t_{THL} = 4 \pm 2$ ns.
3. Item J01-2 is operated in the double pulse mode and uses the positive output. Using item D05 adjust the amplitude and dc offset controls to obtain the required input logic levels of the data and clock input pulses. Before making any measurements per table XIX the data setup and hold time and clock pulse width must be verified.
4. Verify that the SDO output has a high level $\geq$ 2.5 V and a low level $\leq$ 0.4 V before making measurements.
5. The physical length of electrical conductors between resistors R1 and R2 and input pins; between items Y01-1 and Y01-2 and monitored pins; and between the load circuit and output pin, shall be less than 1 inch (25.4 mm).
6. All voltages are referenced to 0 V unless otherwise specified.

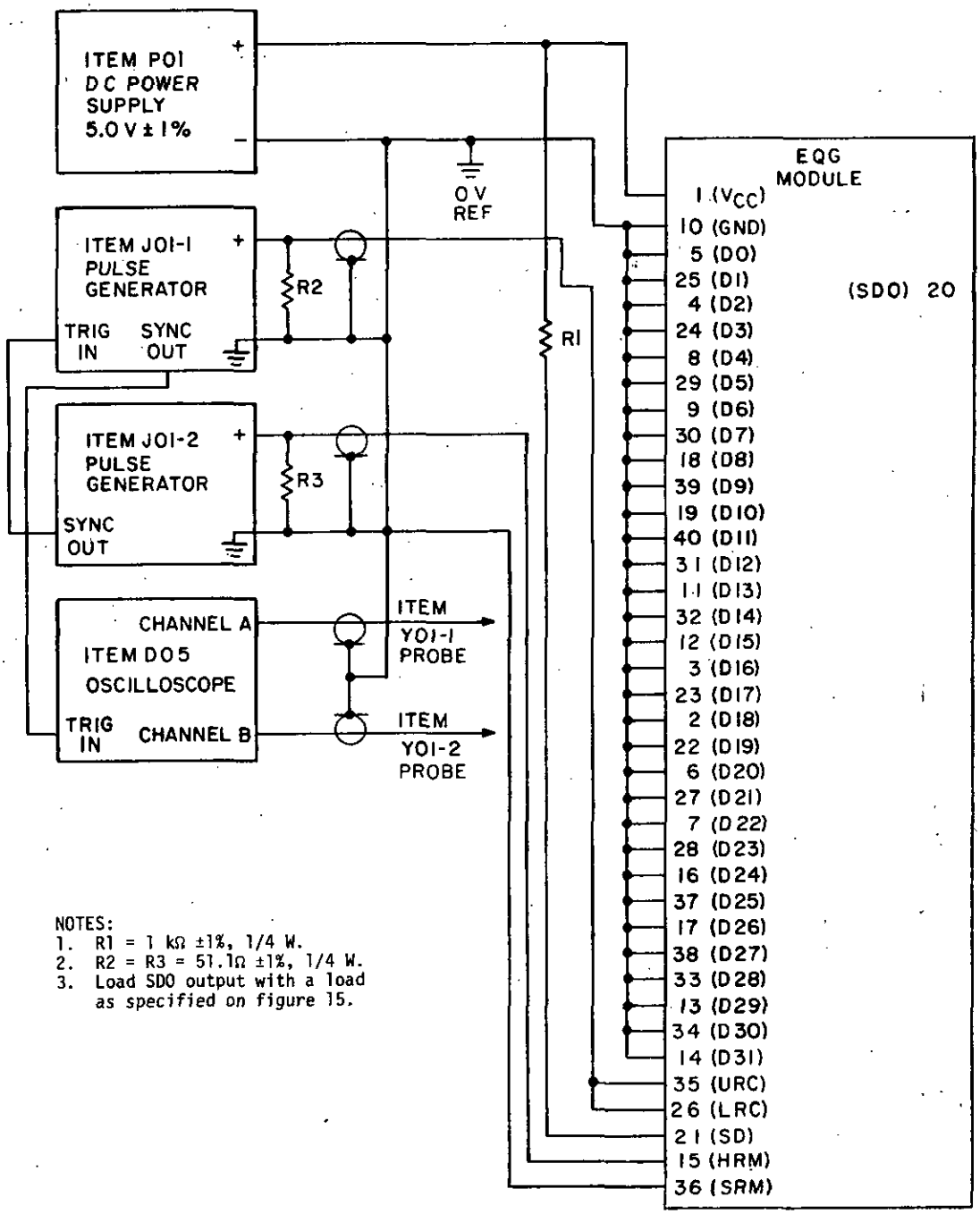
FIGURE 22. Maximum operating frequency check and minimum LRC clock pulse width check (measuring output pulse period t_{PD}) test circuit waveforms for table XIX.

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TABLE XX. HRM/SRM setup and hold time check for parallel load and serial shift.

Test condition	Parameter	Initial limits 25° ±5°C Ambient		Unit	Test figure
		Min	Max		
T1	HRM setup and hold time check for parallel load	0.0	0.4	V	23,24
T2	HRM setup and hold time check for serial shift	2.5	4.5		23,24
T3	SRM setup and hold time check for parallel load	0.0	0.4		25,26
T4	SRM setup and hold time check for serial shift	2.5	4.5	V	25,26

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NOTES:
1. R1 = 1 kΩ ±1%, 1/4 W.
2. R2 = R3 = 51.1Ω ±1%, 1/4 W.
3. Load SDO output with a load as specified on figure 15.

FIGURE 23. HRM setup and hold time check for parallel load and serial shift test circuit for test conditions T1 and T2, table XX.

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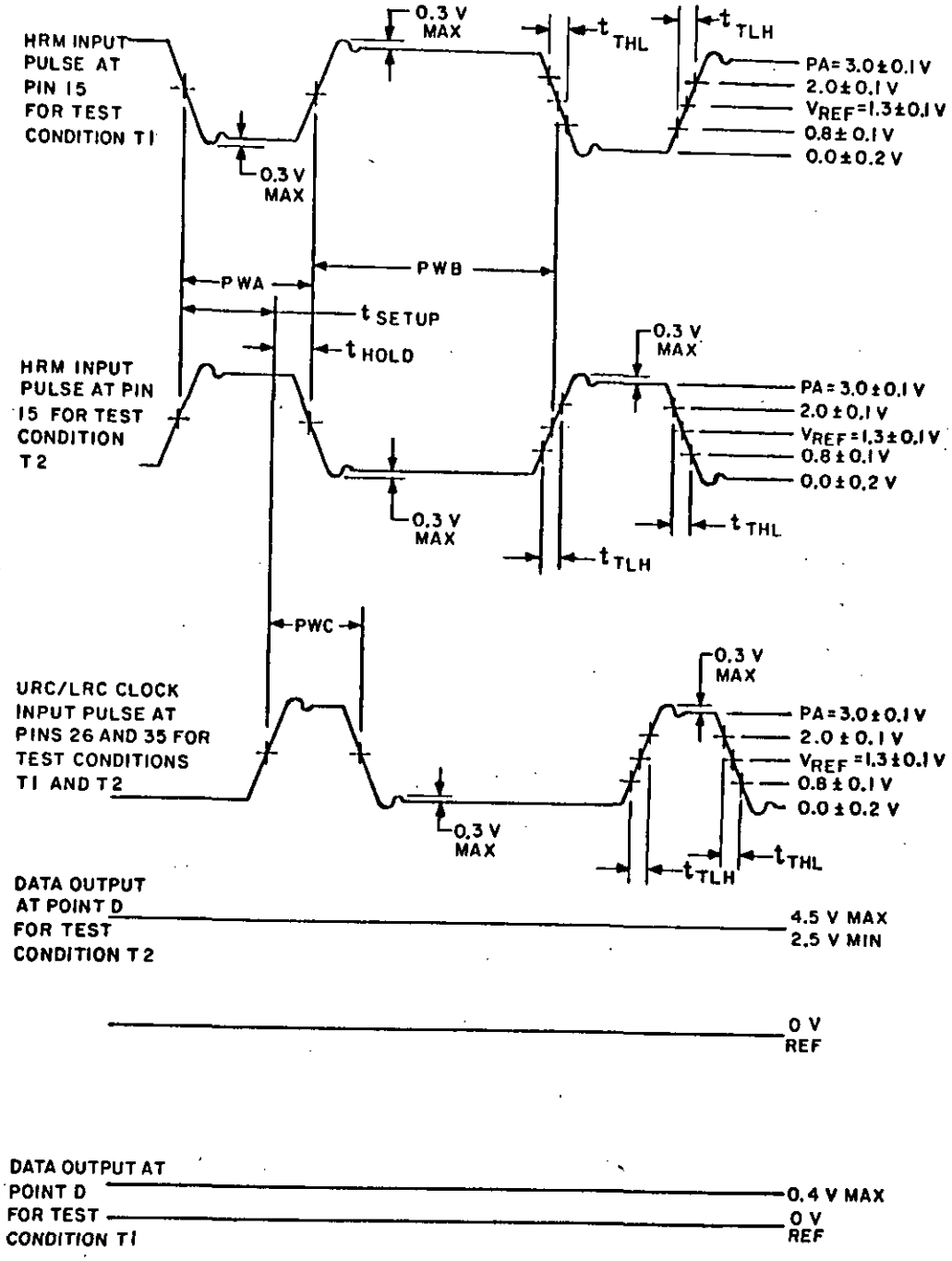


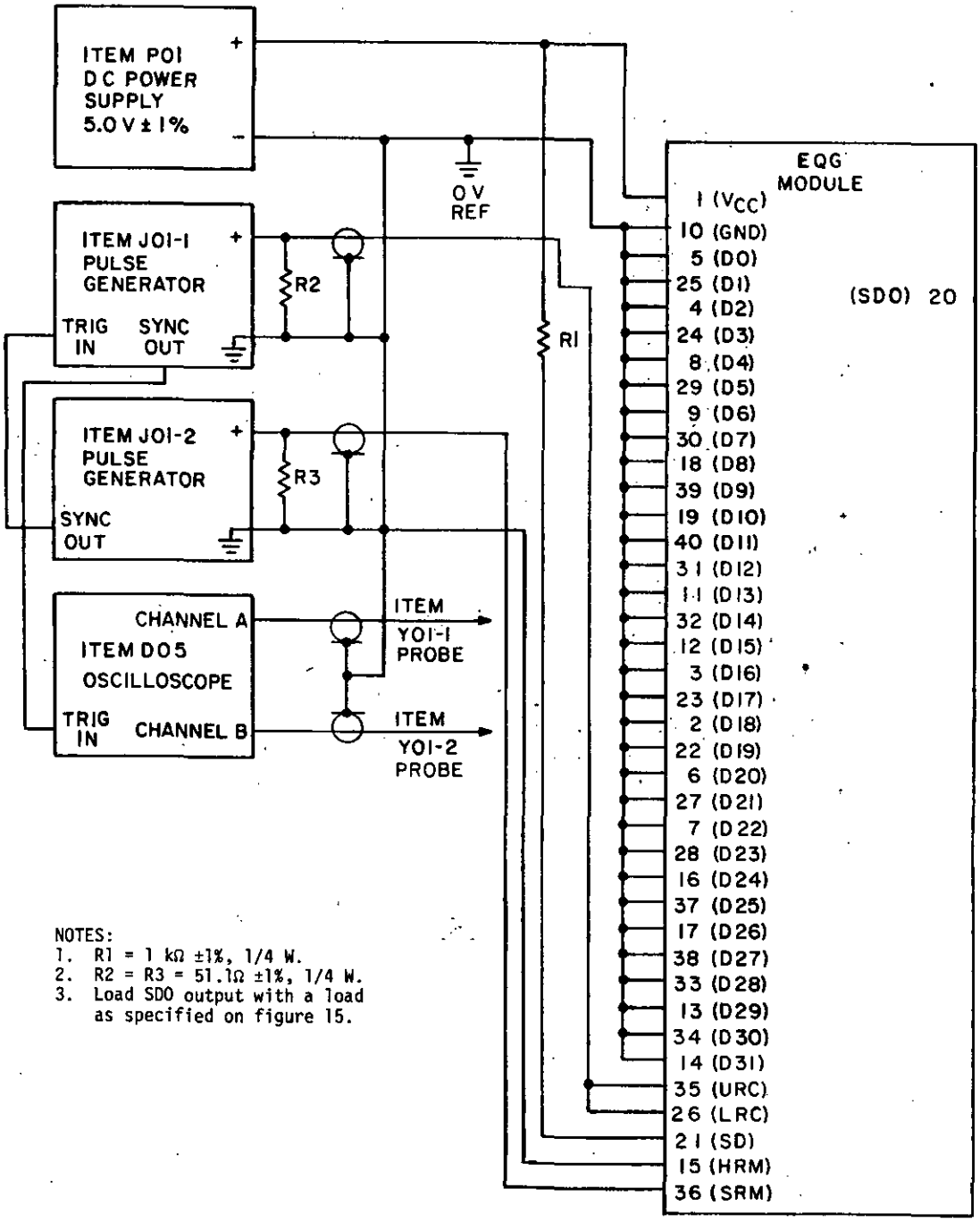
FIGURE 24. HRM setup and hold time check for parallel load and serial shift test circuit waveforms for test conditions T1 and T2, table XX.

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NOTES:
1. HRM input pulse parameters:
   PWA = 45 $^{+1\ ns}_{-0.0\ ns}$
   PWB = 200 ±10 ns.
   $t_{TLH} = t_{THL}$ = 4 ±1 ns.
   t_{SETUP} = 40 $^{+1\ ns}_{-0.0\ ns}$
   t_{HOLD} = 5 $^{+1\ ns}_{-0.0\ ns}$
2. URC/LRC clock input pulse parameters:
   PWC = 25 ±2 ns.
   $t_{TLH} = t_{THL}$ = 4 ±1 ns.
3. Using item D05 adjust the amplitude and dc offset controls to obtain the required input logic levels of HRM and URC/LRC. Using the appropriate delay, pulse width, and frequency controls adjust items J01-1 and J01-2 to obtain the parameters specified in notes 1 and 2.
4. Verify that the SDO output has a high level ≥ 2.5 V and a low level ≤ 0.4 V before making measurements.
5. The physical length of electrical conductors between resistors R2 and R3 and input pins; between items Y01-1 and Y01-2 and monitored pins; and between the load circuit and output pin, shall be less than 1 inch (25.4 mm).
6. All voltages are referenced to 0 V unless otherwise specified.

FIGURE 24. HRM setup and hold time check for parallel load and serial shift test circuit waveforms for test conditions T1 and T2, table XX - Continued.

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Check the source to verify that this is the current version before use.



NOTES:
1. R1 = 1 kΩ ±1%, 1/4 W.
2. R2 = R3 = 51.1Ω ±1%, 1/4 W.
3. Load SDO output with a load as specified on figure 15.

FIGURE 25. SRM setup and hold time check for parallel load and serial shift test circuit for test conditions T3 and T4, table XX.

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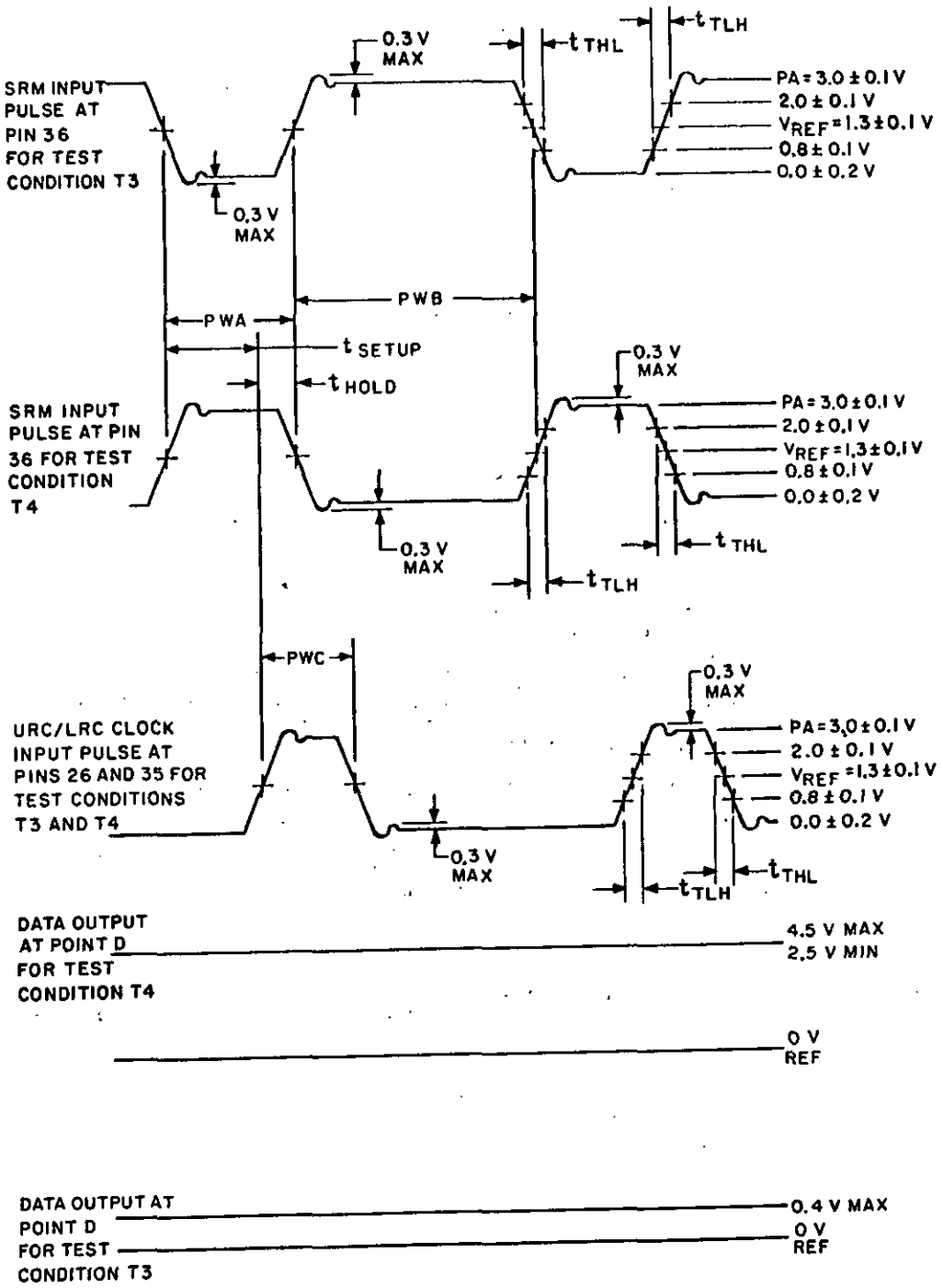


FIGURE 26. SRM setup and hold time check for parallel load and serial shift test circuit waveforms for test conditions T3 and T4, table XX.

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NOTES:
1. SRM input pulse parameters:
   PWA = 45 $^{+1\ ns}_{-0.0\ ns}$.
   PWB = 200 ±10 ns.
   $t_{TLH} = t_{THL}$ = 4 ±1 ns.
   t_{SETUP} = 40 $^{+1\ ns}_{-0.0\ ns}$.
   t_{HOLD} = 5 $^{+1\ ns}_{-0.0\ ns}$.
2. URC/LRC clock input pulse parameters:
   PWC = 25 ±2 ns.
   $t_{TLH} = t_{THL}$ = 4 ±1 ns.
3. Using item D05 adjust the amplitude and dc offset controls to obtain the required input logic levels of SRM and URC/LRC. Using the appropriate delay, pulse width, and frequency controls adjust items J01-1 and J01-2 to obtain the parameters specified in notes 1 and 2.
4. Verify that the SDO output has a high level ≥ 2.5 V and a low level ≤ 0.4 V before making measurements.
5. The physical length of electrical conductors between resistors R2 and R3 and input pins; between items Y01-1 and Y01-2 and monitored pins; and between the load circuit and output pin, shall be less than 1 inch (25.4 mm).
6. All voltages are referenced to 0 V unless otherwise specified.

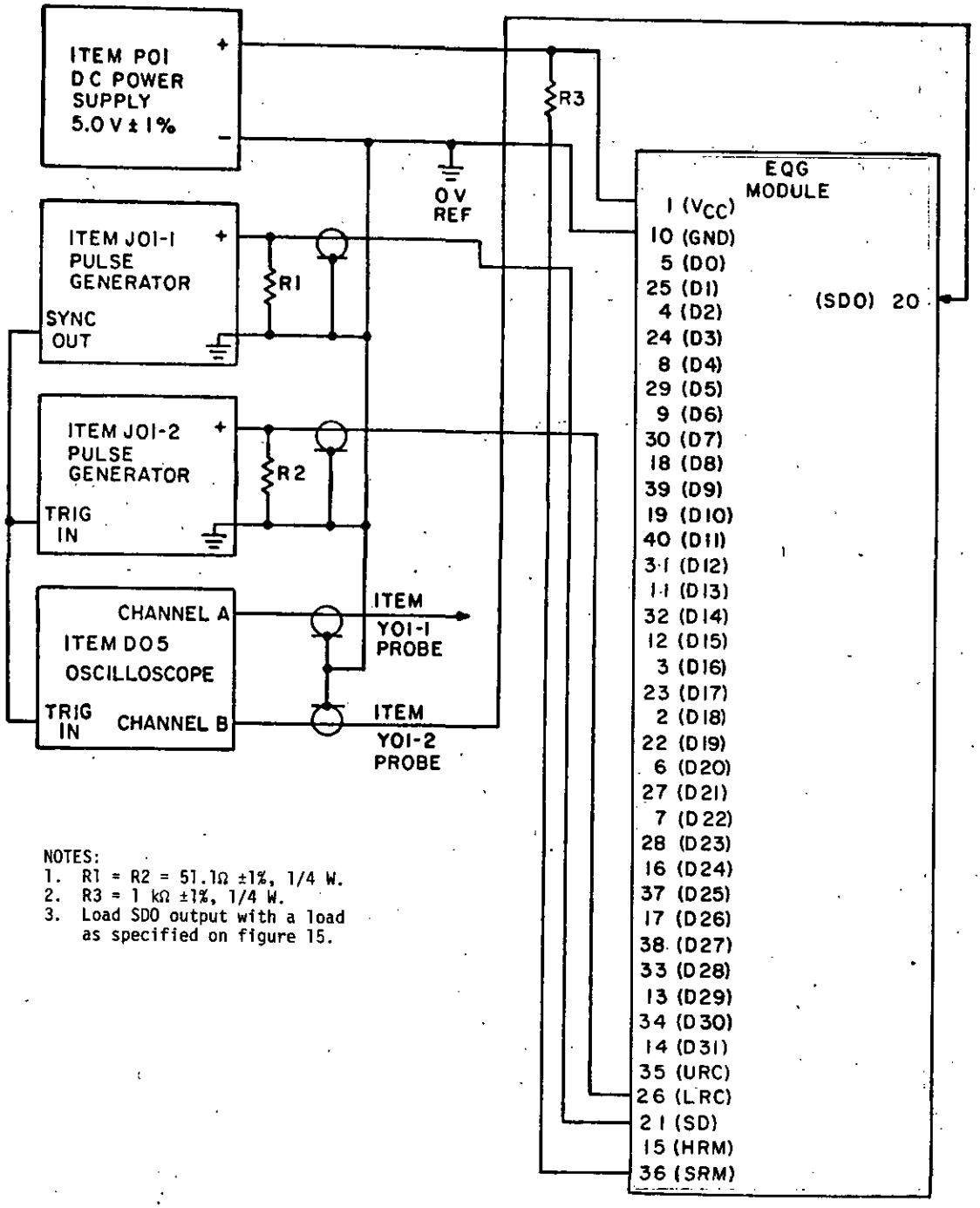
FIGURE 26. SRM setup and hold time check for parallel load and serial shift test circuit waveforms for test conditions T3 and T4, table XX - Continued.

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TABLE XXI. Serial data (high-level and low-level) setup and hold time check (measuring output pulse period, t_{PD} and output pulse width, t_{PW}).

Test condition	Parameter	Initial limits 25° ±5°C Ambient		Unit	Test figure
		Min	Max		
U1	Serial data, high level, setup and hold time check (measuring output pulse period, t_{PD})	130	150	ns	27,28
U2	Serial data, high level, setup and hold time check (measuring output pulse width, t_{PW})	60	80		
U3	Serial data, low level, setup and hold time check (measuring output pulse period, t_{PD})	130	150		
U4	Serial data, low level, setup and hold time check (measuring output pulse width, t_{PW})	60	80	ns	27,28

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NOTES:
1. R1 = R2 = 51.1Ω ±1%, 1/4 W.
2. R3 = 1 kΩ ±1%, 1/4 W.
3. Load SDO output with a load as specified on figure 15.

FIGURE 27. Serial data (high-level and low-level) setup and hold time check (measuring output pulse period t_{PD} and output pulse width t_{PW}) test circuit for table XXI.

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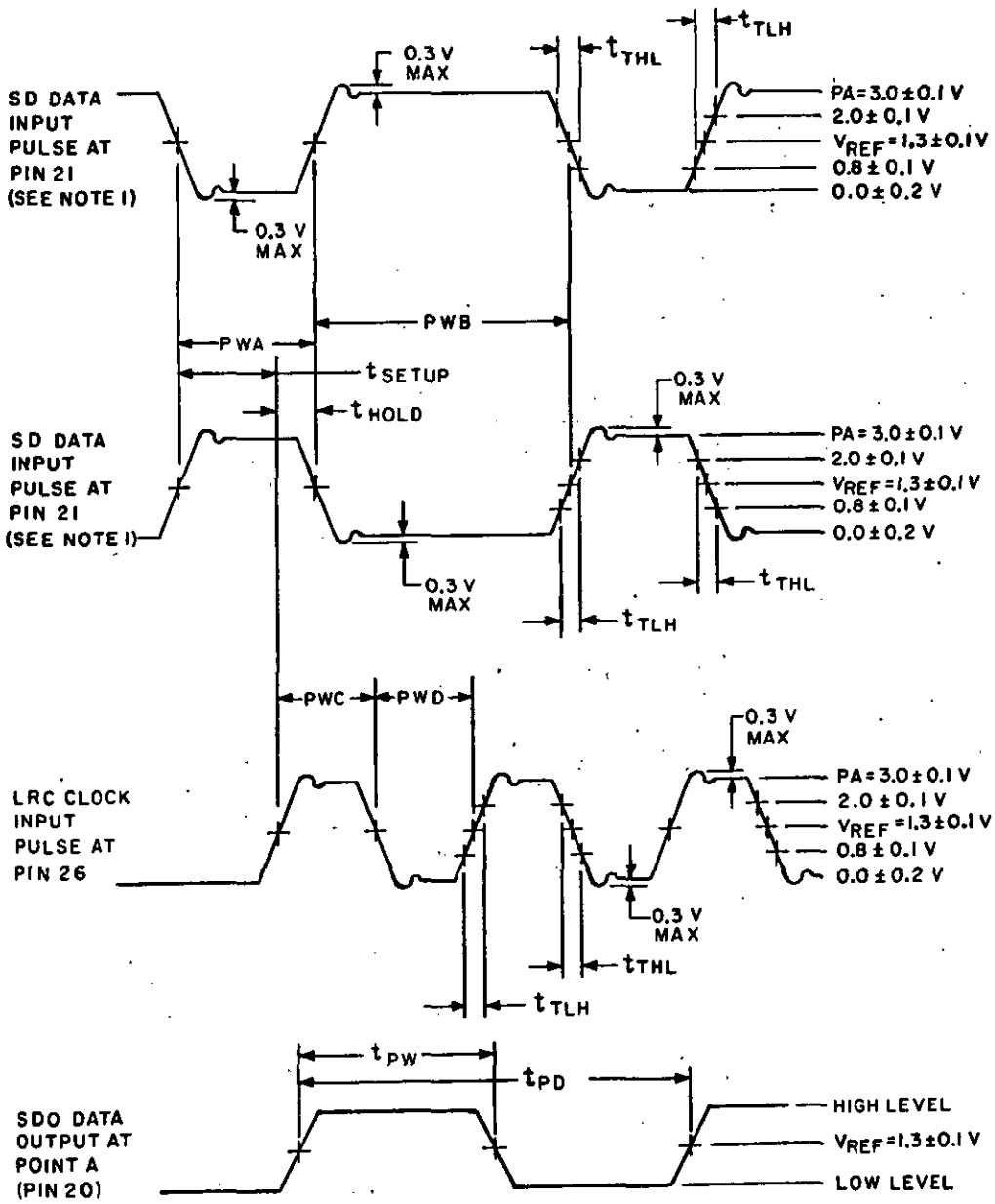


FIGURE 28. Serial data (high-level and low-level) setup and hold time check (measuring output pulse period t_{PD} and output pulse width t_{PW}) test circuit waveforms for table XXI.

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NOTES:
1. *SD data input pulse parameters:*
   PWA = 25 ±1 ns.
   PWB = 115 ±3 ns.
   $t_{TLH} = t_{THL} = 4 \pm 2$ ns.
   The negative data pulses are used for test conditions U3 and U4. The positive data pulses are used for test conditions U1 and U2.
2. LRC clock input pulse parameters:
   PWC = 20 ±2 ns.
   PWD = 50 ±3 ns.
   $t_{TLH} = t_{THL} = 4 \pm 2$ ns.
3. Item J01-2 is operated in the double pulse mode and uses the positive mode. Using item D05 adjust the amplitude and dc offset controls to obtain the required input logic levels of the data and clock input pulses. Using the delay and pulse width controls adjust the pulse relationship of the data input and clock input pulses so that:
   $t_{SETUP} = 20 \begin{smallmatrix}+1.0\text{ ns} \\ -0.0\text{ ns}\end{smallmatrix}$.
   $t_{HOLD} = 5 \begin{smallmatrix}+1.0\text{ ns} \\ -0.0\text{ ns}\end{smallmatrix}$.
   $PWA = t_{SETUP} + t_{HOLD}$.
4. Verify that the SDO output has a high level $\geq$ 2.5 V and a low level $\leq$ 0.4 V before making measurements.
5. The physical length of electrical conductors between resistors R1 and R2 and input pins; between items Y01-1 and Y01-2 and monitored pins; and between the load circuit and output pin, shall be less than 1 inch (25.4 mm).
6. All voltages are referenced to 0 V unless otherwise specified.

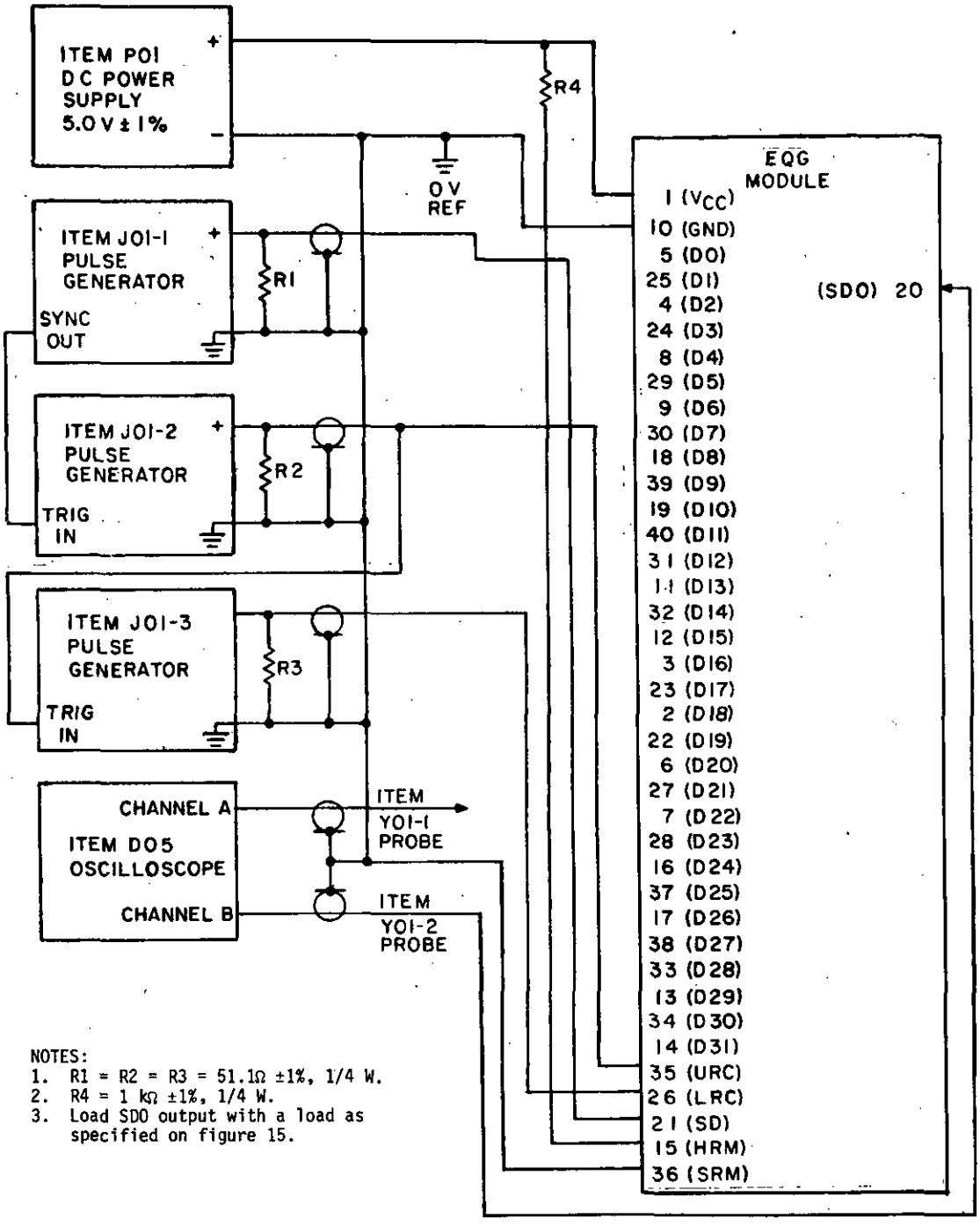
FIGURE 28. Serial data (high-level and low-level) setup and hold time check (measuring output pulse period t_{PD} and output pulse width t_{PW}) test circuit waveforms for table XXI - Continued.

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TABLE XXII. Upper rank clock (URC) and lower rank clock (LRC) minimum pulse width check (measuring output pulse period, t_{PD} and output pulse width, t_{PW}).

Test condition	Parameter	Initial limits 25° ±5°C Ambient		Unit	Test figure
		Min	Max		
V1	Upper rank clock (URC) and lower rank clock (LRC) minimum pulse width check (measuring output pulse period, t_{PD})	130	150	ns	29,30
V2	Upper rank clock (URC) and lower rank clock (LRC) minimum pulse width check (measuring output pulse width, t_{PW})	60	80	ns	29,30

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NOTES:
1. R1 = R2 = R3 = 51.1Ω ±1%, 1/4 W.
2. R4 = 1 kΩ ±1%, 1/4 W.
3. Load SDO output with a load as specified on figure 15.

FIGURE 29. Upper rank clock (URC) and lower rank clock (LRC) minimum pulse width check (measuring output pulse period t_{pp} and output pulse width t_{pw} test circuit for table XXII.

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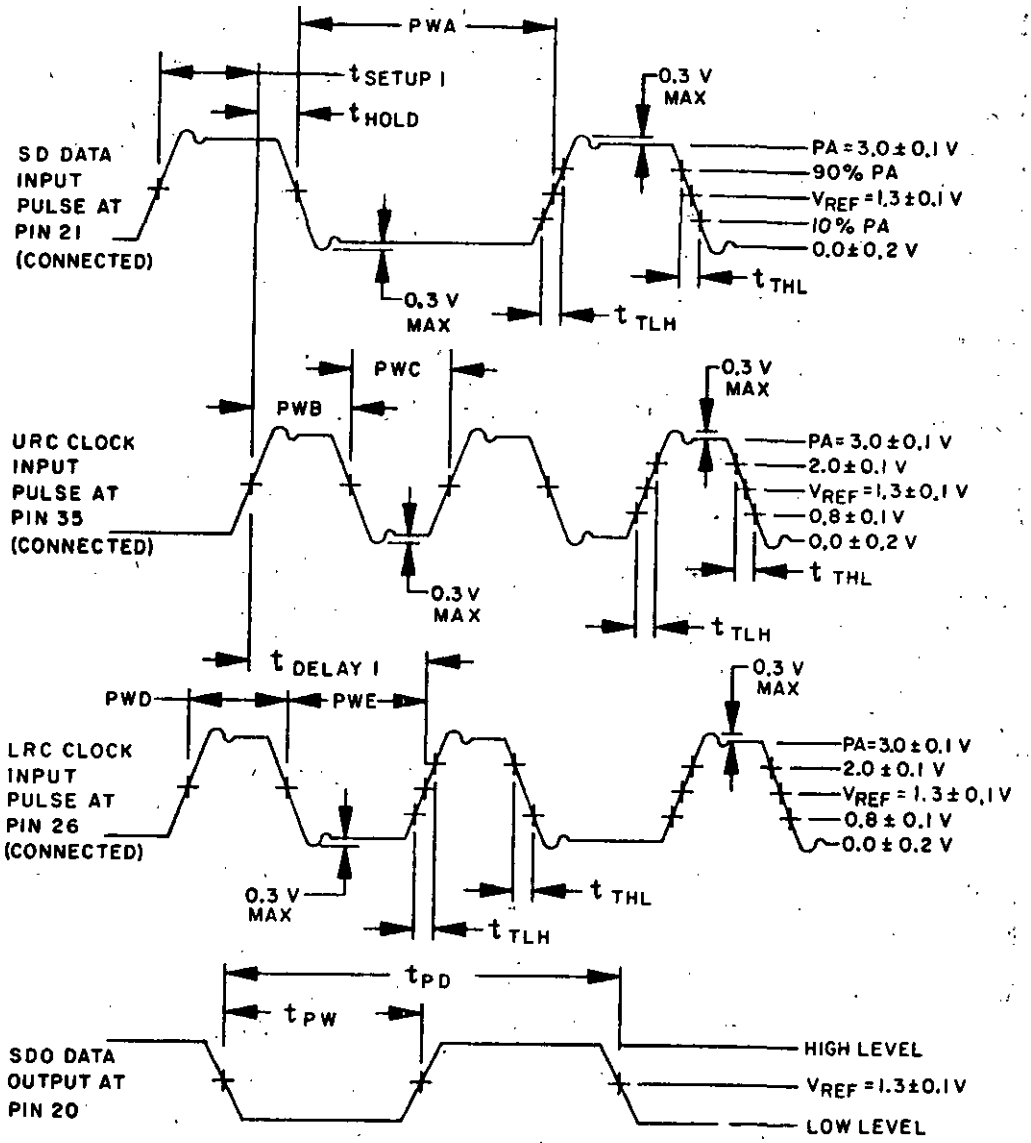


FIGURE 30. Upper rank clock (URC) and lower rank clock (LRC) minimum pulse width check (measuring output pulse period t_{PD} and output pulse width t_{PW}) test circuit waveforms for table XXII.

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NOTES:
1. SD data input pulse parameters:
   PWA = 115 ±5 ns.
   $t_{TLH} = t_{THL}$ = 4 ±2 ns.
   t_{SETUP1} = 20 ±3 ns.
   t_{HOLD} = 5 ±2 ns.
2. URC clock input pulse parameters:
   PWB = 20 $^{+1\ ns}_{-0\ ns}$ (minimum clock pulse width).
   PWC = 50 ±3 ns.
   $t_{TLH} = t_{THL}$ = 4 ±2 ns.
3. LRC clock input pulse parameters:
   PWD = 20 ±3 ns.
   PWE = 50 ±3 ns.
   $t_{TLH} = t_{THL}$ = 4 ±2 ns.
   t_{DELAY1} = 60 ±3 ns.
4. Item J01-2 is operated in the double pulse mode and uses the positive output. Using item D05 adjust the amplitude and dc offset controls to obtain the required input logic levels of data, upper rank clock, and lower rank clock input pulses. Using the appropriate delay and pulse width controls adjust the SD, URC, and LRC input pulses to obtain the parameters specified in notes 1, 2, and 3.
5. Verify that the SDO output has a high level ≥ 2.5 V and a low level ≤ 0.4 V before making measurements.
6. The physical length of conductors between resistors R1, R2, and R3 and input pins; between items Y01-1 and Y01-2 and monitored pins; and between the load circuit and output pin, shall be less than 1 inch (25.4 mm).
7. All voltages are referenced to 0 V unless otherwise specified.

FIGURE 30. Upper rank clock (URC) and lower rank clock (LRC) minimum pulse width check (measuring output pulse period t_{PD} and output pulse width t_{PW}) test circuit waveforms for table XXII - Continued.

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TABLE XXIII. D31 input setup and hold time check and URC to LRC clock delay check (measuring output pulse period, t_{PD} and output pulse width, t_{PW}).

Test condition	Parameter	Initial limits 25° ±5°C Ambient		Unit	Test figure
		Min	Max		
W1	D31 input setup and hold time check and URC to LRC clock delay check (measuring output pulse period, t_{PD})	180	220	ns	31,32
W2	D31 input setup and hold time check and URC to LRC clock delay check (measuring output pulse width, t_{PW})	85	115	ns	31,32

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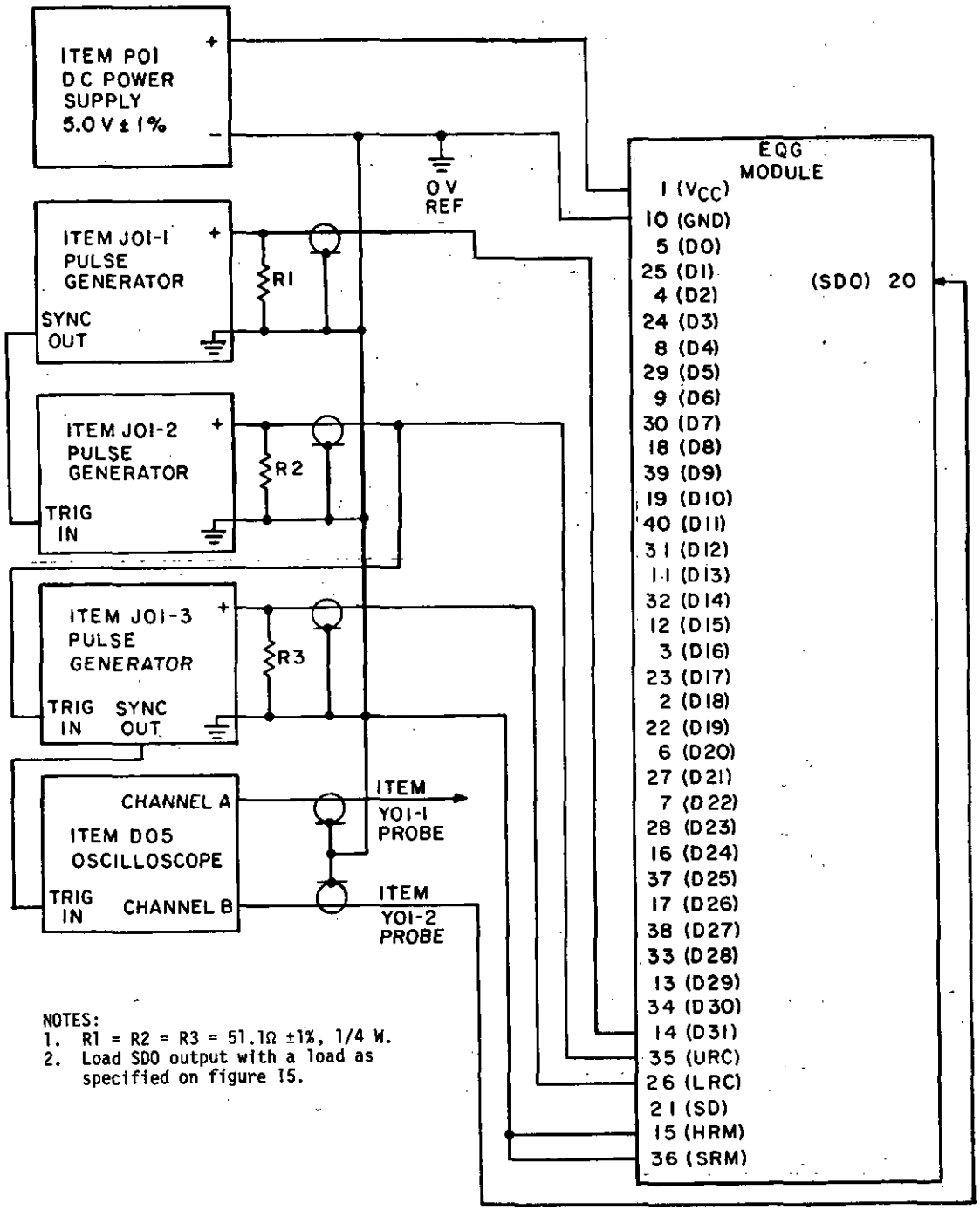


FIGURE 31. D31 input setup and hold time check and URC to LRC clock delay check (measuring output pulse period t_{PD} and output pulse width t_{PW}) test circuit for table XXIII.

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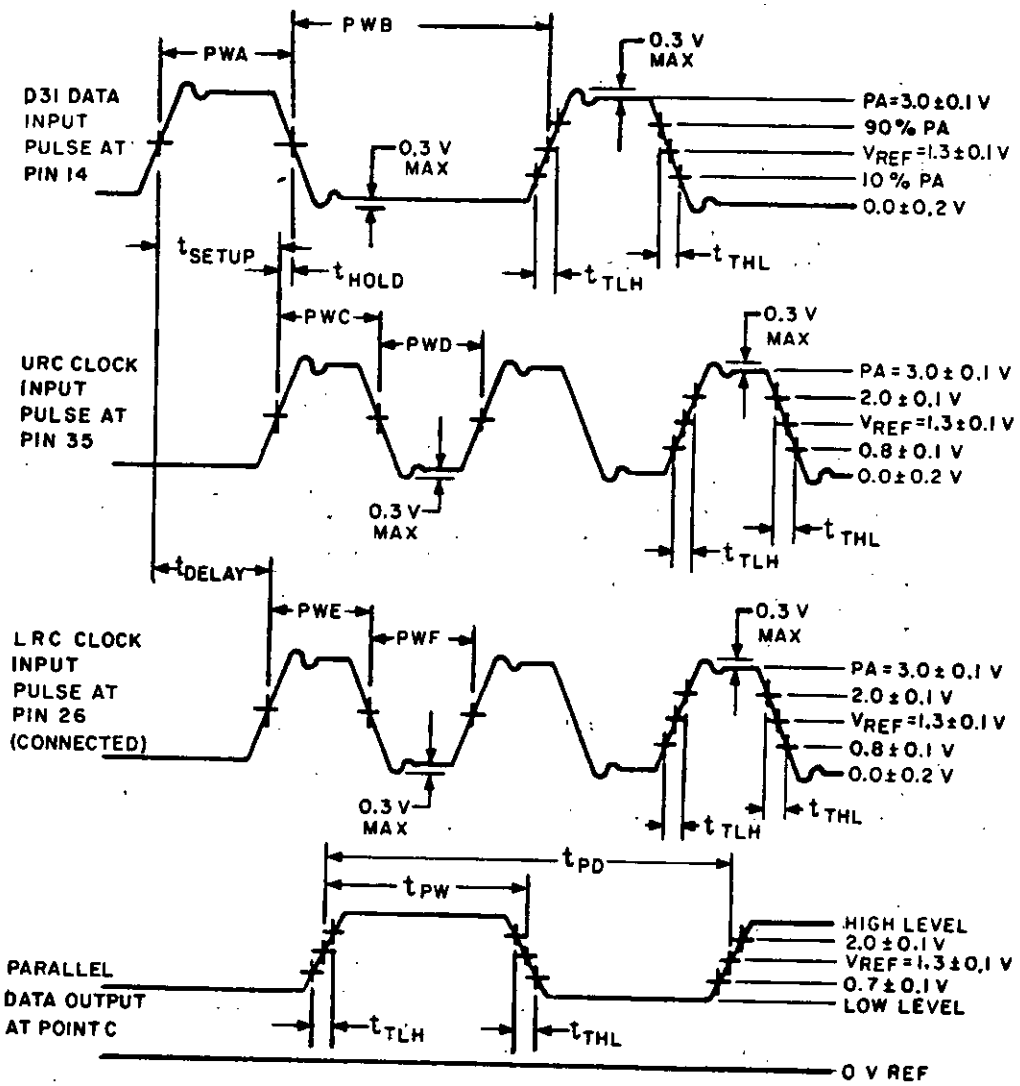


FIGURE 32. D31 input setup and hold time check and URC to LRC clock delay check (measuring output pulse period t_{PD} and output pulse width t_{PW}) test circuit waveforms for table XXIII.

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NOTES:

1. D31 data input pulse parameters:
   PWA = 25 ±1 ns.
   PWB = 175 ±10 ns.
   $t_{TLH} = t_{THL} = 4 \pm 2$ ns.
2. URC clock input pulse parameters:
   PWC = 20 ±2 ns.
   PWD = 80 ±4 ns.
   $t_{TLH} = t_{THL} = 4 \pm 2$ ns.
3. LRC clock input pulse parameters:
   PWE = 20 ±2 ns.
   PWF = 80 ±4 ns.
   $t_{TLH} = t_{THL} = 4 \pm 2$ ns.
4. Item J01-1 is used to drive the D31 input and uses the positive output. Item J01-2 is operated in the double pulse mode and provides the upper rank clock (URC) input and the item J01-3 trigger input which provides the lower rank clock (LRC) input. Using item D05 adjust the amplitude and dc offset controls to obtain the required input logic levels of data and clock input pulses. Using the appropriate delay controls adjust the input pulses so that:
   $t_{SETUP} = 20 \begin{smallmatrix}+1.0 \text{ ns} \\ -0.0 \text{ ns}\end{smallmatrix}$.
   $t_{HOLD} = 5 \begin{smallmatrix}+1.0 \text{ ns} \\ -0.0 \text{ ns}\end{smallmatrix}$.
   $t_{DELAY} = 65 \pm 3$ ns.
5. Verify that the SDO output has a high level $\geq$ 2.5 V and a low level $\leq$ 0.4 V before making measurements.
6. The physical length of electrical conductors between resistors R1, R2, and R3 and input pins; between items Y01-1 and Y01-2 and monitored pins; and between the load circuit and output pin, shall be less than 1 inch (25.4 mm).
7. All voltages are referenced to 0 V unless otherwise specified.

FIGURE 32. D31 input setup and hold time check and URC to LRC clock delay check (measuring output pulse period t_{PD} and output pulse width t_{PW}) test circuit waveforms for table XXIII - Continued.

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TABLE XXIV. Output crosstalk (low-level) test.

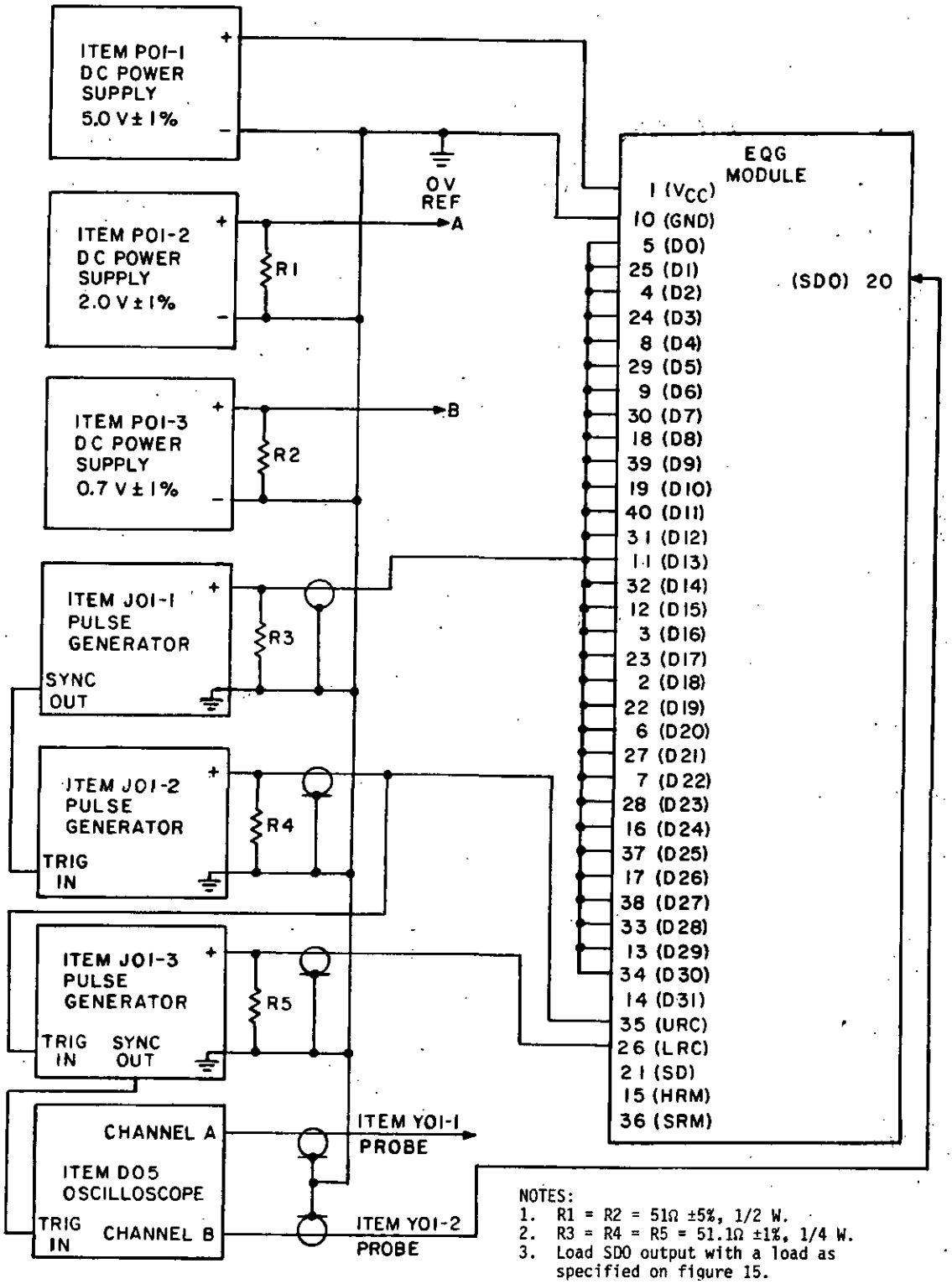
Test condition	Parameter	Connect		Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
X	SDO Output crosstalk, low level	Open	21,15,36,14	0	400	mV	33, 34

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Check the source to verify that this is the current version before use.

TABLE XXV. Output crosstalk (high-level) test.

Test condition	Parameter	Connect		Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
Y	SDO Output crosstalk, high level	14	21,15,36	0	400	mV	33, 34

Source: https://assist.dla.mil -- Downloaded: 202610-20T16:42Z
Check the source to verify that this is the current version before use.



NOTES:
1. R1 = R2 = 51Ω ±5%, 1/2 W.
2. R3 = R4 = R5 = 51.1Ω ±1%, 1/4 W.
3. Load SDO output with a load as specified on figure 15.

FIGURE 33. Output crosstalk (low-level and high-level) test circuit for tables XXIV and XXV.

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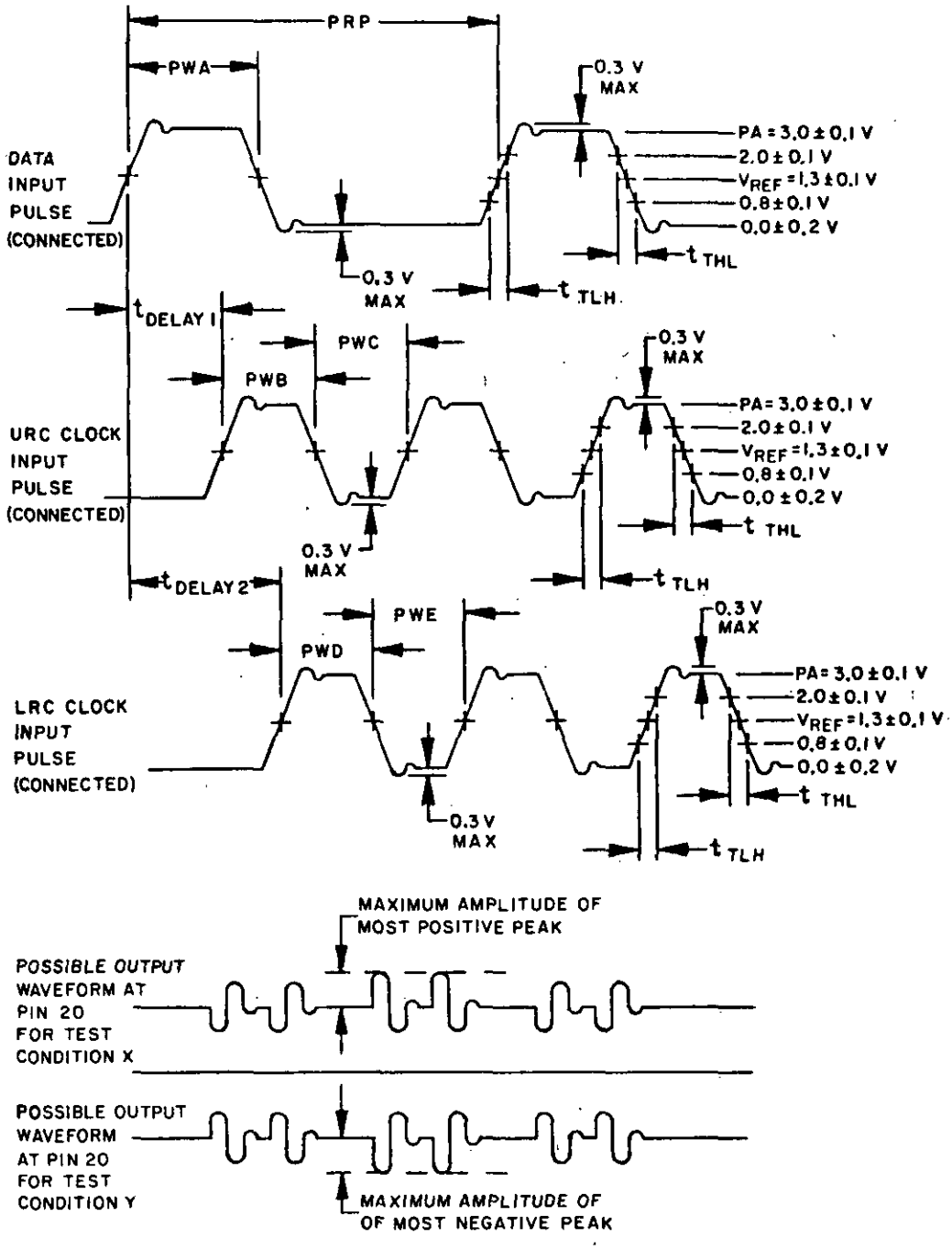


FIGURE 34. Output crosstalk (low-level and high-level) test circuit waveforms for tables XXIV and XXV.

Source: https://assist.dla.mil -- Downloaded: 202640-20T16:42Z
Check the source to verify that this is the current version before use.

NOTES:
1. Data input pulse parameters:
   PWA = 50 ±5 ns.
   PRP = 500 ±50 ns.
   $t_{TLH} = t_{THL} = 4 \pm 2$ ns.
2. URC clock input pulse parameters:
   PWB = 50 ±5 ns.
   PWC = 200 ±20 ns.
   $t_{TLH} = t_{THL} = 4 \pm 2$ ns.
   $t_{DELAY\ 1} = 25 \pm 2$ ns.
3. LRC clock input pulse parameters:
   PWD = 50 ±5 ns.
   PWE = 200 ±20 ns.
   $t_{TLH} = t_{THL} = 4 \pm 2$ ns.
   $t_{DELAY\ 2} = 125 \pm 12$ ns.
4. Item J01-2 is operated in the double pulse mode and uses the positive output. Using item D05 adjust the amplitude and dc offset controls to obtain the required input logic levels of the data, URC clock, and LRC clock input pulses. Using the delay and pulse width controls adjust the data and URC clock input pulses so that $t_{DELAY\ 1} = 25 \pm 2$ ns. Adjust the data and LRC clock input pulses so that $t_{DELAY\ 2} = 125 \pm 12$ ns.
5. The physical length of electrical conductors between resistors R3, R4, and R5 and input pins; between items Y01-1 and Y01-2 and monitored pins; and between the load circuit and the output pin, shall be less than 1 inch (25.4 mm).
6. All voltages are referenced to 0 V unless otherwise specified.

FIGURE 34. Output crosstalk (low-level and high-level) test circuit waveforms for tables XXIV and XXV - Continued.

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TABLE XXVI. Power supply transient amplitude test.

Test condition	Parameter	Initial limits 25° ±5°C Ambient		Unit	Test figure
		Min	Max		
Z	Power supply transient amplitude	0	600	mV	35,36

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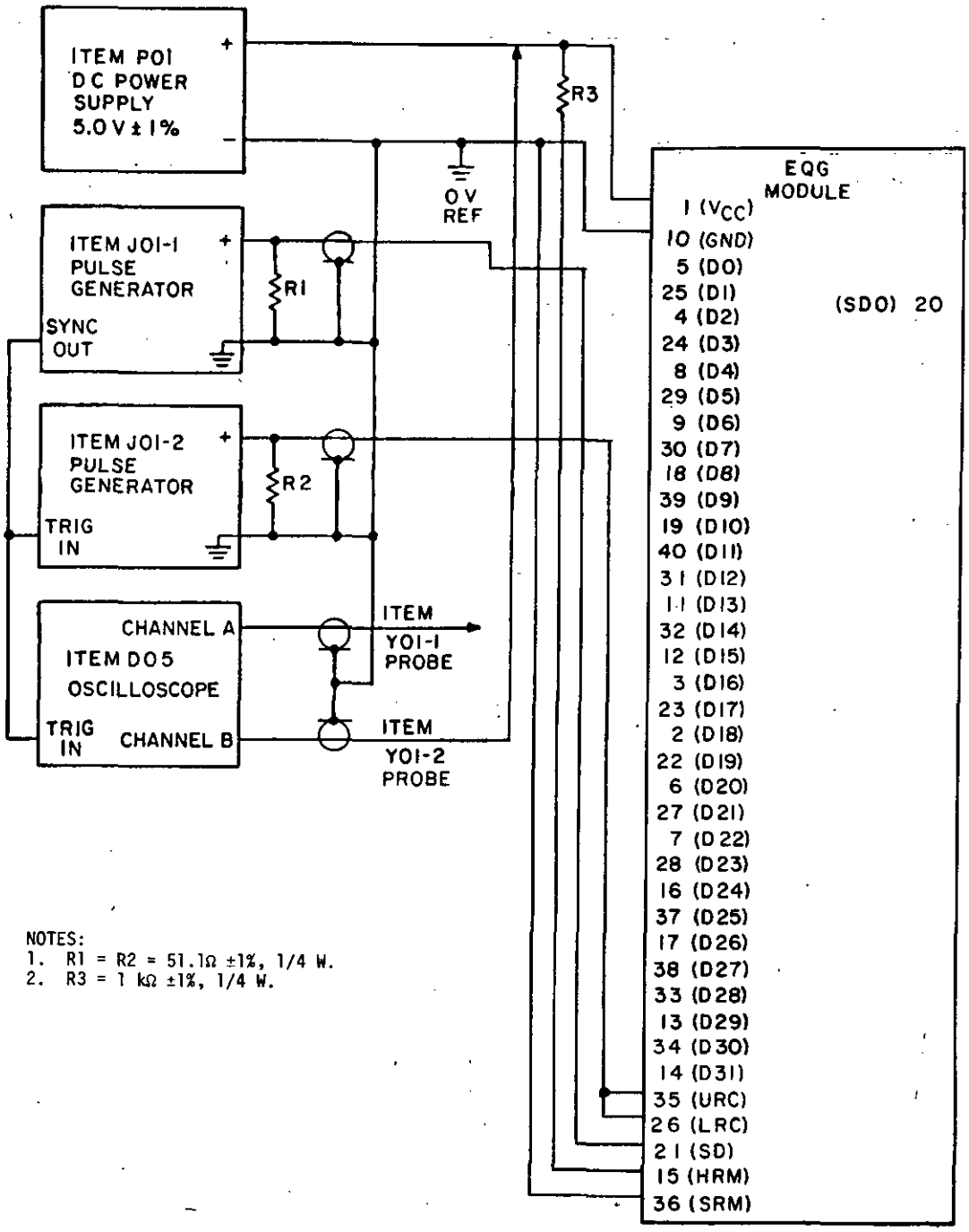
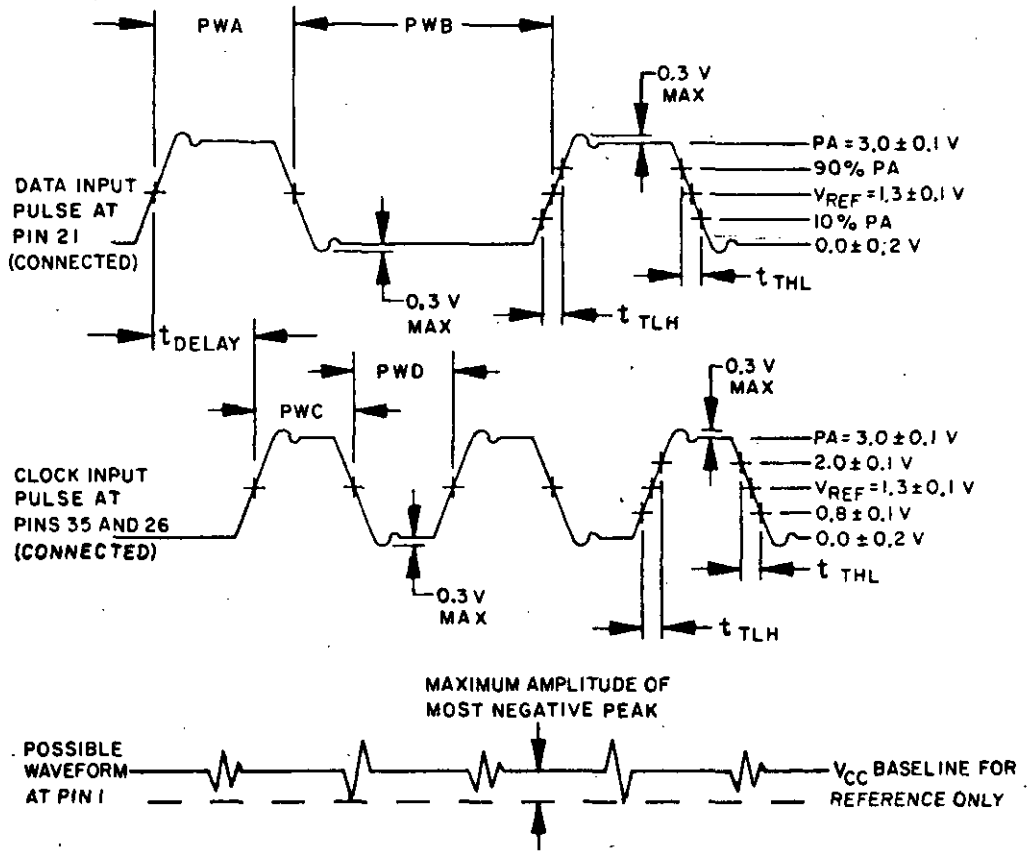


FIGURE 35. Power supply transient amplitude test circuit for table XXVI.

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NOTES:
1. Data input pulse parameters:
   PWA = PWB = 500 ±50 ns.
   $t_{TLH} = t_{THL}$ = 4 ±2 ns.
2. Clock input pulse parameters:
   PWC = PWD = 250 ±25 ns.
   $t_{TLH} = t_{THL}$ = 4 ±2 ns.
   t_{DELAY} = 250 ±25 ns.
3. Item J01-2 is operated in the double pulse mode. Using item D05 adjust the controls of items J01-1 and J01-2 to obtain t_{DELAY} = 250 ±25 ns and the pulse parameters specified in notes 1 and 2.
4. All voltages are referenced to 0 V unless otherwise specified.

FIGURE 36. Power supply transient amplitude test circuit waveforms for table XXVI.

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TABLE XXVII. Power supply filter capacitance test.

Test condition	Parameter	Connect		Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
AA	Power supply filter capacitance	Open	1	0.14	0.26	µF	37

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TABLE XXVIII. Input capacitance test.

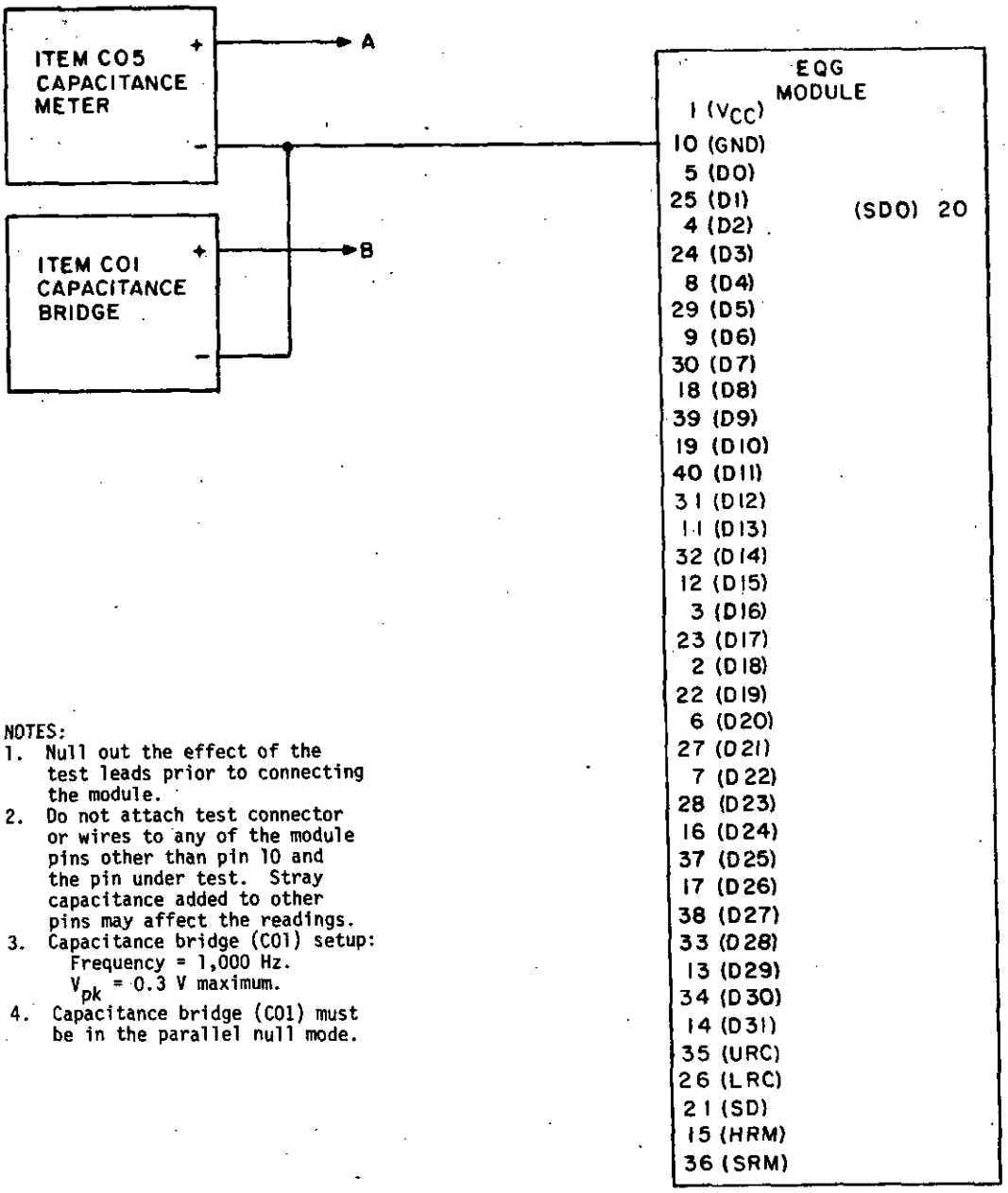
Test condition	Parameter		Connect		Initial limits 25° ±5°C Ambient		Unit	Test figure
			A to pin	B to pin	Min	Max		
BB1	D0	Input capacitance	5	Open	2	20	pF	37
BB2	D1		25					
BB3	D2		4					
BB4	D3		24					
BB5	D4		8					
BB6	D5		29					
BB7	D6		9					
BB8	D7		30					
BB9	D8		18					
BB10	D9		39					
BB11	D10		19					
BB12	D11		40					
BB13	D12		31					
BB14	D13		11					
BB15	D14		32					
BB16	D15		12					
BB17	D16		3					
BB18	D17		23					
BB19	D18	Input capacitance	2	Open	2	20	pF	37

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TABLE XXVIII. Input capacitance test - Continued.

Test condition	Parameter		Connect		Initial limits 25° ±5°C Ambient		Unit	Test figure
			A to pin	B to pin	Min	Max		
BB20	D19	Input capacitance	22	Open	2	20	pF	37
BB21	D20		6					
BB22	D21		27					
BB23	D22		7					
BB24	D23		28					
BB25	D24		16					
BB26	D25		37					
BB27	D26		17					
BB28	D27		38					
BB29	D28		33					
BB30	D29		13					
BB31	D30		34					
BB32	D31		14			20		
BB33	SD		21			50		
BB34	URC		35			100		
BB35	LRC		26					
BB36	HRM		15					
BB37	SRM	Input capacitance	36	Open	2	100	pF	37

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NOTES:
1. Null out the effect of the test leads prior to connecting the module.
2. Do not attach test connector or wires to any of the module pins other than pin 10 and the pin under test. Stray capacitance added to other pins may affect the readings.
3. Capacitance bridge (CO1) setup:
   Frequency = 1,000 Hz.
   V_{pk} = 0.3 V maximum.
4. Capacitance bridge (CO1) must be in the parallel null mode.

FIGURE 37. Power supply filter and input capacitance test circuit for tables XVII and XVIII.

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